04820KA001 Mono Slitter 903_V12_1110_6

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Sysmac Studio Module Version: 1.60.0.64010

1.PLC

1-8.POUs

1-8-1.Programs

1-8-1-1.TaglMonoTRM

1-8-1-1-1.Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Comment
VAR						
SpdDelta	REAL			False	False	Minimum or maximum speed SpdDeltaerence
UwSlowDiamCalc	REAL			False	False	V7c: speed-dependent unwinder slow-down diameter (mm)
SonarRunMem	bool			False	False	
EndStopReq	BOOL			False	False	V12: end-of-roll stop requested (AutoStopCmd or unwinder diameter)
EndRampOK	BOOL			False	False	V12: end-of-roll ramp usable (FinalStopDecel, FF active, running)
EndRampHard	BOOL			False	False	V12: safety net during the ramp -> immediate stop
EndStopNow	BOOL			False	False	V12: immediate end-of-roll stop (Motor_Manager rung 4)
TensTarget	REAL			False	False	Tension Manager: target before rate limit (kg)
TensStep	REAL			False	False	Tension Manager: max change per 4 ms scan (kg)
TensAOReal	REAL			False	False	Tension Manager: AO counts before INT conversion
SonarInit	bool			False	False	
AlarmMem	ARRAY[0..32] OF bool			False	False	
UwSlowDiamLatch	BOOL			False	False	V7c: unwinder diameter slow-down reached (frozen)
FFRate	REAL			False	False	V7f: ramp rate (reference per 4 ms scan)
FFRateSet	REAL			False	False	V7f: rate allowed to still stop at the target
FFJerk	REAL			False	False	V7f: max rate change per scan
SpdScanCnt	INT			False	False	V9: scan counter for the 100 ms speed window
TensGSMMem	REAL			False	False	
TensWidthMem	REAL			False	False	
TensKraftMem	BOOL			False	False	
HornRun	BOOL			False	False	V9: slow-down / final speed running
HornRunMem	BOOL			False	False	V9: edge memory
HornStartMem	BOOL			False	False	V9: start edge memory
HornTicks	INT			False	False	V9: scans since slow-down start
HornCnt	INT			False	False	V9: scans in the beep period
HornPer	INT			False	False	V9: beep period in scans
HornStartScans	INT			False	False	V9: start beep in scans
HornLenScans	INT			False	False	V9: short beep in scans
HornSlowBeep	BOOL			False	False	V9: slow-down beep
HornStopPend	BOOL			False	False	V9: stopped after a slow-down, waiting for standstill
HornStopTicks	INT			False	False	V9: stop beep counter
LongActive	BOOL			False	False	V9: long-baseline learning runni

						ng
LongValid	BOOL			False	False	V9: long baseline >= 20 mm growth, replaces the windows
LongStartPos	DINT			False	False	V9: encoder at the start of the long baseline
LongStartD	REAL			False	False	V9: rewinder diameter at the start (mm)
LongGrow_mm	REAL			False	False	V9: growth since the start (mm)
LongPls	REAL			False	False	V9: pulses since the start
LongK	REAL			False	False	V9: K from the long baseline
RevW	REAL			False	False	V9: encoder pulses summed in the current revolution
RevDP	DINT			False	False	V9: encoder pulses this scan
RevPrevPos	DINT			False	False	V9: encoder last scan
RevSum	REAL			False	False	V9: sum of raw DT35 distance x pulses
RevMin	REAL			False	False	V9: min raw distance in the revolution
RevMax	REAL			False	False	V9: max raw distance in the revolution
RwDiamRevOk	BOOL			False	False	V9: one full revolution averaged since the last fault
UwEmptyTicks	INT			False	False	V9: unwinder empty time (100 ms ticks)
UwFullTicks	INT			False	False	V9: new roll seen time (100 ms ticks)
GsmArmed	BOOL			False	False	V9: roll was empty -> next new roll asks the GSM question
T_JogStartBlk	TOF			False	False	V9: START blocked while JOG held + 2 s
StartHeldBlk	BOOL			False	False	V9: START pressed during the block, wait for release
StartPB_OK	BOOL			False	False	V9: START push-button accepted (replaces I_Start_PB in General 7/8/10)
FFSoftMem	BOOL			False	False	V9: soft stop last scan (rung B)
FFSlowAcc	BOOL			False	False	V9: gentle ramp-up after a cancelled soft stop
FFStartEdge	BOOL			False	False	V9: START push-button edge (rung B)
FFStartMem	BOOL			False	False	V9: START last scan (rung B)
SpdDeltaMin	REAL			False	False	Minimum speed difference
FFSqrtArg	REAL			False	False	SQRT argument (REAL)
SpdDeltaMax	REAL			False	False	Maximum speed SpdDeltaerence
SoftStopMem	BOOL			False	False	v6r: soft stop edge memory
SoftStopRatio	REAL			False	False	v6r: InvRef counts per m/min, frozen at the soft stop start
MemPos	DINT			False	False	Previous encoder count
MatChangeCnt	INT			False	False	v6s: consecutive learning windows far from RwMat_K
DeltaPulse	DINT			False	False	Encoder count difference
SlowMem	BOOL			False	False	SlowDown edge memory
DecelDiam	REAL			False	False	Rewinder diameter used by the decel model (mm, min 100)
TeachSwap	REAL			False	False	Temporary for UNDO swap
SlowStartDiam	REAL			False	False	Diameter at slow-down start (mm)
BrakeHold	BOOL			False	False	Keep T-one in auto during a normal stop (O_Brake)
StopNow	BOOL			False	False	Immediate stop request (replaces

						I_Stop_PB in General rung 9)
StopPB_Mem	BOOL			False	False	STOP button edge memory
T_StopLong	TON			False	False	STOP held 1 s = immediate stop
T_SoftStop	TON			False	False	Soft stop timeout (120 s)
BrakeHoldTicks	INT			False	False	BrakeHold duration after stop (100 ms ticks, max 150)
StopStartDiam	REAL			False	False	Diameter at stop command (mm)
DecelK_Meas	REAL			False	False	Measured decel per mm of diameter
SlowMeasActive	BOOL			False	False	Slow-down measurement running
TeachTicks	INT			False	False	TEACH CORE stable time (100 ms ticks)
SlowStartPos	DINT			False	False	Encoder at slow-down start
SlowStartSpd_Pps	REAL			False	False	Speed at slow-down start (pulses/s)
SlowMeas_Pls	REAL			False	False	Measured slow-down length (pulses)
SlowDecelMeas_mm	REAL			False	False	Measured slow-down decel (mm/s2)
SlowOneRamp	BOOL			False	False	Slow-down uses one ramp (manual or learned)
Velocity	REAL			False	False	Speed in [mm/s]
Alarm_Array	ARRAY[0..32] OF BOOL			False	False	Alarms
T_PlcOn	TON			False	False	PLC active timeout
ControlVoltage	BOOL			False	False	Auxiliaries OK
StartMotor_Cmd	BOOL			False	False	Motor start command
HubExtractEnable	BOOL			False	False	Hub extraction enable command
Brake_Cmd	BOOL			False	False	Brake command
T_StopVent	TOF			False	False	Fans stop timer
TimStopVen	TIME			False	False	Fans stop timeout
MotorCooling	BOOL			False	False	Fans command
OverRider_Auto	BOOL			False	False	OverRider in automatic
OverRider_Hoover	BOOL			False	False	OverRider in Hoover Mode
OverRider_Lock	BOOL			False	False	OverRider lock command
GSM_LearnFault	BOOL			False	False	Paper [Grams / Square Meter] LearnFailCnt ≥ RwLearnFailMax
TickCnt	INT			False	False	100 ms tick scan counter (25 x 4 ms)
RunWasJog	BOOL			False	False	Run used jog / manual / slow button (2nd ramp)
StopLearnValid	BOOL			False	False	This stop may be learned
SpdBand_Pps	REAL			False	False	Speed band of the trim ramp (pulses/s)
TrimDecel_Pps2	REAL			False	False	Trim ramp decel (pulses/s2)
SpdKnee_Pps	REAL			False	False	Speed where coarse ramp hands over to trim ramp (pulses/s)
OverRider_RaiseUp	BOOL			False	False	OverRider raise command
Pot1_Mem	INT			False	False	Work potentiometer value
VelSet_mms	REAL			False	False	Set speed in mm/s
Vfd_SP_Modif	REAL			False	False	Inverter speed modifier
RwAutoLimAct	BOOL			False	False	v6y: rewinder AUTO speed limit active (taken over at standstill)
UwTeachTicks	INT			False	False	Unwinder TEACH CORE samples (100 ms)
T_PosPresOff	TON			False	False	OverRider photocell OFF timer

T_PosPresOn	TON			False	False	OverRider photocell ON timer
TimPosPresOff	TIME			False	False	OverRider photocell OFF timeout
TimPosPresOn	TIME			False	False	OverRider photocell ON timeout
RollEnd_Active	BOOL			False	False	Roll diameter photocell intercepted
Output_Ratio	REAL			False	False	Analog output ratio
Input_Ratio	REAL			False	False	Analog input ratio
T_StartLow	TON			False	False	Slow start timer
UwDiamReached	BOOL			False	False	Roll diameter reached
Index_A	INT			False	False	
AlarmStrobe	BOOL			False	False	Alarm message display strobe
T_Display	TON			False	False	Alarm display timer
RangeOK	BOOL			False	False	Range reached
Vel_Max_mms	REAL			False	False	Maximum speed in mm/s
Slow_Inhibit	BOOL			False	False	Inhibit slow button
Decel_Ramp	REAL			False	False	Deceleration ramp
Accel_Ramp	REAL			False	False	Acceleration ramp
Pot2_Mem	INT			False	False	Final potentiometer value
Current_mm	REAL			False	False	Current millimetres
EncoderPulse_to_mm	REAL			False	False	mm per encoder pulse
Stop	BOOL			False	False	Stop
T_EncoderStill	TON			False	False	(?) Encoder stopped timer
Analog_Res	UINT			True	False	Analog board resolution
Mt_To_mm	REAL			False	False	m/min to mm/s factor
Init_Ramp	BOOL			True	False	(?) Ramp initialisation
SlowByDiam	BOOL			False	False	Slow down by diameter
UwStopByDiam	BOOL			False	False	Stop by diameter
T_Screen_Delay	TON			False	False	(?) Start page timer
T_PulStop	TON			False	False	(?) Stop button timer
Reading_Max	REAL	2000		True	False	Maximum sensor reading (full scale)
DeadZone	REAL	60		True	False	Sensor dead zone
In_UwDiameter	REAL			True	False	UnWinder Diameter Analog input voltage
Tick	BOOL			False	False	100 ms tick
MmPerPulse	REAL			False	False	mm per encoder pulse
RwDistRaw_mm	REAL			False	False	Rewinder Unfiltered distance (mm)
RwFiltInit	BOOL			False	False	Rewinder Filter initialised
SpdInit	BOOL			False	False	Speed init
EncPrev	DINT			False	False	Encoder at previous tick
SpdAct_Pps	REAL			False	False	Actual Speed (pulses/s)
LearnInit	BOOL			False	False	Learn initialised
LearnWinRunning	BOOL			False	False	Motor ran whole window
LearnWinTicks	INT			False	False	Learn window counter
LearnWin_Pls	DINT			False	False	Pulses in window
LearnWinDiamDelta	REAL			False	False	Diameter change in window
MatK_Window	REAL			False	False	Material factor this window
MotorRunMem	BOOL			False	False	Motor start edge memory
StopMeasActive	BOOL			False	False	Stop measurement running
StopStartPos	DINT			False	False	Encoder at stop command
StopStartSpd_Pps	REAL			False	False	Speed at stop command (pulses/s)
StandstillTicks	INT			False	False	Standstill tick counter
LearnWinOk	BOOL			False	False	Learning window gave a valid re

						sult
ThickWin_mm	REAL			False	False	Paper thickness this window (mm)
LearnFailCnt	INT			False	False	Consecutive failed learning windows
StopLearnOk	BOOL			False	False	Stop measurement gave a valid result
StopDecelMeas_mm	REAL			False	False	Measured stop decel (mm/s2)
StopMeas_Pls	REAL			False	False	Measured stop length (pulses)
StopDecelMeas	REAL			False	False	Measured stop decel (pulses/s2)
StopLen_Pls	REAL			False	False	Stop length (pulses)
StopLenFinal_Pls	REAL			False	False	Stop length from slow speed (pulses)
RwSlowLen_Pls	REAL			False	False	Rewinder Slow-down length (pulses)
SlowDecel_Pps2	REAL			False	False	Slow-down decel (pulses/s2)
SpdFinal_Pps	REAL			False	False	Final speed (pulses/s)
RwRemain_Pls	REAL			False	False	Rewinder Remaining length to target (pulses)
LearnWinStartPos	DINT			False	False	Rewinder Position Window 0
LearnWinStartDiam	REAL			False	False	Rewinder Diameter Window 0
T_ByPassProtOff	TOF			False	False	Delay Off Timer
UwEnMem	BOOL			False	False	Edge memory UwStopEnable
RwEnMem	BOOL			False	False	Edge memory RwEnable
SpdRef_Pps	REAL			False	False	Reference speed MAX (actual,setpoint) (pulses/s)
SpdWork_Pps	REAL			False	False	Work speed setpoint (pulses/s)
SlowLenBase_Pls	REAL			False	False	Slow-down length without margin (pulses)
UwSlowLen_Pls	REAL			False	False	Unwinder slow-down length (pulses)
UwMarginUsed_m	REAL			False	False	Unwinder margin at slow speed (m)
SqrtArg	REAL			False	False	Square root argument
AutoSlowCmd	BOOL			False	False	Automatic Slow Command
AutoStopCmd	BOOL			False	False	Automatic Stop Command
RwLastValidDiam	REAL			False	False	Rewinder Last valid diameter (mm)
RwRemainMax_Pls	REAL			False	False	Rewinder Remaining length to max diameter (pulses)
UnWinder_Empty	BOOL			False	False	UnWinder Empty
StopFailCnt	INT			False	False	Consecutive failed stop measurements
UwTeachSum	REAL			False	False	Unwinder TEACH CORE sonar sum (mm)
T_MatReset	TON			False	False	v6s: RwMatReset stays ON 3 s
RwAutoMaxSpdAct	REAL			False	False	v6y: active rewinder AUTO max speed (taken over at standstill)
UwEmptyMem	BOOL			False	False	v6x: UnWinder_Empty edge memory
LearnWinLen_Pls	REAL			False	False	v6w: learning window length in encoder pulses
WebRatio	REAL			False	False	V7: drive counts per m/min of web at the current diameter
FFActive	BOOL			False	False	V7: feed-forward loop active

	FFWasActive	BOOL			False	False	V7: edge memory
	FFTarget	REAL			False	False	V7: reference for the set speed (mm/s of the 0-10 V scale)
	RampRef	REAL			False	False	V7: ramped reference (mm/s of the 0-10 V scale)
	StepAcc	REAL			False	False	V7: accel step per scan
	StepDec	REAL			False	False	V7: decel step per scan
	FFSteadyTicks	INT			False	False	V7: scans at target
	FFErr	REAL			False	False	V7: relative speed error
	Clock_100ms	bool			False	False	Clock100ms
	AlarmCnt	INT			False	False	
	AlarmCntMem	INT			False	False	
	RstEdge	BOOL			False	False	
	RstMem	BOOL			False	False	
	SilenceTicks	INT			False	False	
	WarnCnt	INT			False	False	
	WarnCntMem	INT			False	False	
	WarnBeepTicks	INT			False	False	
	TensIdeal_N	REAL			False	False	
	TensionSeen	BOOL	False		False	False	Tension reached once since START
	T_TensOn	TON			False	False	Tension ON debounce 0.5 s
	T_TensOff	TON			False	False	Tension OFF debounce 2 s
	T_UwSensFlt	TON			False	False	Unwinder sonar fault debounce 1 s
	AS_RunMem	BOOL			False	False	
	AutoStopSeen	BOOL			False	False	
	T_AutoStopDisp	TON			False	False	
	RwArmMem	BOOL			False	False	
	UwArmMem	BOOL			False	False	
	T_AutoEnRef	TON			False	False	
	TensRef	REAL			False	False	
	WebBreakNow	BOOL			False	False	
VAR_EXTERNAL							
	VelSet	REAL				False	Set speed
	OverSpeed	BOOL				False	Speed too high
	SpdDelta_Percent	REAL				False	Speed difference percentage
	Encoder_Pos	DINT				False	Encoder counter
	FinalStopDecel	REAL				False	
	FinalStopRamp	BOOL				False	
	Velocity_M_min	REAL				False	Speed in m/min
	Current_Metre	REAL				False	Current metres
	PaperKraft	BOOL				False	
	TensFactorBleached	REAL				False	
	TensFactorKraft	REAL				False	
	TensAuto	BOOL				False	
	TensAOEnable	BOOL				False	
	TensManual_kg	REAL				False	
	TensTaper	REAL				False	
	TensCoreBoost	REAL				False	
	TensRate_kg_s	REAL				False	
	TensMin_kg	REAL				False	
	TensMax_kg	REAL				False	
	TensSet_kg	REAL				False	
	TensAtCore	REAL				False	
	TensAtMid	REAL				False	
	TensAtMax	REAL				False	Metre counter reset button

TensDiaFrac	REAL				False	
SonarAskYes	BOOL				False	
WorkSpd_Max_Set	REAL				False	HMI request: work pot full scale (m/min), applied only with motor stopped
FinalSpd_Max_Set	REAL				False	HMI request: final pot full scale (m/min), applied only with motor stopped
SpdMax_Pending	BOOL				False	New pot range waiting for the motor to stop
RwAxisDist_Prev	REAL				False	Rewinder: RwAxisDist before the last TEACH (for UNDO)
UwCoreDist_Prev	REAL				False	Unwinder: UwCoreDistance before the last TEACH (for UNDO)
RwTeachUndo	BOOL				False	HMI: UNDO rewinder TEACH (PLC resets it)
UwTeachUndo	BOOL				False	HMI: UNDO unwinder TEACH (PLC resets it)
RwLearnReset	BOOL				False	HMI / Watch: clear learned decels RwStopDecelK, RwStopDecel, RwSlowDecelK (PLC resets it)
UnderSpeed	BOOL				False	Speed too low
I_ControlVoltage	BOOL				False	Auxiliaries ON
I_Side_Door_Closed	BOOL				False	Roll diameter photocell
RwCoreDiameter	REAL				False	Rewinder core outside diameter for TEACH CORE (mm)
UwTeachState	INT				False	Unwinder TEACH CORE: 0 idle 1 teaching 2 done 3 refused
SoftStop	BOOL				False	Soft stop in progress (HMI)
SoftStopEnable	BOOL				False	STOP button = soft stop (False = immediate as before)
SoftStopEndSpd	REAL				False	Soft stop: run command drops below this speed (m/min)
SoftStopDecel	REAL				False	Soft stop decel (mm/s2), 0 = DecelCoarse
RwTeachCore	BOOL				False	HMI: TEACH CORE request (PLC resets it)
RwTeachState	INT				False	TEACH CORE: 0 idle 1 teaching 2 done 3 refused
O_MotorStart	BOOL				False	Motor start
RwSlowDecelK	REAL				False	Learned slow-down decel per mm of rewinder diameter (mm/s2 per mm)
RwStopDecelK	REAL				False	Learned stop decel per mm of rewinder diameter (mm/s2 per mm)
I_OR_Position_Sens	BOOL				False	Rider roll position sensor
WorkSpd_Max	REAL				False	Work speed pot full scale (m/min), 50..VelMax
FinalSpd_Max	REAL				False	Final speed pot full scale (m/min), 5..WorkSpd_Max
I_OR_Hoover_Mode	BOOL				False	Rider roll pressure switch
I_OR_Auto_OS	BOOL				False	Rider roll in automatic
I_OR_Cont_OS	BOOL				False	Rider roll in continuous
I_Start_PB	BOOL				False	Start button
I_Motor_Therma	BOOL				False	Motor thermal switch

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I_FanTemp_Ok	BOOL				False	Motor (fan) thermal OK
I_Stop_PB	BOOL				False	Stop button
I_Hub_Position	BOOL				False	Hub/chuck in extraction position
I_FrontGuard_Closed	BOOL				False	Front guard closed
RwMaxSlowFallback	REAL				False	Rewinder max: slow-down diameter while K not learned (mm)
RwMaxSlowAtDiam	REAL				False	Rewinder max: slow-down starts at diameter (mm)
O_Alarm_Array	BOOL				False	Alarm lamp
O_RwUnload	BOOL				False	Unloaders enable - opening
I_Prot_ByPass	BOOL				False	Guards-bypassed selector switch
O_Prot_ByPass_Lamp	BOOL				False	Guards bypassed lamp
O_Brake	BOOL				False	Brake command
Fan_Timeout	REAL				False	Fans stop timeout (from panel)
O_CoolingFan	BOOL				False	Brake and motor fans
O_OR_Auto_Lamp	BOOL				False	Rider roll auto/continuous lamp
O_OverRider_Up	BOOL				False	Raise rider roll
O_OverRider_Hold	BOOL				False	Lock rider roll
AO_TOne_SetPoint	INT				False	Fixed 10 Vdc generator
InvRef	INT				False	Inverter speed reference
Encoder_On	BOOL				False	Encoder enable
Encoder_Reset	BOOL				False	Encoder reset
OverRider_TimeOFF	REAL				False	Rider roll photocell OFF timeout (panel)
OverRider_TimeON	REAL				False	Rider roll photocell ON timeout (panel)
VelMax	REAL				False	Maximum speed
I_Jog_PB	BOOL				False	Slow speed button
O_Motor_Lamp	BOOL				False	Motor ON lamp
AlarmOn	BOOL				False	Alarm active
BuzzerOn	BOOL				False	Panel buzzer command
I_VFD_Alarm	BOOL				False	Inverter in alarm
Msg	INT				False	Message
Index_B	INT				False	Active alarm index
MotStop	BOOL				False	Motor stopped
Clock1s	BOOL				False	1 s clock
ManMode	BOOL				False	Manual mode
Counter_Reset	BOOL				False	Panel counter reset button
I_FrontGuard_Open	BOOL				False	Front guard open
O_RwLoading_Enable	BOOL				False	Unloaders enable - closing
JogSpd_SP	REAL				False	Speed in manual
O_AccDec2	BOOL				False	Second acc/dec ramp
AccelTrim	REAL				False	Acceleration coefficient in run
DecelTrim	REAL				False	Deceleration coefficient in run
DecelCoarse	REAL				False	Acceleration coefficient at start (orig. comment; name suggests decel)
AccelCoarse	REAL				False	Deceleration coefficient at start (orig. comment; name suggests accel)

Speed_Display	REAL				False	Set speed (displayed)
O_AcousticSignal	BOOL				False	Horn / acoustic signal
Pot1_Work_Velocity	INT				False	Potentiometer 1 value - Work speed -
O_FrontGuard_ByPass	BOOL				False	Front guard bypass
Pot2_Slow_Velocity	INT				False	Potentiometer 2 value - Final speed -
LowSpd_Display	REAL				False	Slow-down speed (displayed)
Encoder_Pulse	INT				False	Encoder pulses
RwSlowDecelLrn	REAL				False	Learned real slow-down decel (mm/s2) - 0 = not learned
UwSlowMargin_m	REAL				False	Slow-down metres
UwStop_at_M_Val	REAL				False	Total metres
I_AutoStop_OS	BOOL				False	Stop on metre counter
I_Input_11_Spare_1255	BOOL				False	Slow down at end of roll
I_T_one_TensionOK	BOOL				False	Slow-down photocell
I_WebBreak_Monitor	BOOL				False	Paper break monitoring ON
I_WebBreak_PH	BOOL				False	Paper break photocell
I_Vacuum_OK	BOOL				False	Extractor fan thermal OK
CountReset_Req	BOOL				False	Metre counter reset request
Psw	INT				False	Password
I_RsetAlarm	BOOL				False	Alarm reset
EncoderWheel_Circumf	REAL				False	Encoder wheel diameter
AlarmCodeSV	ARRAY[0..32] OF BOOL				False	Panel alarms
SafeGuard_ByPass	BOOL				False	Guards bypassed
FrontGuard_Closed	BOOL				False	Front guard closed
Core_Measure	BOOL				False	(?) Stored core value
UwDiam_Measure	REAL				False	Measured distance
UwCoreDistance	REAL				False	(?) Core measurement
RwSlowFallbackMm	REAL				False	
AI_Diam_Unwinder	INT				False	Analog value - parent roll measurement
RwMatChangePct	REAL				False	
UwCoreDiameter	REAL				False	(?) Shaft diameter
Measure_ON	BOOL				False	(?) Measurement enabled
UwDiam_Act	REAL				False	(?) Measured roll diameter
UwStopDiam_Limit	REAL				False	(?) Stop diameter
UwSlow_at_D_Val	REAL				False	(?) Slow-down diameter
Alarm_Reset	BOOL				False	(?) Alarm reset from panel
HMI_Screen	ARRAY[0..1] OF DINT				False	HMI panel page
Psw22	BOOL				False	(?) Password 22
RwMatReset	BOOL				False	

Psw44	BOOL				False	(?) Password 44
AI_Diam_Rewinder	INT				False	Rewinder DT35 raw - I/O Map Unit2 Ch2
RwStopEnable	BOOL				False	Rewinder diameter stop enabled
RwTargetDiam	REAL				False	Rewinder final diameter (mm)
RwUseSlow	BOOL				False	Slow down before target
RwSlowMargin_m	REAL				False	Metres at slow speed before stop
UwRawMin	INT				False	
UwRawMax	INT				False	
RwSlowDecel	REAL				False	Slow-down decel mm/s2 - 0 = auto from DecStart
RwResetHyst	REAL				False	Reset requests below target - this (mm)
RwDist0V	REAL				False	DT35 distance at 0 V (mm)
RwDist10V	REAL				False	DT35 distance at 10 V (mm)
RwRawMin	INT				False	Raw below = sensor fault
RwRawMax	INT				False	Raw above = sensor fault
RwStopOnFault	BOOL				False	Stop on sensor fault
RwAxisDist	REAL				False	Sensor to shaft centre (mm) - teach
RwFilter_K	REAL				False	Distance filter per 100 ms (0..1)
RwMinPulses	DINT				False	Min pulses in window to learn
RwLearn_K	REAL				False	Material factor filter weight
RwMat_K	REAL				False	Learned material factor
RwLearnStop	BOOL				False	Learn stop decel on every stop
RwMinLearnSpd	REAL				False	Min speed to learn a stop (m/min)
RwStopDecel	REAL				False	Learned stop decel (pulses/s2)
RwTreact	REAL				False	Stop reaction time (s)
RwTstopDef	REAL				False	Stop time until decel learned (s)
RwTrim	REAL				False	Stop length trim (>1 = earlier)
RwDist	REAL				False	Filtered distance (mm)
RwDiam	REAL				False	Rewinder diameter (mm)
RwSpeed	REAL				False	Line speed from encoder (m/min)
RwRemain_m	REAL				False	Metres to target diameter
RwStopLen_m	REAL				False	Stopping length now (m)
RwSlowLen_m	REAL				False	Slow-down length now (m)
RwStop_at_D_Req	BOOL				False	Rewinder diameter STOP request
RwSlow_at_D_Req	BOOL				False	Rewinder diameter SLOW request
RwSensFault	BOOL				False	Rewinder DT35 sensor fault
UwStopEnable	BOOL				False	Unwinder metre-counter stop ON (HMI)
SlowAtDisp	REAL				False	HMI: slow-down starts at (m or mm, see SlowAtMode)
UwSlow_at_D_Req	BOOL				False	
UwSlowDiamAct	REAL				False	
RwJogMaxDiam	REAL				False	
JogInhibit	BOOL				False	
SCurveTime	REAL				False	
TOneDecel	BOOL				False	
HornStartBeep_s	REAL				False	
HornBeepLen_s	REAL				False	
HornBeepPeriod	REAL				False	
HornStopBeep_s	REAL				False	
HornTest	BOOL				False	

RwLockDiam	REAL				False	
UwSensTest	BOOL				False	
RwLockMinGrow_mm	REAL				False	
RwMatLocked	BOOL				False	
LongThick_mm	REAL				False	
RwDiamRev	REAL				False	
RwWave_mm	REAL				False	
UwStopLenShow	BOOL				False	
UwSlowDiamShow	BOOL				False	
SoftCancelAccel	REAL				False	
SpdHold	BOOL				False	
SlowAtMode	INT				False	HMI: 0=none 1=unwinder [m] 2=rewinder [mm]
SlowAtShow	BOOL				False	HMI: show slow-down display
UwSlow_at_M_Val	REAL				False	Unwinder: slow-down starts at (m)
UwSlow_at_M_Req	BOOL				False	Unwinder metre slow-down request (latched)
RwSlow_at_D_Val	REAL				False	Rewinder: slow-down starts at diameter (mm)
RwStop_at_D_Val	REAL				False	Rewinder: stop command at diameter (mm)
RwMaxDiam	REAL				False	Rewinder max diameter protection (mm)
RwMaxStop	BOOL				False	Rewinder max diameter STOP (latched)
RwMaxSlow	BOOL				False	Rewinder max diameter SLOW (latched)
UwEmptyHyst	REAL				False	
RwThickMin_mm	REAL				False	Learning: min plausible paper thickness (mm)
RwThickMax_mm	REAL				False	Learning: max plausible paper thickness (mm)
RwDecelMin_mm	REAL				False	Learning: min plausible stop decel (mm/s2)
RwDecelMax_mm	REAL				False	Learning: max plausible stop decel (mm/s2)
RwLearnFailMax	INT				False	Learning: failed windows before alarm (12 = 2 min)
RwStopFailMax	INT				False	Learning: failed stops before alarm
GSM_Learned_OK	BOOL				False	Rewinder learning fault (alarm 16)
RwThick_mm	REAL				False	Learned paper thickness (mm)
Psw33	BOOL				False	
RwAutoMaxSpd	REAL				False	
RwWinLen_m	REAL				False	
RwWinGrow_mm	REAL				False	
GsmAsk	BOOL				False	
GsmAskYes	BOOL				False	
DecelWeb	REAL				False	
AccelWeb	REAL				False	
RwRatioD	REAL				False	
FFEnable	BOOL				False	
UwStopDiamShow	BOOL				False	

	UwStop_at_D_Val	REAL				False	
	Clock10ms	BOOL				False	
	Sec_to_Min	REAL				False	
	Psw48	BOOL				False	
	JogRun	BOOL				False	
	GuardStopOK	BOOL				False	V11: stopped for real -> guard bypass allowed (Output 13)
	SlowDown	BOOL				False	Slow down
	WarningOn	bool				False	
	AlarmSilenced	BOOL				False	
	PaperGSM	REAL				False	
	WebWidth_mm	REAL				False	
	TensFactor	REAL				False	
	TensIdeal_Disp	REAL				False	
	TensCheckEnable	BOOL				False	
	TensLostWarn	BOOL				False	
	TensWaitSpd	REAL				False	
	TensWait	BOOL				False	
	TensionOK	BOOL				False	
	UwSensFault	BOOL				False	
	Thickness_Fault	BOOL				False	
	AutoStopDone	BOOL				False	
	AutoStopWhy	INT				False	
	AutoEnRefused	BOOL				False	
	AutoEnWhy	INT				False	
	TensWaitHz	REAL				False	
	StartBlocked	BOOL				False	
	GuardOpen	BOOL				False	
	SideDoorOpen	BOOL				False	
	SonarAsk	BOOL				False	
	Psw40	BOOL				False	
	TensJogBoost	REAL				False	
	TensTrim_kg	REAL				False	
	TensJogLatch	BOOL				False	

1-8-1-1-2.General

0 Remote 10.146.233.11 ... 1s clock.



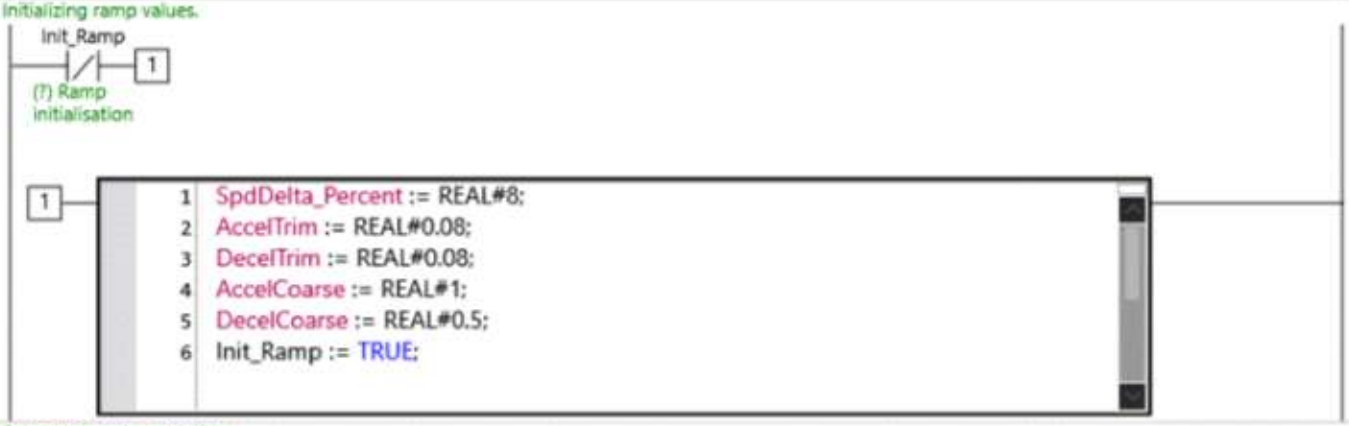
1 100ms clock.



2 10ms clock.



3 Initializing ramp values.



4 Set analog scale resolution.



5 Variable for converting m/min to mm/sec and vice versa.



6 10 V DC source.



7 START blocked by JOG

```
// ==== V11: REPLACE the code of General rung 7 ("START blocked by JOG", printed V9).
// START push-button is ignored while the JOG (slow speed) button is held and for
// 2 s after it is released. A START held during the block must be released and
// pressed again (no start by itself when the 2 s are over).
// Without this: START pressed during a jog sets Slow_Inhibit -> releasing the jog
// no longer stops (General rung 11) and VelSet jumps to the work speed.

T_JogStartBlk(In := I_Jog_PB, PT := T#2s);      // TOF: ON while held + 2 s
StartHeldBlk := (StartHeldBlk OR (I_Start_PB AND T_JogStartBlk.Q)) AND I_Start_PB;
StartPB_OK := I_Start_PB AND NOT T_JogStartBlk.Q AND NOT StartHeldBlk;

// Side door open (no key bypass): no JOG, and stop any JOG / run in progress
SideDoorOpen := NOT I_Side_Door_Closed AND NOT I_Prot_ByPass;

// V10: JogRun = a JOG really running: JOG button held, NOT a normal START run
// (Slow_Inhibit = normal run, set by START) and NOT a soft stop. Used instead of
// the raw I_Jog_PB for: guard bypass (Output 13), 2nd drive ramp (Output 12),
// pot reading (Motor_Manager 8), feed-forward (Motor_Manager 14/17), AutoStop learning.
// Without it, holding JOG during a 500 m/min run bypassed the front gate / lateral
// doors safety (23KA4 // 5SA1) and switched the drive to its 2nd ramp.

JogRun := I_Jog_PB AND NOT Slow_Inhibit AND NOT SoftStop AND NOT SideDoorOpen;

// Guard bypass (Output_Manager rung 13, 23KA4 // 5SA1): ON when the run command is
// off (machine stopped by the operator).

GuardStopOK := NOT StartMotor_Cmd;

// START refused:
//  - no paper at the web break photocell (monitoring ON)
//  - front gate or side door open, unless the key bypass 5SA1 (I_Prot_ByPass) is ON
// JOG: allowed with the front gate open (guard bypass), NOT with the side door open

WebBreakNow := I_WebBreak_Monitor AND I_WebBreak_PH;
GuardOpen   := (NOT I_FrontGuard_Closed OR NOT I_Side_Door_Closed) AND NOT I_Prot_ByPass;
StartBlocked := WebBreakNow OR GuardOpen;
```

8

```
// ==== General: NEW ST rung (no contact), directly BELOW rung 7 (JogRun / GuardStopOK)
// ==== Comment: "T-one Alarm 0: tension established"
// T-one: F.9 = 0 (alarm when tension ABOVE threshold), F.10 = threshold (kg), F.11 = full scale
// Relay pins 15-16 (Alarm 0): +24 V -> pin 15, pin 16 -> Inline IB IL 24 DI 8 IN05, pin 1.3
// I_T_one_TensionOK = node#[1.0] _IN05 (Pin 1.3) (was I_SlowSens_PH, used in no rung)
// Relay closed = tension > F.10 (not latched). Broken wire / T-one off = "no tension" (fail-safe).

T_TensOn(In := I_T_one_TensionOK, PT := T#500ms);       // stable 0.5 s
T_TensOff(In := NOT I_T_one_TensionOK, PT := T#2s);     // ignore short dips (splice, wave, chatter)

IF T_TensOn.Q OR NOT TensCheckEnable THEN               // HMI switch OFF = old behaviour
    TensionOK := TRUE;
ELSIF T_TensOff.Q THEN
    TensionOK := FALSE;
END_IF;

// Latch: tension reached once since START

IF NOT StartMotor_Cmd THEN
    TensionSeen := FALSE;
ELSIF TensionOK THEN
    TensionSeen := TRUE;
END_IF;

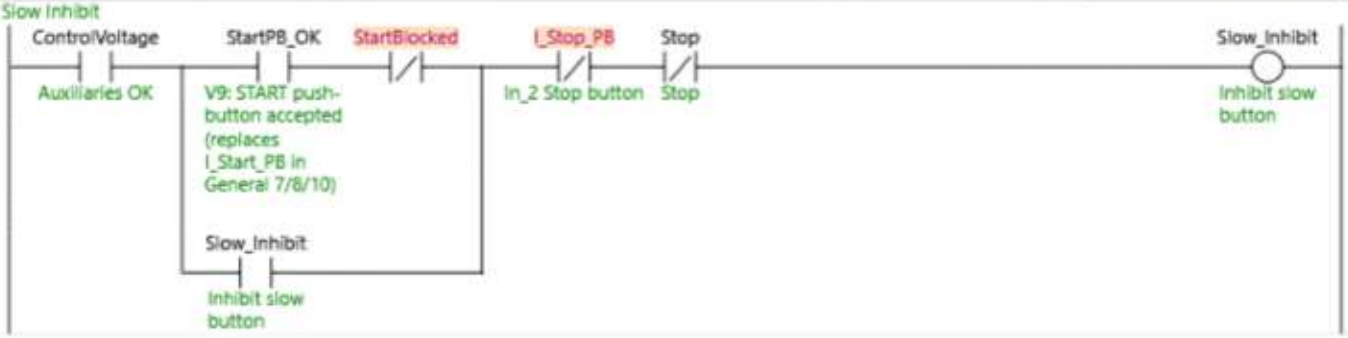
// After START: hold at TensWaitSpd until the tension is there (JOG not affected)

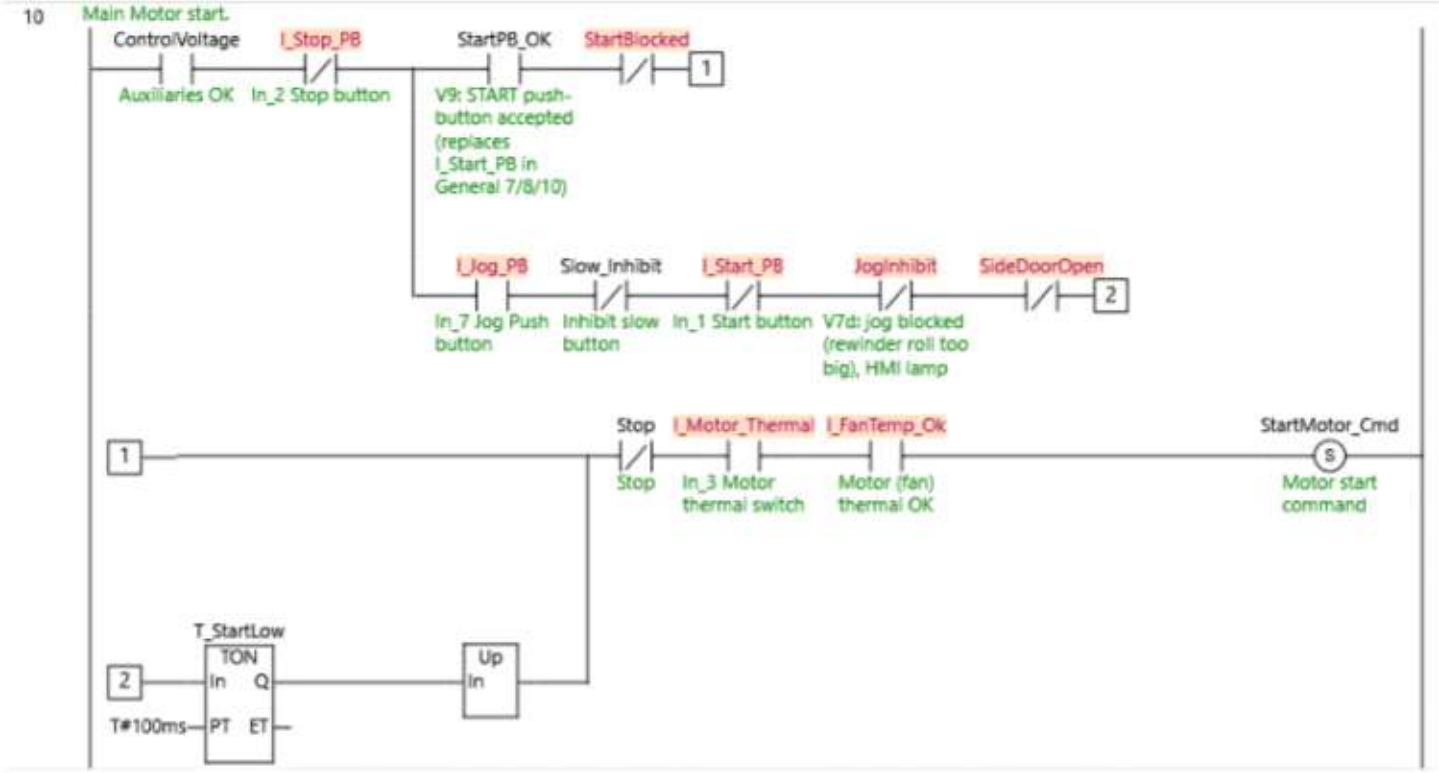
TensWait := StartMotor_Cmd AND NOT TensionSeen AND NOT JogRun;

// Tension lost later while running -> warning only (no slow-down)

TensLostWarn := StartMotor_Cmd AND TensionSeen AND NOT TensionOK;
```

9 Slow Inhibit




10
Main Motor start.
ControlVoltage
Auxiliaries OK
I_Stop_PB
In_2 Stop button
StartPB_OK
V9: START push-button accepted (replaces I_Start_PB in General 7/8/10)
StartBlocked
1
I_Jog_PB
In_7 Jog Push button
Slow_Inhibit
Inhibit slow button
I_Start_PB
In_1 Start button
JogInhibit
V7d: jog blocked (rewinder roll too big), HMI lamp
SideDoorOpen
2
1
Stop
Stop
I_Motor_Thermal
In_3 Motor thermal switch
I_FanTemp_Ok
Motor (fan) thermal OK
StartMotor_Cmd
S
Motor start command
T_StartLow
TON
2
In
Q
T#100ms
PT
ET
Up
In

11 Soft Stop Process

```
// ==== V12: REPLACE the code of General rung 10 ("Soft Stop Process", printed V9).
// v6m: STOP button = soft stop. StopNow replaces I_Stop_PB in rung 9.
// V12: an end-of-roll stop (FinalStopRamp, AutoStop section 11b) also starts the
// soft stop, at FinalStopDecel (Motor_Manager rung 17): START cannot cancel it.

StopNow := FALSE;
T_StopLong(In := I_Stop_PB, PT := T#1s);

IF NOT StartMotor_Cmd THEN
    SoftStop := FALSE;
ELSIF I_Stop_PB AND NOT StopPB_Mem THEN            // STOP pressed (edge)

    IF SoftStop OR NOT SoftStopEnable OR I_Jog_PB
      OR (Velocity_M_min <= SoftStopEndSpd) THEN
        StopNow := TRUE;                           // 2nd press / jog / slow / disabled
    ELSE
        SoftStop := TRUE;                          // start soft stop
    END_IF;
END_IF;

StopPB_Mem := I_Stop_PB;

IF T_StopLong.Q THEN
    StopNow := TRUE;                               // held 1 s = immediate
END_IF;

// V12: end-of-roll stop ramp (FinalStopRamp set by AutoStop section 11b)

IF FinalStopRamp AND StartMotor_Cmd AND NOT SoftStop THEN
    SoftStop := TRUE;
END_IF;

// Side door opened while running (not JOG): soft stop. START cannot cancel it

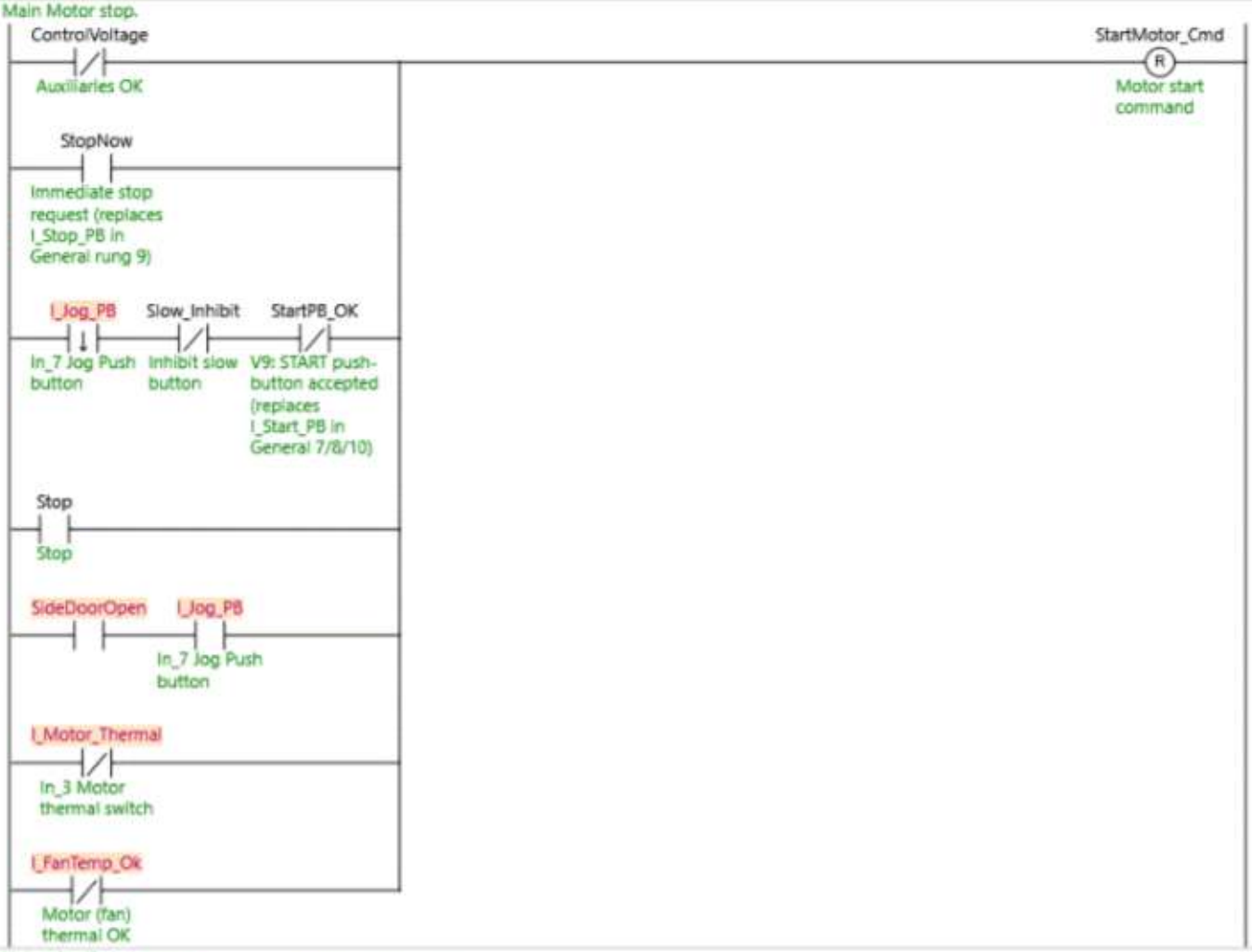
IF SideDoorOpen AND StartMotor_Cmd AND NOT I_Jog_PB AND NOT SoftStop THEN
    SoftStop := TRUE;
END_IF;

IF SoftStop AND I_Start_PB AND NOT FinalStopRamp AND NOT SideDoorOpen THEN
    SoftStop := FALSE;                             // START cancels the soft stop
END_IF;

T_SoftStop(In := SoftStop, PT := T#120s);

IF SoftStop AND ((Velocity_M_min <= SoftStopEndSpd) OR T_SoftStop.Q) THEN
    StopNow := TRUE;                               // almost stopped (or timeout)
END_IF;
```

12 Main Motor stop.



13 Hub extraction clearance.



14 Brake control.

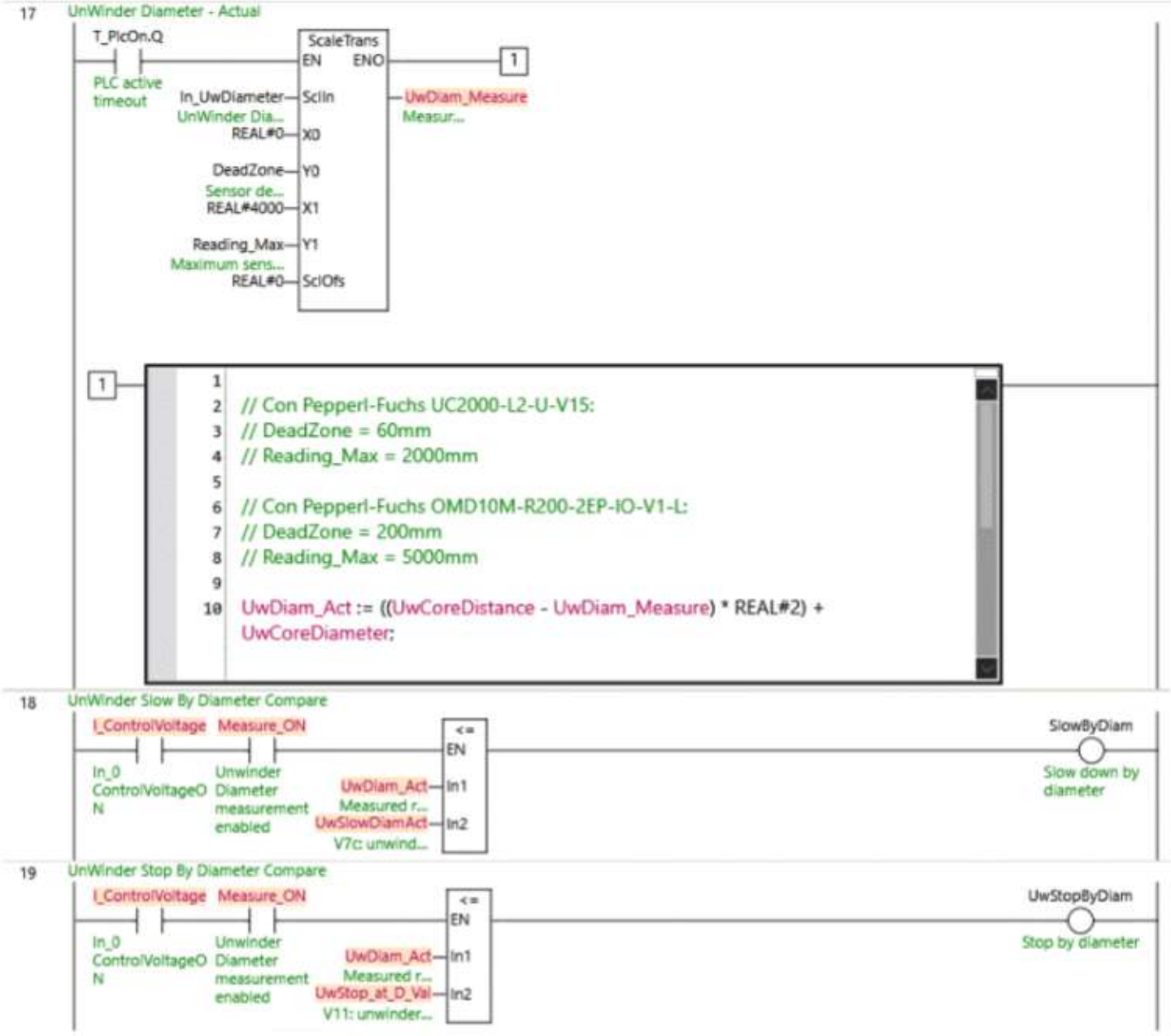


15 Cooling Fan control.



16 UnWinder Diameter - Input

```
1  In_UwDiameter := INT_TO_REAL(AI_Diam_Unwinder)/REAL#1;
```


17
UnWinder Diameter - Actual
T_PlcOn.Q
PLC active timeout
ScaleTrans
EN
ENO
1
In_UwDiameter
UnWinder Dia...
REAL#0
SclIn
X0
DeadZone
Sensor de...
REAL#4000
Y0
X1
Reading_Max
Maximum sens...
REAL#0
Y1
SclOfs
UwDiam_Measure
Measur...
1
// Con Pepperl-Fuchs UC2000-L2-U-V15:
// DeadZone = 60mm
// Reading_Max = 2000mm
// Con Pepperl-Fuchs OMD10M-R200-2EP-IO-V1-L:
// DeadZone = 200mm
// Reading_Max = 5000mm
UwDiam_Act := ((UwCoreDistance - UwDiam_Measure) * REAL#2) +
UwCoreDiameter;
18
UnWinder Slow By Diameter Compare
I_ControlVoltage
In_0 ControlVoltageON
Measure_ON
Unwinder Diameter measurement enabled
<=
EN
UwDiam_Act
Measured r...
In1
UwSlowDiamAct
V7c: unwind...
In2
SlowByDiam
Slow down by diameter
19
UnWinder Stop By Diameter Compare
I_ControlVoltage
In_0 ControlVoltageON
Measure_ON
Unwinder Diameter measurement enabled
<=
EN
UwDiam_Act
Measured r...
In1
UwStop_at_D_Val
V11: unwinder...
In2
UwStopByDiam
Stop by diameter

1-8-1-1-3.Motor_Manager

0 Enable encoder reading.



1 Calculation of mm per encoder pulse.

```
IF Encoder_Pulse = INT#0 THEN;
    Encoder_Pulse :=INT#1;
END_IF;
// Encoder Standard 400mm
IF  EncoderWheel_Circumf = REAL#0 THEN;
    EncoderWheel_Circumf := REAL#400;
END_IF;

EncoderPulse_to_mm := EncoderWheel_Circumf / (INT_TO_REAL(Encoder_Pulse) *
REAL#4);
```

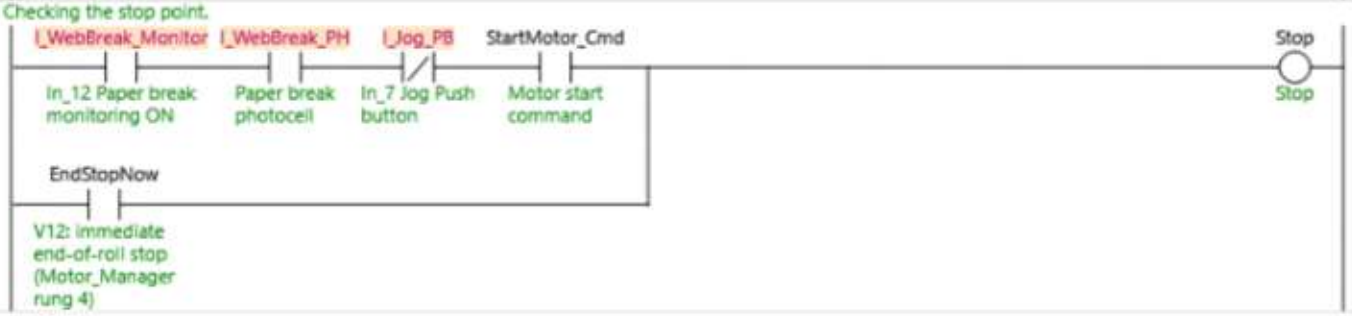
2 Reset encoder counter.



3 Checking the slowdown point.



4 Checking the stop point.



5 Check the maximum and minimum values for working speed and slow speed.

```
IF VelMax < REAL#600 THEN;
    VelMax := REAL#600;
ELSIF VelMax > REAL#1000 THEN;
    VelMax := REAL#1000;
END_IF;

Vel_Max_mms := VelMax * REAL#1000 / REAL#60;

IF JogSpd_SP < REAL#10 THEN;
    JogSpd_SP := REAL#10;
ELSIF JogSpd_SP > REAL#70 THEN;
    JogSpd_SP := REAL#70;
END_IF;

// v6y: rewinder AUTO max speed, 0 = no limit

IF RwAutoMaxSpd > VelMax THEN
    RwAutoMaxSpd := VelMax;
ELSIF (RwAutoMaxSpd > REAL#0) AND (RwAutoMaxSpd < REAL#50) THEN
    RwAutoMaxSpd := REAL#50;
END_IF;

IF NOT StartMotor_Cmd THEN          // taken over at standstill only
    RwAutoLimAct := RwStopEnable AND (RwAutoMaxSpd > REAL#0);
    RwAutoMaxSpdAct := RwAutoMaxSpd;
END_IF;

// ---- V7: add at the END of Motor_Manager rung 5 ("Check the maximum and
minimum values...")
// ---- after the JogSpd_SP clamp, before Output_Ratio. Same style as the VelMax /
JogSpd clamps.
// V7: limits on settings that stop the machine or break the auto stop if 0 / absurd

IF SpdDelta_Percent < REAL#2 THEN                // fine band [m/min]: 0 = speed
hunting
    SpdDelta_Percent := REAL#2;
ELSIF SpdDelta_Percent > REAL#15 THEN
    SpdDelta_Percent := REAL#15;
END_IF;

IF AccelTrim < REAL#0.02 THEN                  // trim ramps: 0 = no speed correction
    AccelTrim := REAL#0.02;
ELSIF AccelTrim > REAL#0.5 THEN
    AccelTrim := REAL#0.5;
END_IF;

IF DecelTrim < REAL#0.02 THEN
    DecelTrim := REAL#0.02;
ELSIF DecelTrim > REAL#0.5 THEN
    DecelTrim := REAL#0.5;
END_IF;

IF AccelCoarse < REAL#0.2 THEN                  // coarse ramps: 0 = never reaches
speed
    AccelCoarse := REAL#0.2;
ELSIF AccelCoarse > REAL#4 THEN
    AccelCoarse := REAL#4;
```

```
55  END_IF;
56
57  IF DecelCoarse < REAL#0.2 THEN
58      DecelCoarse := REAL#0.2;
59  ELSIF DecelCoarse > REAL#4 THEN
60      DecelCoarse := REAL#4;
61  END_IF;
62
63  IF DecelWeb > REAL#0 THEN                    // V7 web ramps: 0 = old coarse ramp
    (allowed)
64      IF DecelWeb < REAL#50 THEN
65          DecelWeb := REAL#50;
66      ELSIF DecelWeb > REAL#1000 THEN
67          DecelWeb := REAL#1000;
68      END_IF;
69  END_IF;
70
71  IF AccelWeb > REAL#0 THEN
72      IF AccelWeb < REAL#50 THEN
73          AccelWeb := REAL#50;
74      ELSIF AccelWeb > REAL#1000 THEN
75          AccelWeb := REAL#1000;
76      END_IF;
77  END_IF;
78
79  IF SoftStopDecel > REAL#1500 THEN            // 0 = DecelCoarse (allowed)
80      SoftStopDecel := REAL#1500;
81  END_IF;
82
83  IF RwMaxDiam > REAL#1450 THEN                // rewinder MECHANICAL limit
    [mm]
84      RwMaxDiam := REAL#1450;
85  ELSIF RwMaxDiam < REAL#300 THEN
86      RwMaxDiam := REAL#300;
87  END_IF;
88
89  IF RwMaxSlowFallback >= RwMaxDiam THEN       // fallback slow-down below
    the max
90      RwMaxSlowFallback := RwMaxDiam - REAL#30;
91  END_IF;
92
93  // ---- V9 (V7d): add at the END of Motor_Manager rung 5, after the V7 clamps
94  // ---- (last lines now: RwMaxSlowFallback clamp), before Output_Ratio /
    Input_Ratio.
95  // V7d: jog / slow start button blocked on a big rewinder roll.
96  // RwJogMaxDiam (mm, HMI, Retain): 0 = no limit, else min 150.
97  // DT35 fault -> jog allowed (needed to thread / clear the machine).
98
99  IF (RwJogMaxDiam > REAL#0) AND (RwJogMaxDiam < REAL#150) THEN
100     RwJogMaxDiam := REAL#150;
101 END_IF;
102
103 JogInhibit := (RwJogMaxDiam > REAL#0) AND (RwDiam > RwJogMaxDiam) AND
    NOT RwSensFault;
104
105 Output_Ratio := (VelMax * Mt_To_mm) / UINT_TO_REAL(Analog_Res); //REAL#4000;
106 Input_Ratio := UINT_TO_REAL(Analog_Res) / VelMax ;
```

6 Reading the current value of the potentiometers. v6n: requested pot ranges, limited, applied only with the motor stopped !

```
IF WorkSpd_Max_Set < 50.0 THEN
    WorkSpd_Max_Set := 50.0;
ELSIF WorkSpd_Max_Set > VelMax THEN
    WorkSpd_Max_Set := VelMax;
END_IF;

IF FinalSpd_Max_Set < 5.0 THEN
    FinalSpd_Max_Set := 5.0;
ELSIF FinalSpd_Max_Set > WorkSpd_Max_Set THEN
    FinalSpd_Max_Set := WorkSpd_Max_Set;
END_IF;

IF NOT StartMotor_Cmd THEN
    WorkSpd_Max := WorkSpd_Max_Set;          // motor stopped: apply
    FinalSpd_Max := FinalSpd_Max_Set;
END_IF;

SpdMax_Pending := (WorkSpd_Max <> WorkSpd_Max_Set)
            OR (FinalSpd_Max <> FinalSpd_Max_Set)
            OR (RwAutoLimAct <> (RwStopEnable AND (RwAutoMaxSpd > REAL#0)))
            OR (RwAutoLimAct AND (RwAutoMaxSpdAct <> RwAutoMaxSpd));

// v6i: pots scaled to their own full scale (m/min)

IF Pot1_Work_Velocity > INT#5 THEN
    Pot1_Mem := REAL_TO_INT(INT_TO_REAL(Pot1_Work_Velocity) * WorkSpd_Max /
VelMax);
ELSE
    Pot1_Mem := INT#0;
END_IF;

IF Pot2_Slow_Velocity > INT#5 THEN
    Pot2_Mem := REAL_TO_INT(INT_TO_REAL(Pot2_Slow_Velocity) * FinalSpd_Max /
VelMax);
ELSE
    Pot2_Mem := INT#0;
END_IF;
```

7 Updating the set speed.

```
IF ManMode = FALSE THEN;

    Speed_Display := UINT_TO_REAL(Pot1_Mem) / Input_Ratio;
    LowSpd_Display := UINT_TO_REAL(Pot2_Mem) / Input_Ratio;

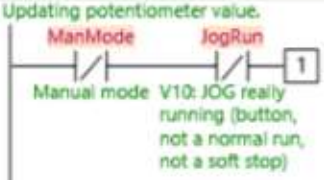
  // v6y: rewinder AUTO stop -> work speed limited

  IF RwAutoLimAct AND (Speed_Display > RwAutoMaxSpdAct) THEN
     Speed_Display := RwAutoMaxSpdAct;
  END_IF;

ELSE;
  Speed_Display := Pot1_Work_Velocity / (Analog_Res / UINT#40); // v6i: raw pot. %
  LowSpd_Display := Pot2_Slow_Velocity / (Analog_Res / UINT#40);

END_IF;
```

8 Updating potentiometer value.




```
IF SoftStop THEN
   VelSet := 0.0;                    // v6m: soft stop
ELSIF SlowDown = FALSE  THEN
   VelSet := UINT_TO_REAL(Pot1_Mem) / Input_Ratio;

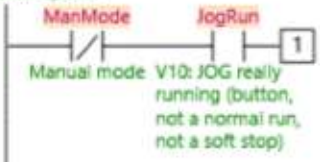
   // v6y: rewinder AUTO stop -> work speed limited

   IF RwAutoLimAct AND (VelSet > RwAutoMaxSpdAct) THEN
      VelSet := RwAutoMaxSpdAct;
   END_IF;

ELSE
   VelSet := UINT_TO_REAL(Pot2_Mem) / Input_Ratio;
END_IF;

VelSet_mms := VelSet * Mt_To_mm;
```

9 Slow speed.




```
VelSet := JogSpd_SP;
VelSet_mms := VelSet * Mt_To_mm;
```

10 Check the difference between the set speed and the detected speed.

```
//SpdDelta := VelSet_mms * SpdDelta_Percent / REAL#100;

SpdDelta := SpdDelta_Percent;
SpdDeltaMax := VelSet + SpdDelta;
SpdDeltaMin := VelSet - SpdDelta;

// Too fast

IF Velocity_M_min > SpdDeltaMax THEN
    OverSpeed := TRUE;
ELSE
    OverSpeed := FALSE;
END_IF;

// Too slow

IF Velocity_M_min < SpdDeltaMin THEN
    UnderSpeed := TRUE;
ELSE
    UnderSpeed := FALSE;
END_IF;
```

11 Speed calculation.

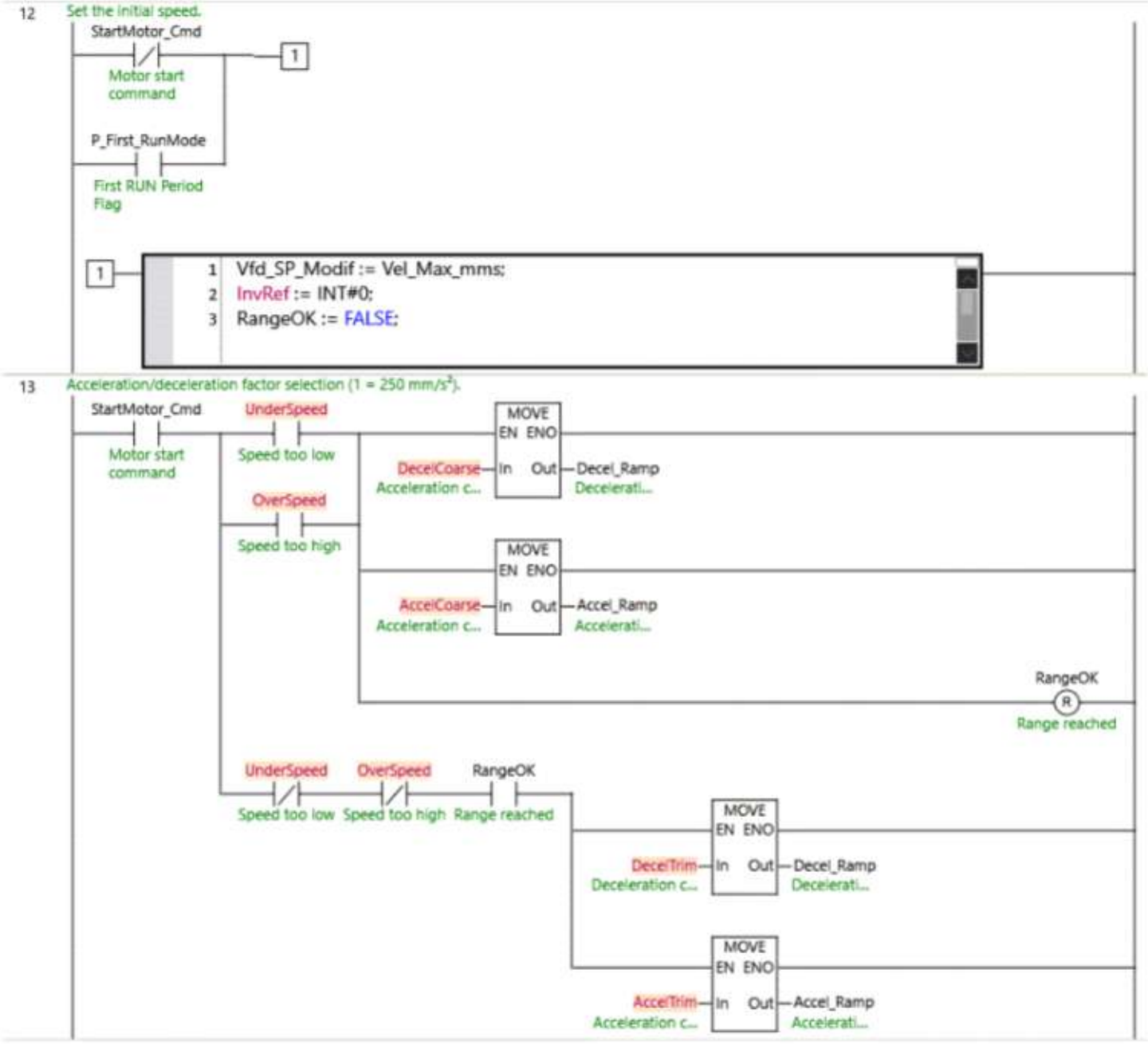
```
// ==== V9: REPLACE Motor_Manager rungs 11 ("Read the current value of the encoder", MOVE on
// ==== Clock100ms rising edge) and 12 ("Speed calculation", Clock100ms falling edge) by ONE
// ==== ST rung with this code (no contact in front). Comment: "V9: speed over exactly 100 ms".
// Old: window = time between the rising and falling edge of Clock100ms, assumed 50 ms,
// but with the 4 ms PrimaryTask it is 12 or 13 scans (48 / 52 ms) -> the speed jumps
// -4 % / +4 % (418 set: display 397 / 431).
// New: exactly 25 scans = 100 ms (same as AutoStop SpdAct_Pps), updated 10 times/s.

SpdScanCnt := SpdScanCnt + 1;

IF SpdScanCnt >= 25 THEN
    SpdScanCnt := 0;
    DeltaPulse := Encoder_Pos - MemPos;
    MemPos := Encoder_Pos;

    IF ABS(DeltaPulse) > 100000 THEN            // counter reset (Encoder_Reset)
        DeltaPulse := 0;
    END_IF;

    Velocity := DINT_TO_REAL(DeltaPulse) * EncoderPulse_to_mm * REAL#10;   // mm/s
    Velocity_M_min := Velocity / Mt_To_mm;
END_IF;
```


12
Set the initial speed.
StartMotor_Cmd
Motor start command
1
P_First_RunMode
First RUN Period Flag
1
1 Vfd_SP_Modif := Vel_Max_mms;
2 InvRef := INT#0;
3 RangeOK := FALSE;
13
Acceleration/deceleration factor selection (1 = 250 mm/s²).
StartMotor_Cmd
Motor start command
UnderSpeed
Speed too low
OverSpeed
Speed too high
MOVE
EN ENO
DecelCoarse
Acceleration c...
In Out
Decel_Ramp
Decelerati...
MOVE
EN ENO
AccelCoarse
Acceleration c...
In Out
Accel_Ramp
Accelerati...
RangeOK
R
Range reached
UnderSpeed
Speed too low
OverSpeed
Speed too high
RangeOK
Range reached
MOVE
EN ENO
DecelTrim
Deceleration c...
In Out
Decel_Ramp
Decelerati...
MOVE
EN ENO
AccelTrim
Acceleration c...
In Out
Accel_Ramp
Accelerati...

14 Ratio drive counts per m/min of web at the current rewinder diameter.

```
// ==== V10: REPLACE the code of Motor_Manager rung 14 (printed V9) ("Ratio drive counts per m/min of web
// ==== at the current rewinder diameter." = V7 rung A) with this file. (V7b jog line inside.)
// Ratio drive counts per m/min of web at the current rewinder diameter.
// The motor drives the rewinder: web speed per Hz grows with the diameter, so
// counts per m/min = RwRatioD / diameter. RwRatioD (Retain) is adapted by rung B.

IF (RwRatioD > 0.0) AND (DecelDiam >= 100.0) THEN
    WebRatio := RwRatioD / DecelDiam;
ELSE
    WebRatio := 0.0;
END_IF;

// Fallback only (FFEnable = FALSE): old integrating loop (rung 16) with the coarse
// ramps in real web mm/s2. With FFEnable = TRUE, rung B drives the reference.
// V7b: never in jog -> jog keeps the rung 13 ramps exactly as v6y. V10: JogRun (jog really running).

IF NOT FFEnable AND NOT JogRun AND StartMotor_Cmd AND (UnderSpeed OR OverSpeed)
   AND (WebRatio > 0.0) THEN

    IF DecelWeb > 0.0 THEN
        Decel_Ramp := DecelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
        IF Decel_Ramp > 8.0 THEN
            Decel_Ramp := 8.0;
        END_IF;
    END_IF;

    IF AccelWeb > 0.0 THEN
        Accel_Ramp := AccelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;

        IF Accel_Ramp > 8.0 THEN
            Accel_Ramp := 8.0;
        END_IF;
    END_IF;
END_IF;
```

15 Soft Stop - SoftStopDecel (mm/s2), same for every roll.

```
// v6r: soft stop at a REAL web decel SoftStopDecel (mm/s2), same for every roll.
// Ratio drive counts / web m/min frozen at the soft stop start (encoder = exact).

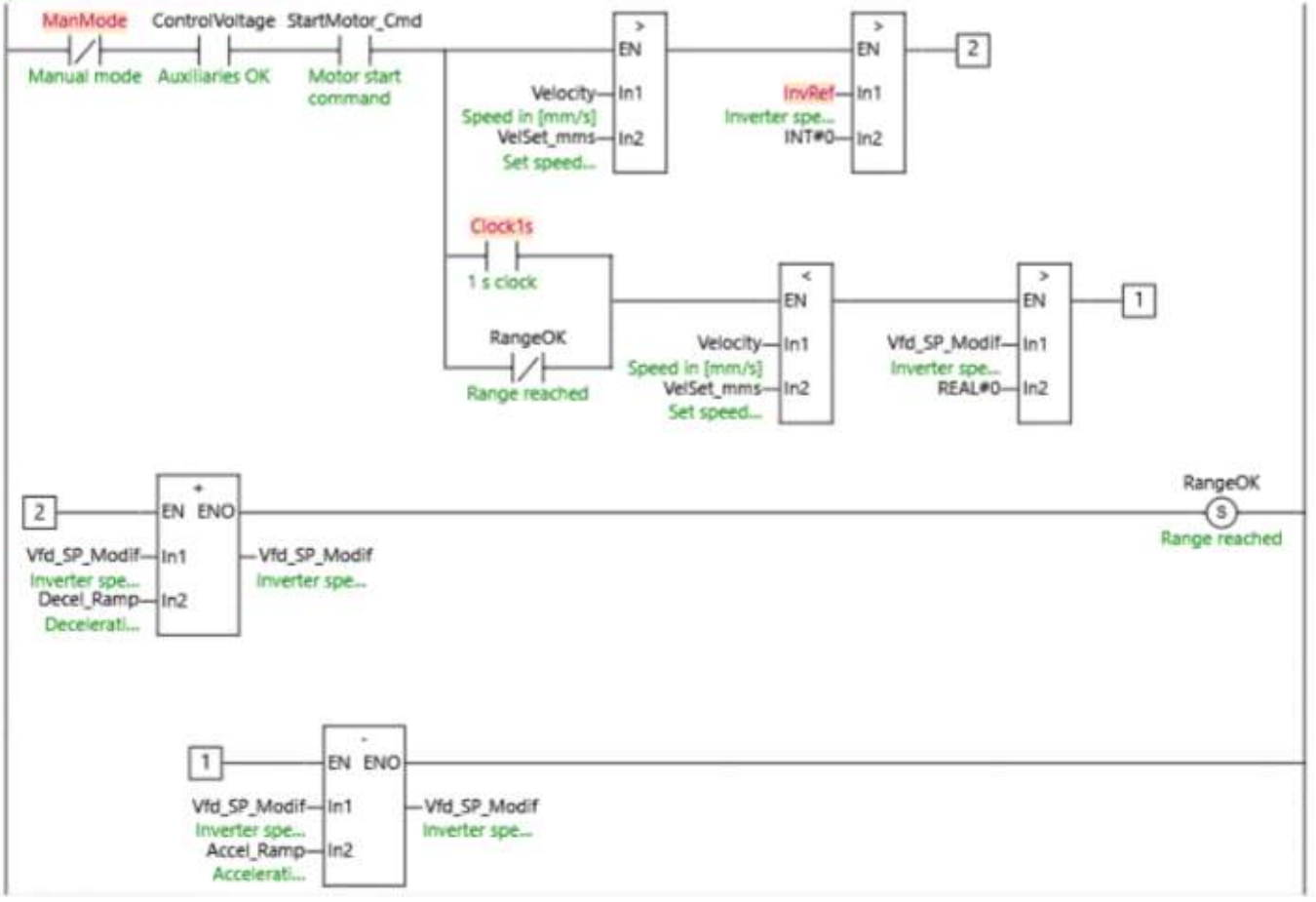
IF SoftStop AND NOT SoftStopMem THEN
    IF Velocity_M_min > 1.0 THEN
        SoftStopRatio := INT_TO_REAL(InvRef) / Velocity_M_min;
    ELSE
        SoftStopRatio := 0.0;
    END_IF;
END_IF;

SoftStopMem := SoftStop;

IF SoftStop AND (SoftStopDecel > 0.0) THEN
    IF SoftStopRatio > 0.0 THEN
        Decel_Ramp := SoftStopDecel * 0.06 * SoftStopRatio * Vel_Max_mms /
1000000.0;
    ELSE
        Decel_Ramp := SoftStopDecel / 250.0;          // fallback (old v6m behaviour)
    END_IF;

    IF Decel_Ramp > 8.0 THEN
        Decel_Ramp := 8.0;                        // max 9 Hz/s asked from the PLC
    END_IF;
END_IF;
```

16 Speed correction factor calculation.



17 Slow adaptation of RwRatioD on the relative speed error.

```
// ==== V12: REPLACE the code of Motor_Manager rung 17 (printed V9) ("Slow adaptation of RwRatioD on the
// ==== relative speed error." = V7 rung B) with this file. (V7a + V7b + V7f + V9 + V10 + V12 inside.)
// The PLC sends the drive STRAIGHT to the frequency of the set speed
// (VelSet x WebRatio), along a ramp in REAL web mm/s2 that does not depend on the
// measured speed -> no hunting, whatever the roll inertia / web delay.
// The measured speed only adapts RwRatioD slowly (~10 s), once at target.
// It overwrites Vfd_SP_Modif after rung 16, so rung 16 has no effect in AUTO.
// FFEnable = FALSE -> old loop (rung 16) exactly as before.
// V7b: not in jog (V10: JogRun = jog really running): at 10 m/min the feed-forward gives
// only ~0.4 Hz, too little to move the web -> the old loop (rung 16) runs the jog.

FFActive := FFEnable AND NOT ManMode AND NOT JogRun AND StartMotor_Cmd
        AND ControlVoltage AND (WebRatio > 0.0) AND (Vel_Max_mms > 0.0);

IF NOT StartMotor_Cmd THEN
   RampRef := 0.0;
   FFRate := 0.0;
   FFSteadyTicks := 0;
   FFSlowAcc := FALSE;
END_IF;

// V9: START during a soft stop: the decel flattens and the speed is HELD
// (SpdHold, HMI lamp) until START is pushed AGAIN -> gentle ramp up at
// SoftCancelAccel (mm/s2) to the set speed. The hold only blocks speeding up:
// a lower target (pot down, auto slow-down, RwAutoMaxSpd) still slows down.
// STOP during the hold = new soft stop (General rung 9).

IF (SoftCancelAccel > 0.0) AND (SoftCancelAccel < 20.0) THEN
   SoftCancelAccel := 20.0;
ELSIF SoftCancelAccel > 1000.0 THEN
   SoftCancelAccel := 1000.0;
END_IF;

FFStartEdge := I_Start_PB AND NOT FFStartMem;
FFStartMem := I_Start_PB;

IF FFSoftMem AND NOT SoftStop AND StartMotor_Cmd AND FFActive THEN   // soft stop cancelled
   SpdHold := TRUE;
   FFSlowAcc := FALSE;
ELSIF SpdHold AND FFStartEdge THEN                    // 2nd START: resume
   SpdHold := FALSE;
   FFSlowAcc := TRUE;
END_IF;

IF SoftStop OR NOT StartMotor_Cmd OR NOT FFActive THEN    // FFEnable OFF: old loop
   SpdHold := FALSE;
   FFSlowAcc := FALSE;
END_IF;

FFSoftMem := SoftStop;

// V7f: S-curve time (s). 0 = off, max 2 s
```

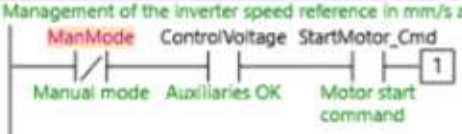
```
53
54  IF SCurveTime < 0.0 THEN
55      SCurveTime := 0.0;
56  ELSIF SCurveTime > 2.0 THEN
57      SCurveTime := 2.0;
58  END_IF;
59
60  IF FFActive THEN
61      IF NOT FFWasActive THEN
62          RampRef := Vel_Max_mms - Vfd_SP_Modif;          // bumpless start / switch-
    over
63          FFRate := 0.0;
64      END_IF;
65
66      // Target reference (mm/s of the 0..10 V scale) for the set speed
67      FFTarget := VelSet * WebRatio * Output_Ratio;
68
69      IF FFTarget > Vel_Max_mms THEN
70          FFTarget := Vel_Max_mms;
71      ELSIF FFTarget < 0.0 THEN
72          FFTarget := 0.0;
73      END_IF;
74
75      // Tension: hold at TensWaitSpd until the T-one reports tension
    (I_T_one_TensionOK)
76
77  //    IF TensWait AND (VelSet > TensWaitSpd) THEN
78  //        FFTarget := TensWaitSpd * WebRatio * Output_Ratio;
79  //    END_IF;
80
81      // Tension: hold at a minimum DRIVE FREQUENCY until the T-one reports tension
82      // (independent of the rewinder diameter). TensWaitHz = 0 -> old m/min hold.
83
84      IF TensWait THEN
85
86          IF TensWaitHz > REAL#0.0 THEN
87              TensRef := TensWaitHz * REAL#4000.0 / REAL#60.0 * Output_Ratio;
88          ELSE
89              TensRef := TensWaitSpd * WebRatio * Output_Ratio;
90          END_IF;
91
92          IF FFTarget > TensRef THEN
93              FFTarget := TensRef;
94          END_IF;
95      END_IF;
96
97
98      // V9: hold after a cancelled soft stop (the S-curve flattens it smoothly):
99      // never above the present reference, lower targets still allowed
100
101     IF SpdHold THEN
102         IF FFTarget > RampRef THEN
103             FFTarget := RampRef;
104         END_IF;
105
106     ELSIF FFSlowAcc AND (ABS(FFTarget - RampRef) <= FFTarget * 0.005 + 2.0) THEN
107         FFSlowAcc := FALSE;                          // back at the set speed
108     END_IF;
109
```

```
110     // Ramp steps per 4 ms scan: web mm/s2 -> reference mm/s per scan
111
112     IF FFSlowAcc AND (SoftCancelAccel > 0.0) THEN
113         StepAcc := SoftCancelAccel * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;  //
    V9
114     ELSIF AccelWeb > 0.0 THEN
115         StepAcc := AccelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
116     ELSE
117         StepAcc := AccelCoarse;
118     END_IF;
119
120     IF SoftStop AND FinalStopRamp AND (FinalStopDecel > 0.0) THEN
121         StepDec := FinalStopDecel * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;  //
    V12 end-of-roll
122     ELSIF SoftStop AND (SoftStopDecel > 0.0) THEN
123         StepDec := SoftStopDecel * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
124     ELSIF DecelWeb > 0.0 THEN
125         StepDec := DecelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
126     ELSE
127         StepDec := DecelCoarse;
128     END_IF;
129
130     IF StepAcc > 8.0 THEN
131         StepAcc := 8.0;
132     END_IF;
133
134     IF StepDec > 8.0 THEN
135         StepDec := 8.0;
136     END_IF;
137
138     // Ramp generator
139     // V7f: S-curve. FFRate (reference per scan) changes at most FFJerk per scan,
140     // so the full ramp is reached after SCurveTime (s) and is rounded off the
141     // same way before the target -> the T-one / brake air can follow.
142     // SCurveTime = 0 -> plain linear ramp as V7a.
143
144     IF SCurveTime > 0.0 THEN
145         IF (FFRate > 0.0) OR ((FFRate = 0.0) AND (FFTarget > RampRef)) THEN
146             FFJerk := StepAcc * 0.004 / SCurveTime;
147         ELSE
148             FFJerk := StepDec * 0.004 / SCurveTime;
149         END_IF;
150
151         // rate allowed so that it can still fall to 0 at the target
152         // (no MIN(), argument in a REAL variable first: Sysmac data type rules,
153         //  same pattern as SqrtArg / SQRT(SqrtArg) in AutoStop)
154
155         IF FFTarget >= RampRef THEN
156             FFSqrtArg := REAL#2.0 * FFJerk * (FFTarget - RampRef);
157             FFRateSet := SQRT(FFSqrtArg);
158
159             IF FFRateSet > StepAcc THEN
160                 FFRateSet := StepAcc;
161             END_IF;
162         ELSE
163             FFSqrtArg := REAL#2.0 * FFJerk * (RampRef - FFTarget);
164             FFRateSet := SQRT(FFSqrtArg);
165
166             IF FFRateSet > StepDec THEN
```

```
167             FFRateSet := StepDec;
168           END_IF;
169
170           FFRateSet := -FFRateSet;
171         END_IF;
172
173         IF FFRateSet > FFRate + FFJerk THEN
174           FFRate := FFRate + FFJerk;
175         ELSIF FFRateSet < FFRate - FFJerk THEN
176           FFRate := FFRate - FFJerk;
177         ELSE
178           FFRate := FFRateSet;
179         END_IF;
180
181         RampRef := RampRef + FFRate;
182
183         IF (ABS(FFTarget - RampRef) <= 3.0) AND (ABS(FFRate) <= 2.0 * FFJerk) THEN
184           RampRef := FFTarget;                    // at target (pot noise)
185           FFRate := 0.0;
186         END_IF;
187       ELSE
188         FFRate := 0.0;
189
190         IF RampRef < FFTarget - StepAcc THEN
191           RampRef := RampRef + StepAcc;
192         ELSIF RampRef > FFTarget + StepDec THEN
193           RampRef := RampRef - StepDec;
194         ELSE
195           RampRef := FFTarget;                    // at target
196         END_IF;
197       END_IF;
198
199       // V7a: "at target" = within 0.5 % of the target (the pot and RwDiam noise
200       // move FFTarget by a few counts every scan -> the counter never reached 500)
201
202       IF (ABS(FFTarget - RampRef) <= FFTarget * 0.005 + 2.0)
203         AND NOT SpdHold AND NOT FFSlowAcc AND NOT TensWait THEN          // V9:
    not during the hold
204
205         IF FFSteadyTicks < 30000 THEN
206           FFSteadyTicks := FFSteadyTicks + 1;
207         END_IF;
208       ELSE
209         FFSteadyTicks := 0;
210       END_IF;
211
212       // Slow adaptation of RwRatioD on the relative speed error (2 s after the
213       // target is reached, not at the 60 Hz limit, not during soft stop / jog)
214
215       IF (FFSteadyTicks >= 500) AND NOT SoftStop AND NOT JogRun
216         AND (VelSet > 15.0) AND (Velocity_M_min > 10.0) AND NOT RwSensFault
217         AND (FFTarget < Vel_Max_mms * 0.98) THEN
218         FFErr := (VelSet - Velocity_M_min) / VelSet;
219
220         IF FFErr > 0.05 THEN
221           FFErr := 0.05;
222         ELSIF FFErr < -0.05 THEN
223           FFErr := -0.05;
```

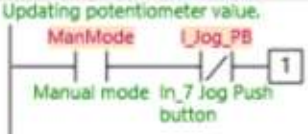
```
224       END_IF;
225
226       IF ABS(FFErr) > 0.003 THEN                  // dead band 0.3 % (encoder noise)
227         RwRatioD := RwRatioD * (1.0 + 0.0004 * FFErr);  // ~10 s time constant
228       END_IF;
229
230       IF RwRatioD < 1000.0 THEN
231         RwRatioD := 1000.0;
232       ELSIF RwRatioD > 8000.0 THEN
233         RwRatioD := 8000.0;
234       END_IF;
235     END_IF;
236
237     Vfd_SP_Modif := Vel_Max_mms - RampRef;            // rung 17 -> InvRef
238 END_IF;
239
240 FFWasActive := FFActive;
241
242 // V7f: T-one decel signal (for a T-one "stop / boost" input, spare PLC output):
243 // ON while the feed-forward ramps down (more than 1 % above the target, so the
244 // slow drift of the target with the diameter does not pulse it) or soft stop.
245
246 TOneDecel := (FFActive AND (RampRef - FFTarget > FFTarget * 0.01 + 5.0))
247           OR (SoftStop AND StartMotor_Cmd);
```

18 Management of the inverter speed reference in mm/s and conversion to analog points (0–4000 points = 0–10 VDC = 0–10,000 mm/s).

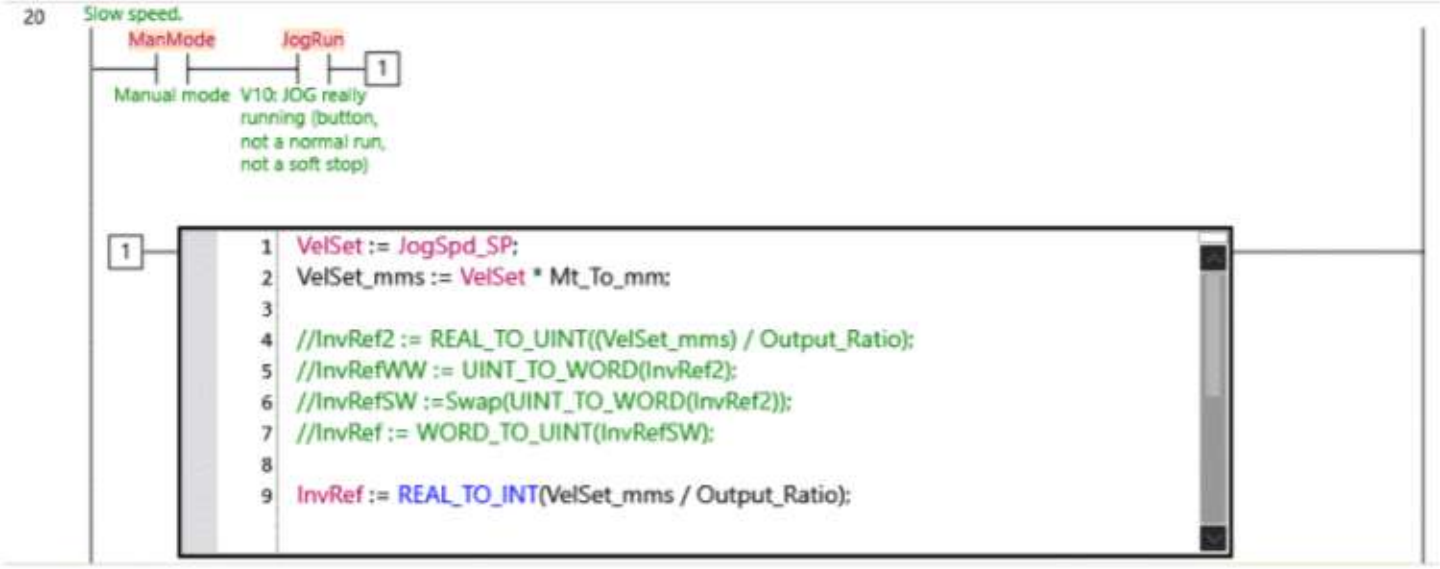



```
1
1   //InvRef2 := REAL_TO_UINT((Vel_Max_mms - Vfd_SP_Modif ) / Output_Ratio);
2   //InvRefWW := UINT_TO_WORD(InvRef2);
3   //InvRefSW :=Swap(UINT_TO_WORD(InvRef2));
4   //InvRef := WORD_TO_UINT(InvRefSW);
5
6   InvRef := REAL_TO_INT((Vel_Max_mms - Vfd_SP_Modif ) / Output_Ratio);
```

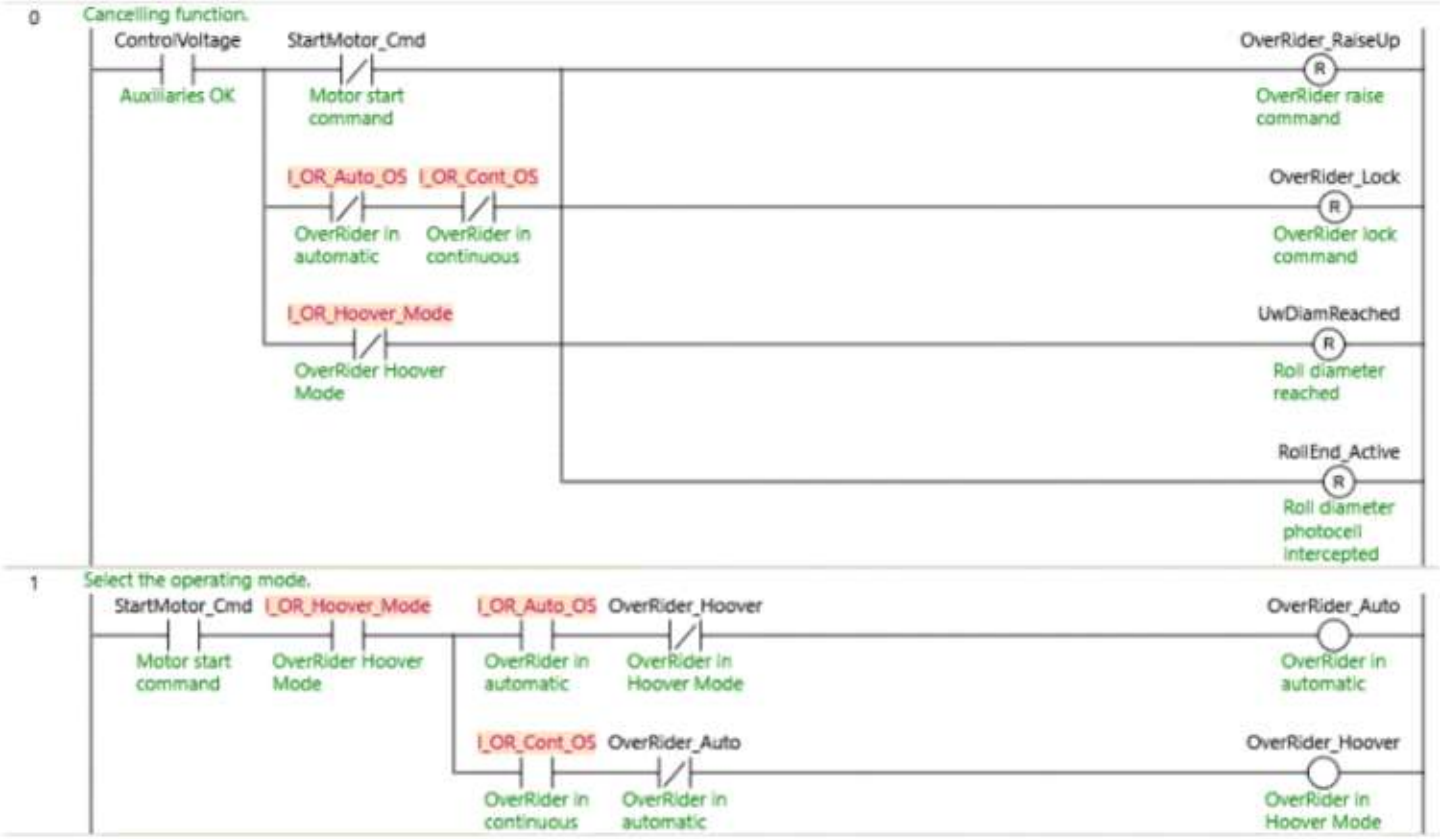
19 Updating potentiometer value.

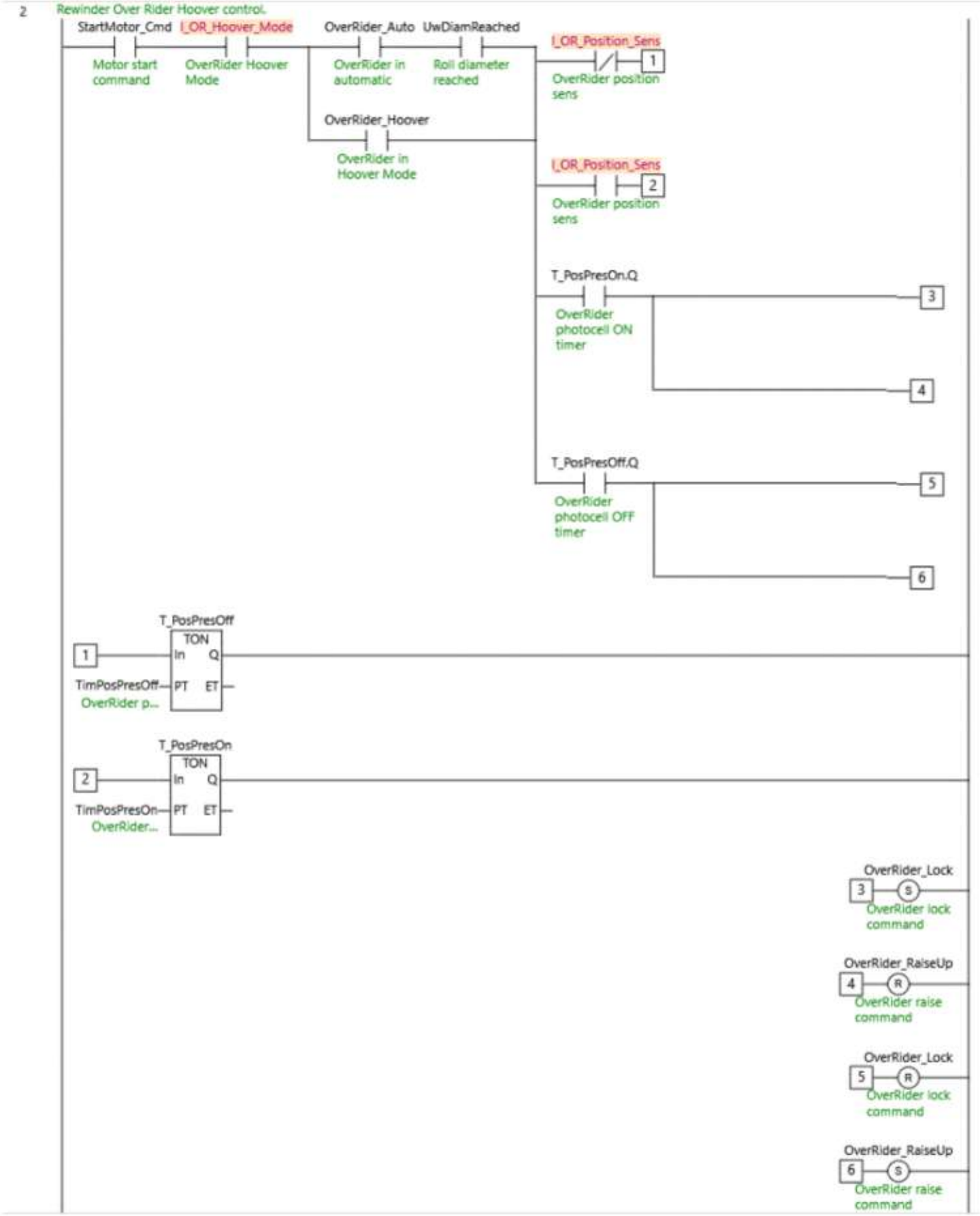



```
1
1 IF SoftStop THEN
2     InvRef := INT#0;                    // v6m: drive ramps down on its own
3 ELSIF SlowDown = FALSE THEN
4     InvRef := Pot1_Work_Velocity;
5 ELSE
6     InvRef := Pot2_Slow_Velocity;
7 END_IF;
```


20
Slow speed.
ManMode
JogRun
1
Manual mode
V10: JOG really running (button, not a normal run, not a soft stop)
1
1 VelSet := JogSpd_SP;
2 VelSet_mms := VelSet * Mt_To_mm;
3
4 //InvRef2 := REAL_TO_UINT((VelSet_mms) / Output_Ratio);
5 //InvRefWW := UINT_TO_WORD(InvRef2);
6 //InvRefSW :=Swap(UINT_TO_WORD(InvRef2));
7 //InvRef := WORD_TO_UINT(InvRefSW);
8
9 InvRef := REAL_TO_INT(VelSet_mms / Output_Ratio);

1-8-1-1-4.OverRider_Manager




2
Rewinder Over Rider Hoover control.
StartMotor_Cmd
Motor start command
I_OR_Hoover_Mode
OverRider Hoover Mode
OverRider_Auto
OverRider in automatic
UwDiamReached
Roll diameter reached
I_OR_Position_Sens
1
OverRider position sens
OverRider_Hoover
OverRider in Hoover Mode
I_OR_Position_Sens
2
OverRider position sens
T_PosPresOn.Q
OverRider photocell ON timer
3
4
T_PosPresOff.Q
OverRider photocell OFF timer
5
6
T_PosPresOff
TON
In
Q
PT
ET
TimPosPresOff
OverRider p...
T_PosPresOn
TON
In
Q
PT
ET
TimPosPresOn
OverRider...
OverRider_Lock
S
OverRider lock command
OverRider_RaiseUp
R
OverRider raise command
OverRider_Lock
R
OverRider lock command
OverRider_RaiseUp
S
OverRider raise command

3 Rider roll raise command.

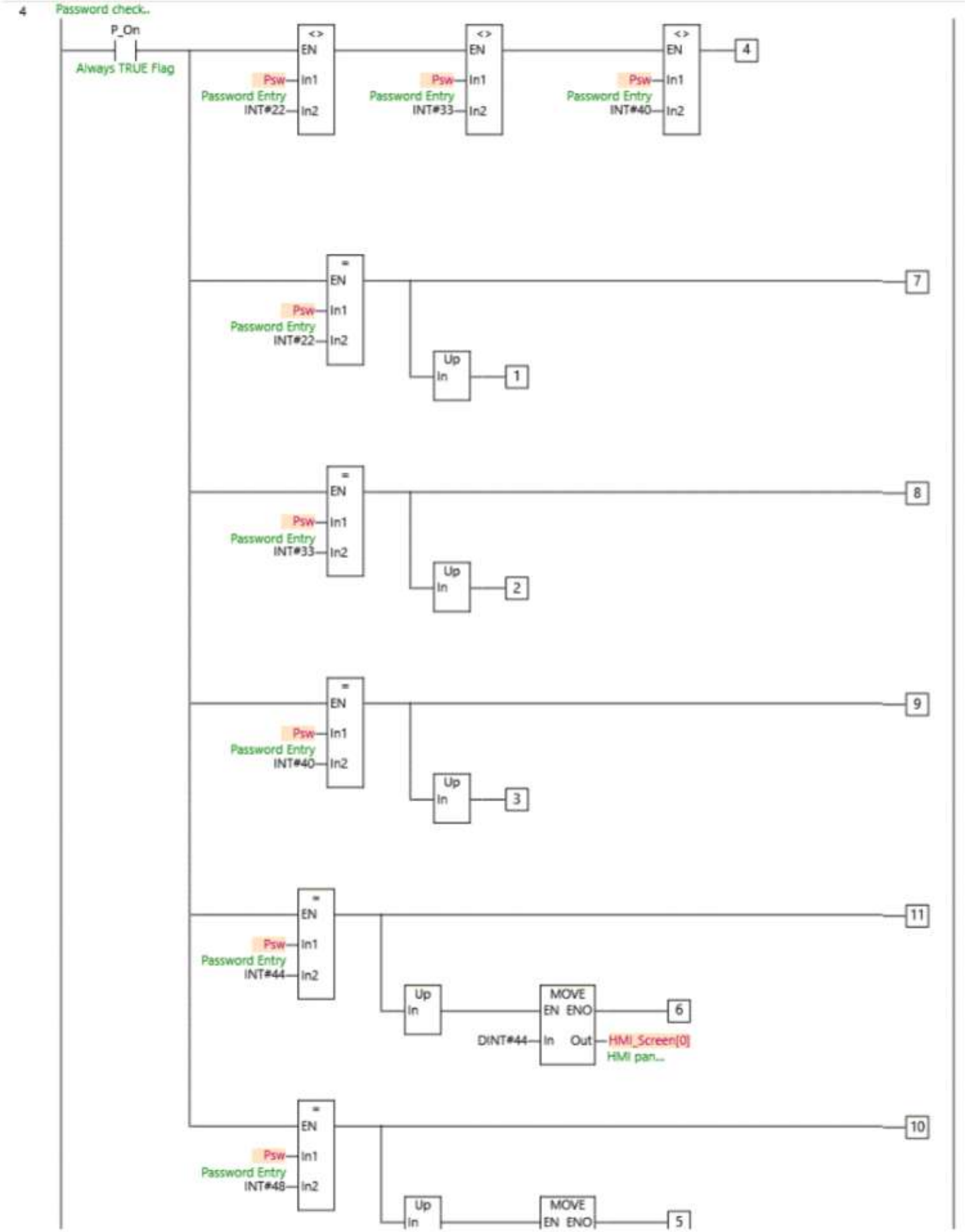


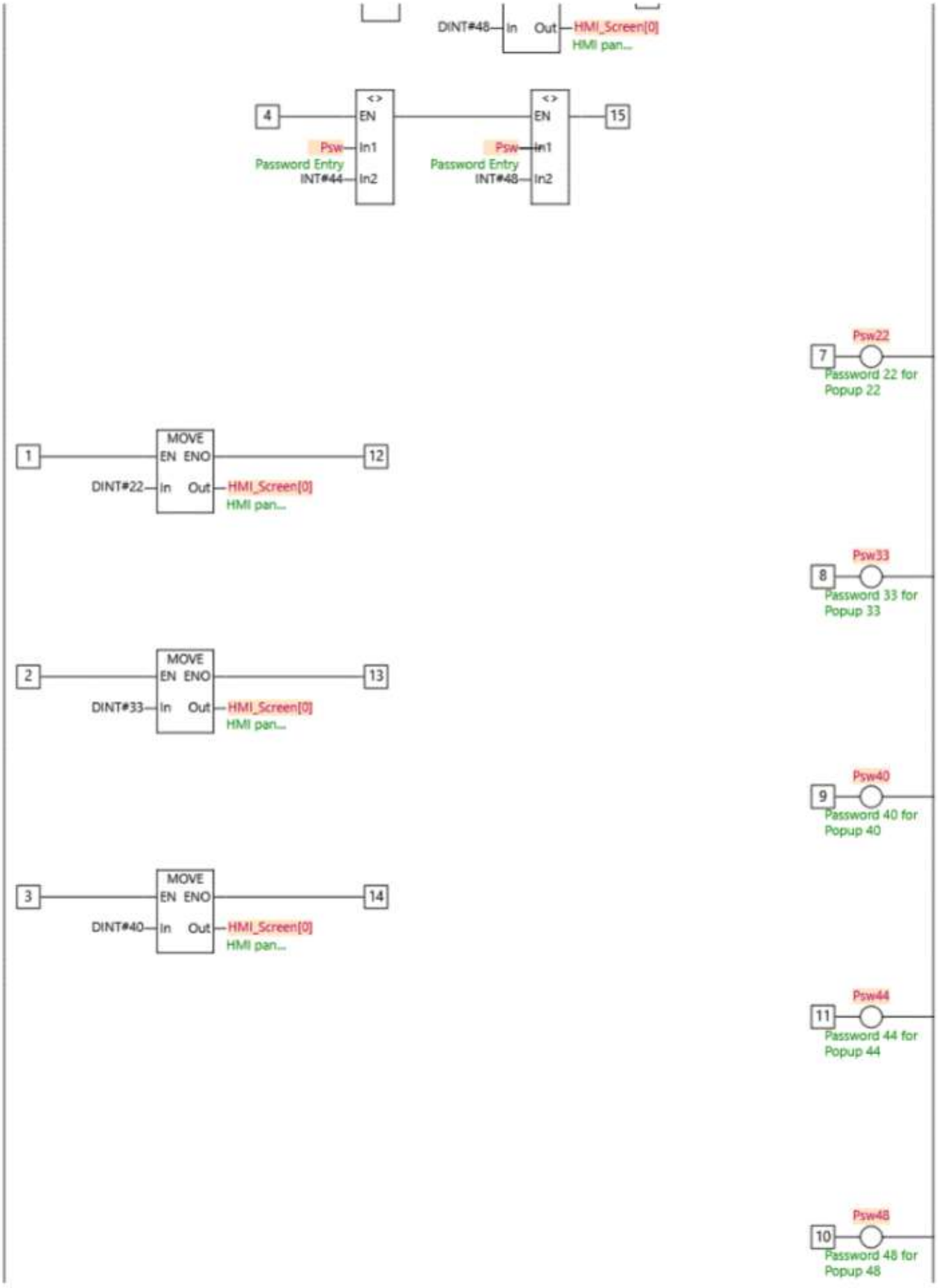
4 Rider roll raise command Reset

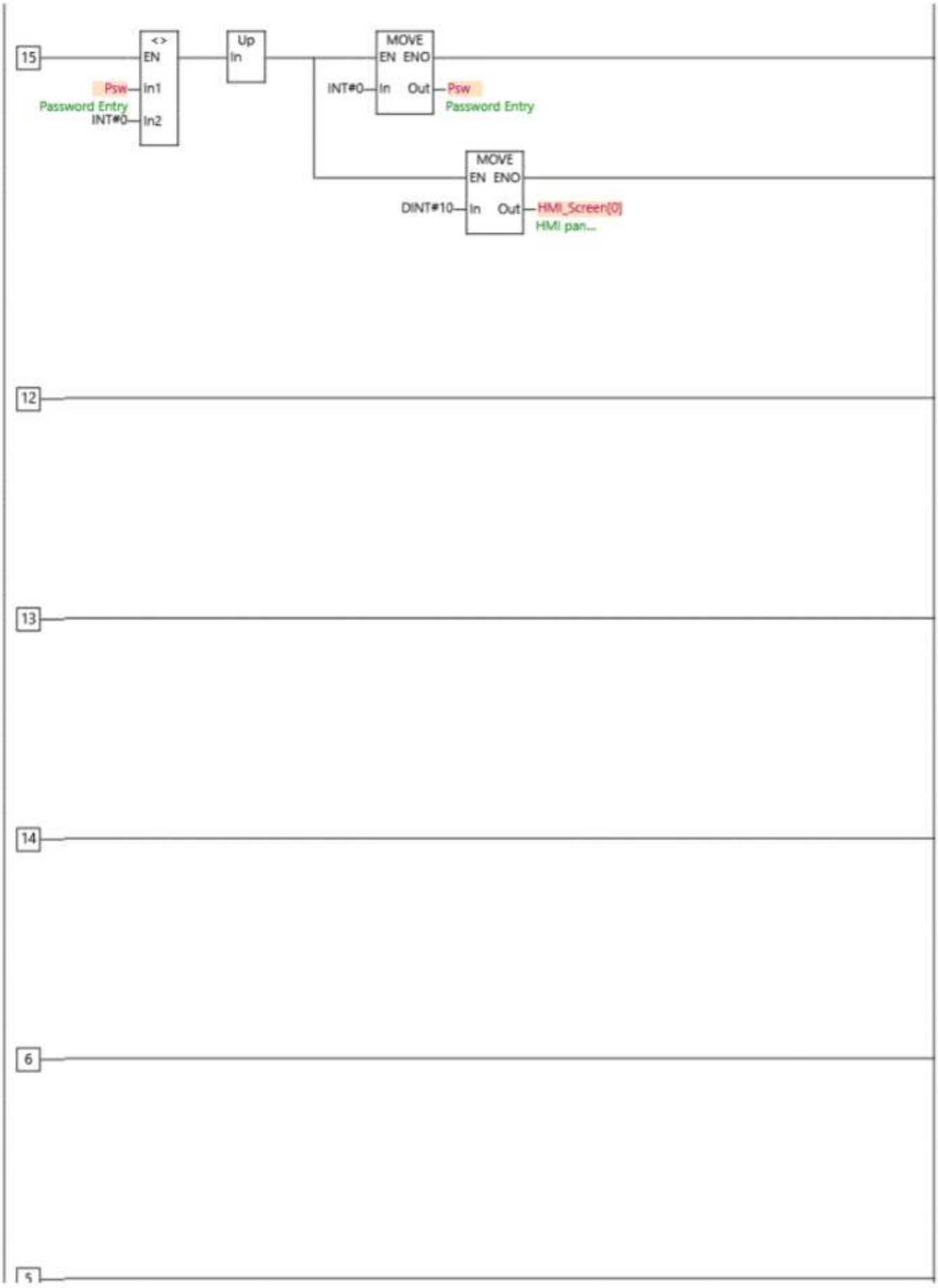


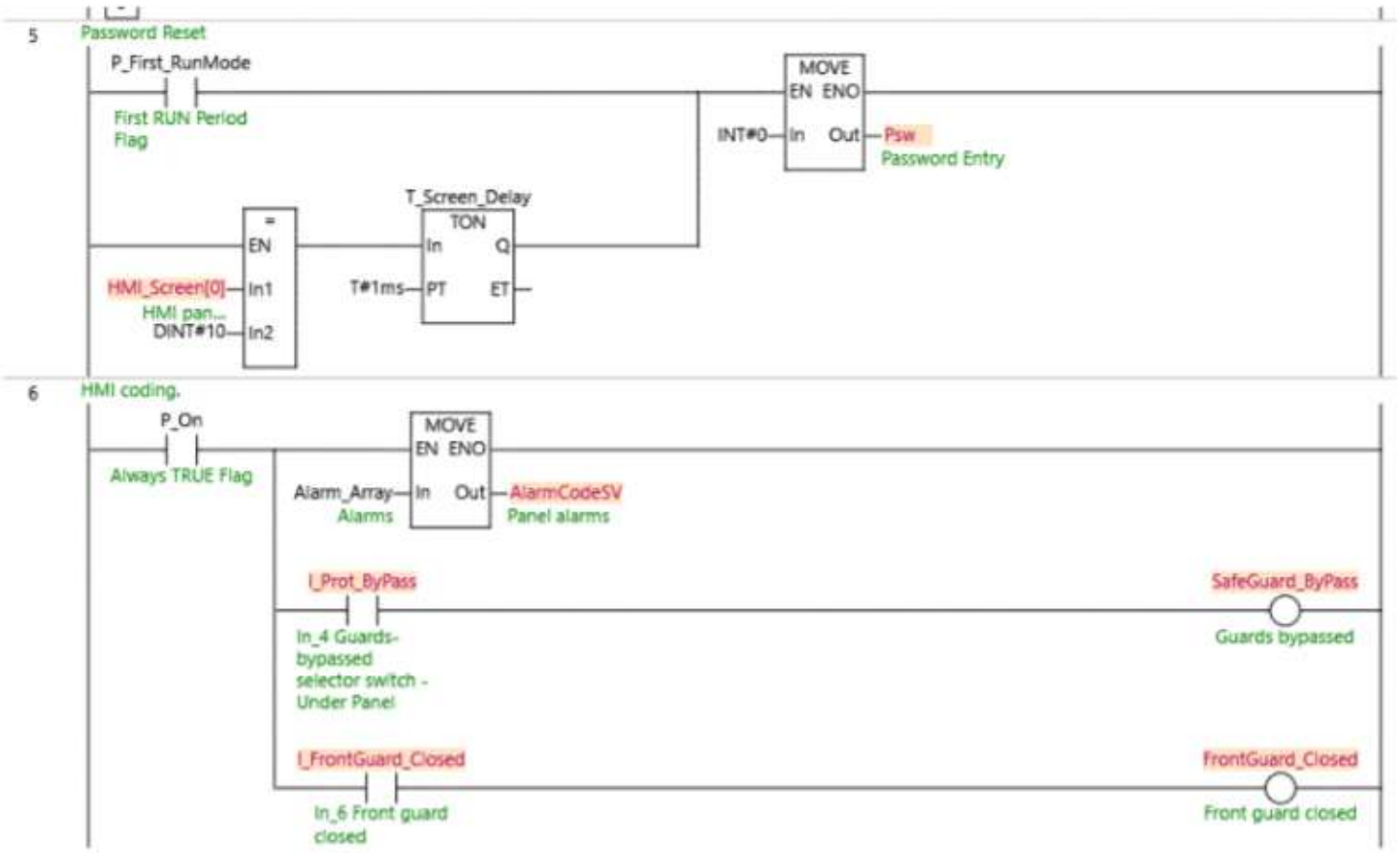
1-8-1-1-5.Console




4
Password check..
P_On
Always TRUE Flag
<>
EN
Psw
Password Entry
In1
In2
INT#22
INT#33
INT#40
4
=
7
1
8
2
9
3
11
6
10
5
Up
In
MOVE
EN ENO
DINT#44
Out
HMI_Screen[0]
HMI pan...


DINT#48
In
Out
HMI_Screen[0]
HMI pan...
4
EN
Psw
Password Entry
INT#44
In1
In2
15
INT#48
Psw22
7
Password 22 for Popup 22
MOVE
EN ENO
1
12
DINT#22
Psw33
8
Password 33 for Popup 33
2
13
DINT#33
Psw40
9
Password 40 for Popup 40
3
14
DINT#40
Psw44
11
Password 44 for Popup 44
Psw48
10
Password 48 for Popup 48


15
<>
EN
Psw
Password Entry
In1
INT#0
In2
Up
In
MOVE
EN ENO
INT#0
In
Out
Psw
Password Entry
MOVE
EN ENO
DINT#10
In
Out
HMI_Screen[0]
HMI pan...
12
13
14
6
5


5
Password Reset
P_First_RunMode
First RUN Period Flag
MOVE
EN ENO
INT#0
In
Out
Psw
Password Entry
T_Screen_Delay
TON
In
Q
PT
ET
T#1ms
=
EN
In1
In2
HMI_Screen[0]
HMI pan...
DINT#10
6
HMI coding.
P_On
Always TRUE Flag
MOVE
EN ENO
Alarm_Array
Alarms
In
Out
AlarmCodeSV
Panel alarms
I_Prot_ByPass
In_4 Guards-bypassed selector switch - Under Panel
SafeGuard_ByPass
Guards bypassed
I_FrontGuard_Closed
In_6 Front guard closed
FrontGuard_Closed
Front guard closed

1-8-1-1-6.Alarms

0 Check Present alarms.

```
// Alarm / warning handling + panel buzzer (BuzzerOn).
// Alarms: buzzer pulses (Clock1s). ALARM RESET (panel or HMI) silences it 1 min;
//         a NEW alarm during the silence sounds again at once.
// Warnings: ONE 1 s beep each time a new warning appears.
// PrimaryTask 4 ms: 250 scans = 1 s, 15000 scans = 60 s.

AlarmOn := FALSE;
WarningOn := FALSE;
AlarmCnt := 0;
WarnCnt := 0;

FOR Index_A := 0 TO 32 DO

   IF Alarm_Array[Index_A] THEN

      CASE Index_A OF
         5, 13, 15, 16, 17, 18, 19, 20, 21:        // warnings
            WarningOn := TRUE;
            WarnCnt := WarnCnt + 1;

            IF NOT AlarmMem[Index_A] THEN
               WarnBeepTicks := 250;              // NEW warning -> 1 s beep
            END_IF;
      ELSE                                        // alarms
         AlarmOn := TRUE;
         AlarmCnt := AlarmCnt + 1;
      END_CASE;

   END_IF;
   AlarmMem[Index_A] := Alarm_Array[Index_A];
END_FOR;

// ALARM RESET button (panel I_RsetAlarm or HMI Alarm_Reset): silence 1 min

RstEdge := (I_RsetAlarm OR Alarm_Reset) AND NOT RstMem;
RstMem := I_RsetAlarm OR Alarm_Reset;

IF RstEdge AND AlarmOn THEN
   SilenceTicks := 15000;
END_IF;

IF (AlarmCnt > AlarmCntMem) OR NOT AlarmOn THEN
   SilenceTicks := 0;                             // new alarm / no alarm: not silenced
END_IF;

IF SilenceTicks > 0 THEN
   SilenceTicks := SilenceTicks - 1;
END_IF;

IF WarnBeepTicks > 0 THEN
   WarnBeepTicks := WarnBeepTicks - 1;
END_IF;

AlarmCntMem := AlarmCnt;
WarnCntMem := WarnCnt;

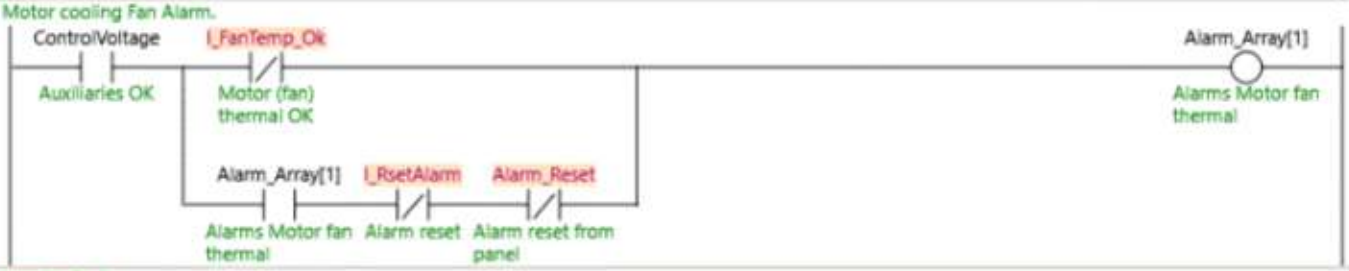
```

```
57  AlarmSilenced := AlarmOn AND (SilenceTicks > 0);
58
59  BuzzerOn := ControlVoltage
60          AND ((AlarmOn AND NOT AlarmSilenced AND Clock1s) OR (WarnBeepTicks
    > 0));
```

1

```
1   // ---- Alarm 14 (ALARM): unwinder sonar (NX-AD2603 Ch1, P+F UC2000) fault ----
2   // Raw value out of range for 1 s (no echo / broken wire / sensor off).
3   // Set when the measurement is used (unwinder AutoStop ON or motor running).
4   // latched. RESET only once the sensor is OK again (same as alarm 10, DT35).
5
6   T_UwSensFlt(In := (AI_Diam_Unwinder < UwRawMin) OR (AI_Diam_Unwinder >
    UwRawMax), PT := T#1s);
7   UwSensTest := T_UwSensFlt.Q;
8
9   IF ControlVoltage AND Measure_ON AND UwSensTest AND (UwStopEnable OR
    StartMotor_Cmd) THEN
10      UwSensFault := TRUE;
11  ELSIF NOT UwSensTest AND (I_RsetAlarm OR Alarm_Reset) THEN
12      UwSensFault := FALSE;
13  END_IF;
14
15  // ---- Alarm 17 (WARNING): Paper thickness not learned ----
16  // RwMat_K = 0 -> diameter / stop-length targets use the fixed fallback.
17  // Only above 300 mm rewinder diameter (learning needs ~20 mm growth after
    NEW PAPER),
18  // so it does not beep at every new roll. Not latched.
19
20  Thickness_Fault := ControlVoltage AND StartMotor_Cmd
21              AND (RwMat_K <= 0.0) AND (RwDiam > 300.0);
```

2 Motor cooling Fan Alarm.



3 Inverter fault.



4 Machine in emergency mode.



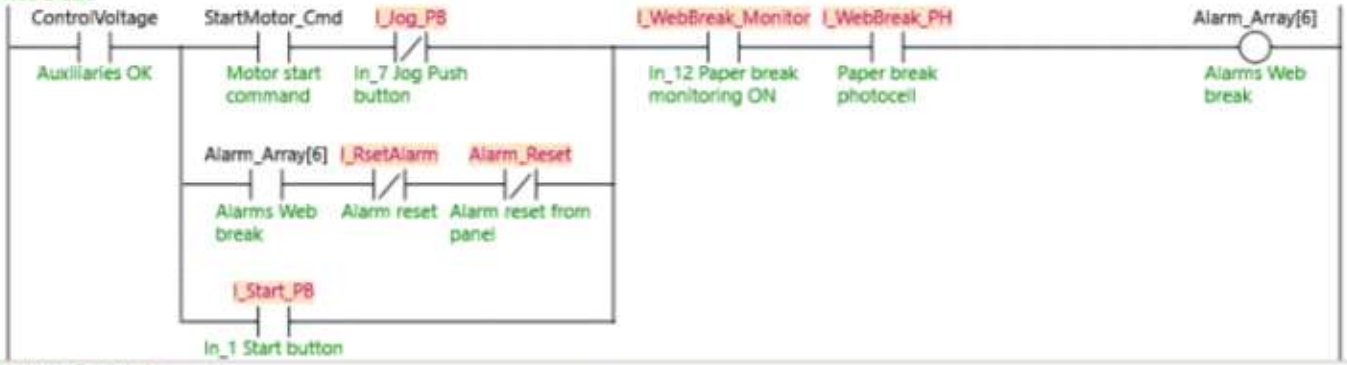
5 Main Motor thermal cutout.



6 Front Guard - Warning



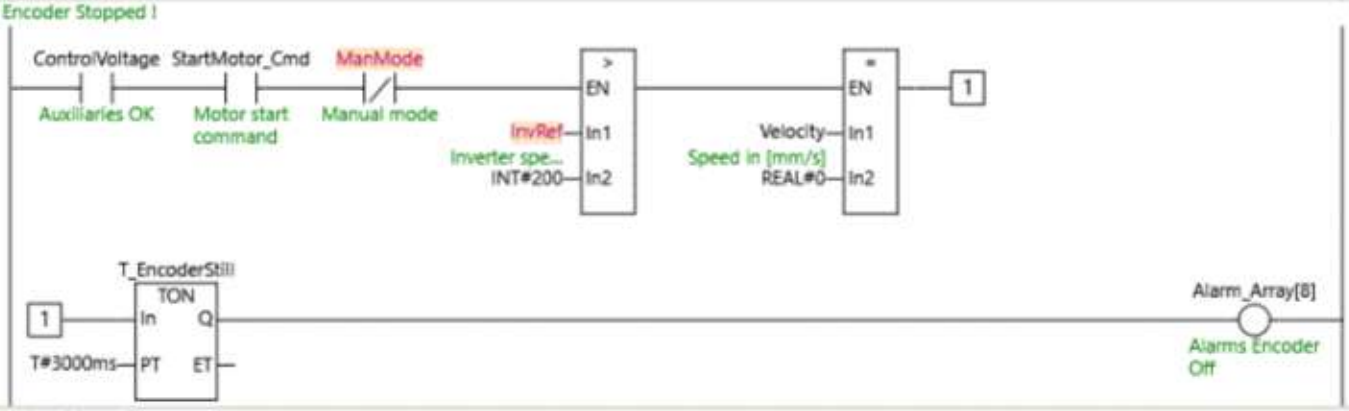
7 Web Break.



8 Vacuum Cooling fan.



9 Encoder Stopped !



10 Hub not in position.



11 Rewinder Diameter Sensor Fault.



12 Rewinder Diameter Sensor Fault - Reset



13 Rewinder max diameter Machine protection.



14 Rewinder max diameter Machine protection - Reset



15 Emergency Stop by Holding Stop PB 2s



16 Web Tension Monitor - Warning



17 Unwinder Diameter Sensor Fault.



18 Unwinder Roll is Empty - Warning



19 Unwinder Roll is Empty - Reset



20 Rewinder GSM learning fault



21 Rewinder learning fault - Reset



22 Paper Thickness Learn Process - Not Done ! - Warning



23 Side Door Opened - Warning

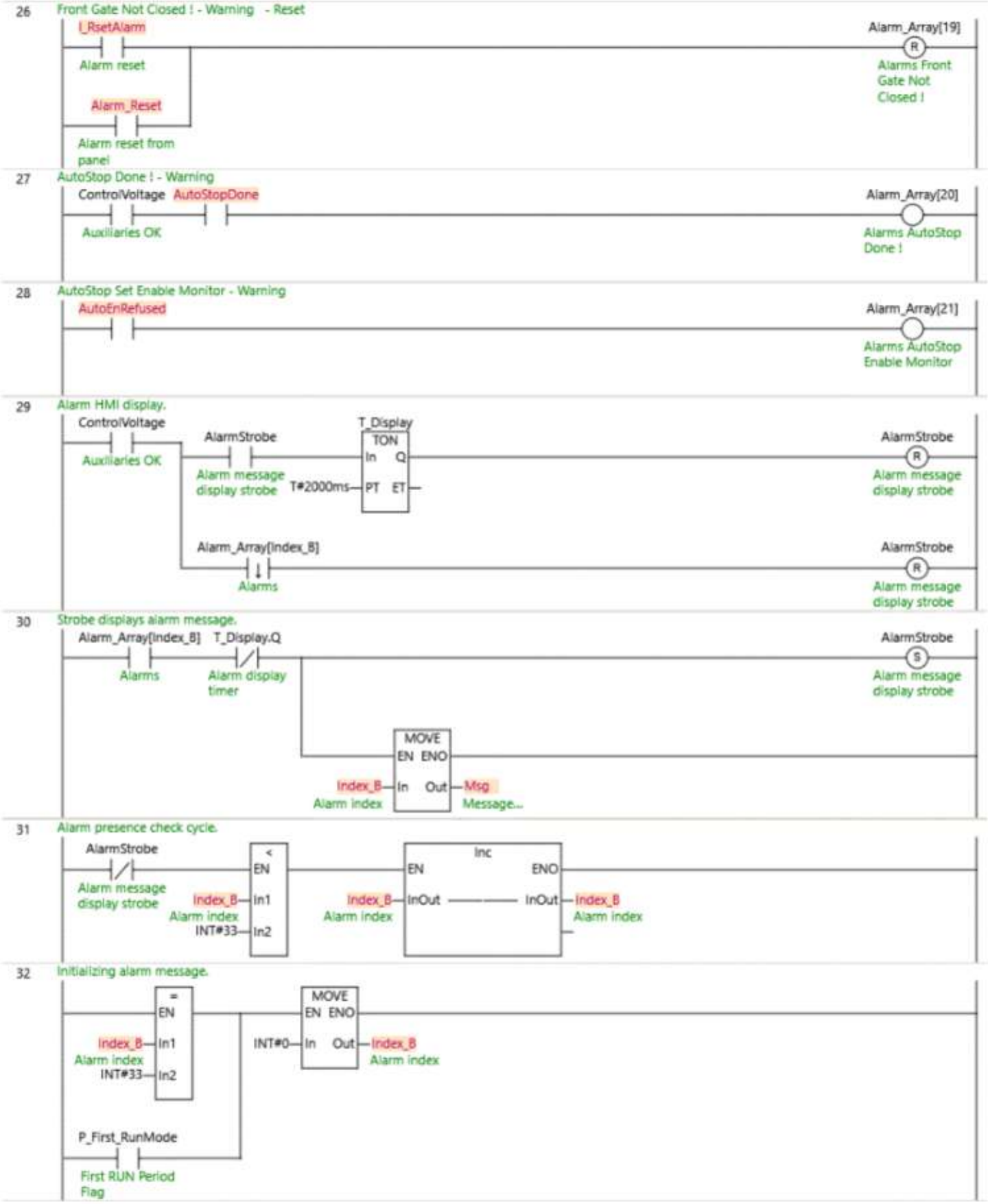


24 Side Door Opened - Warning - Reset



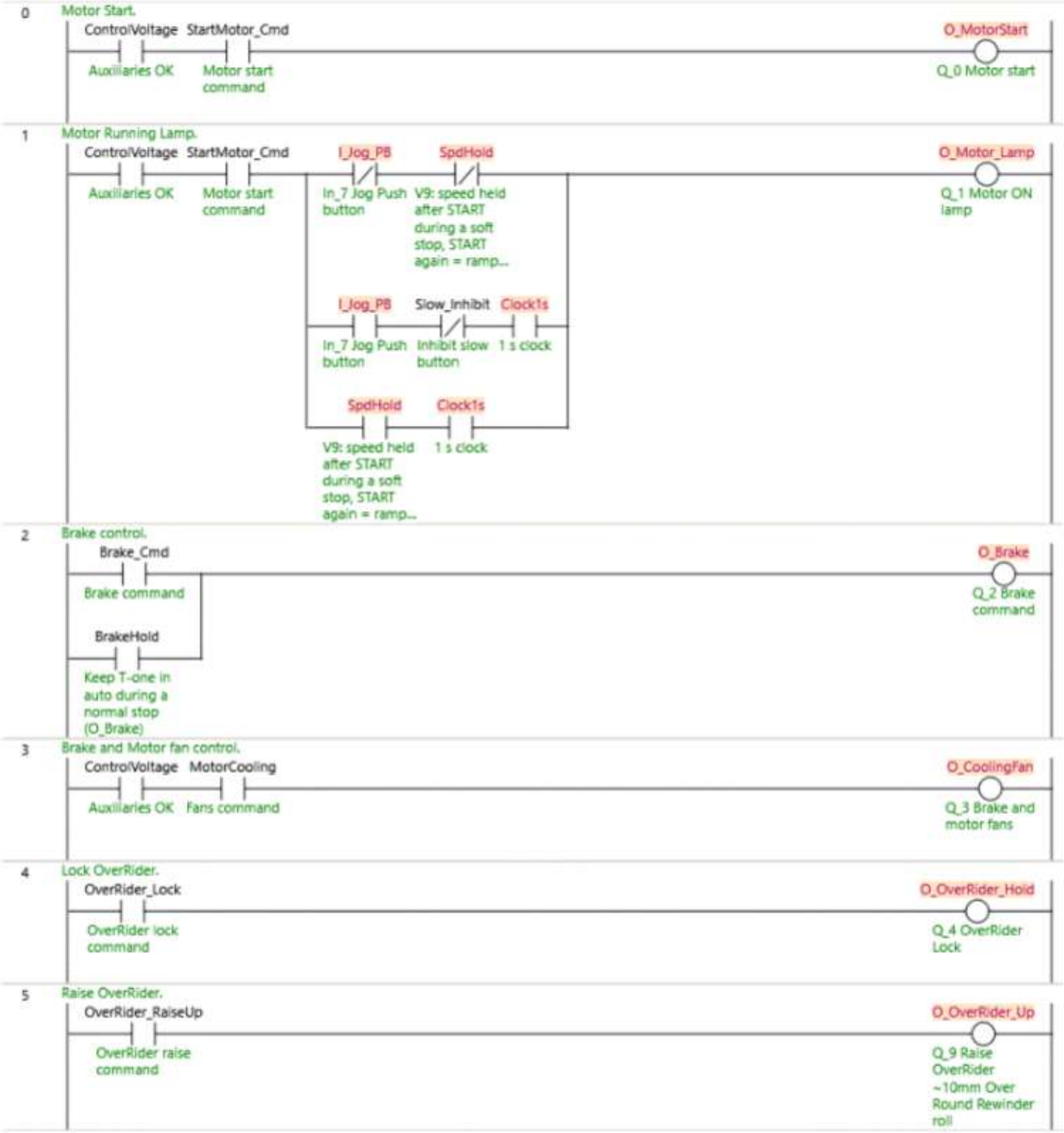
25 Front Gate Not Closed ! - Warning

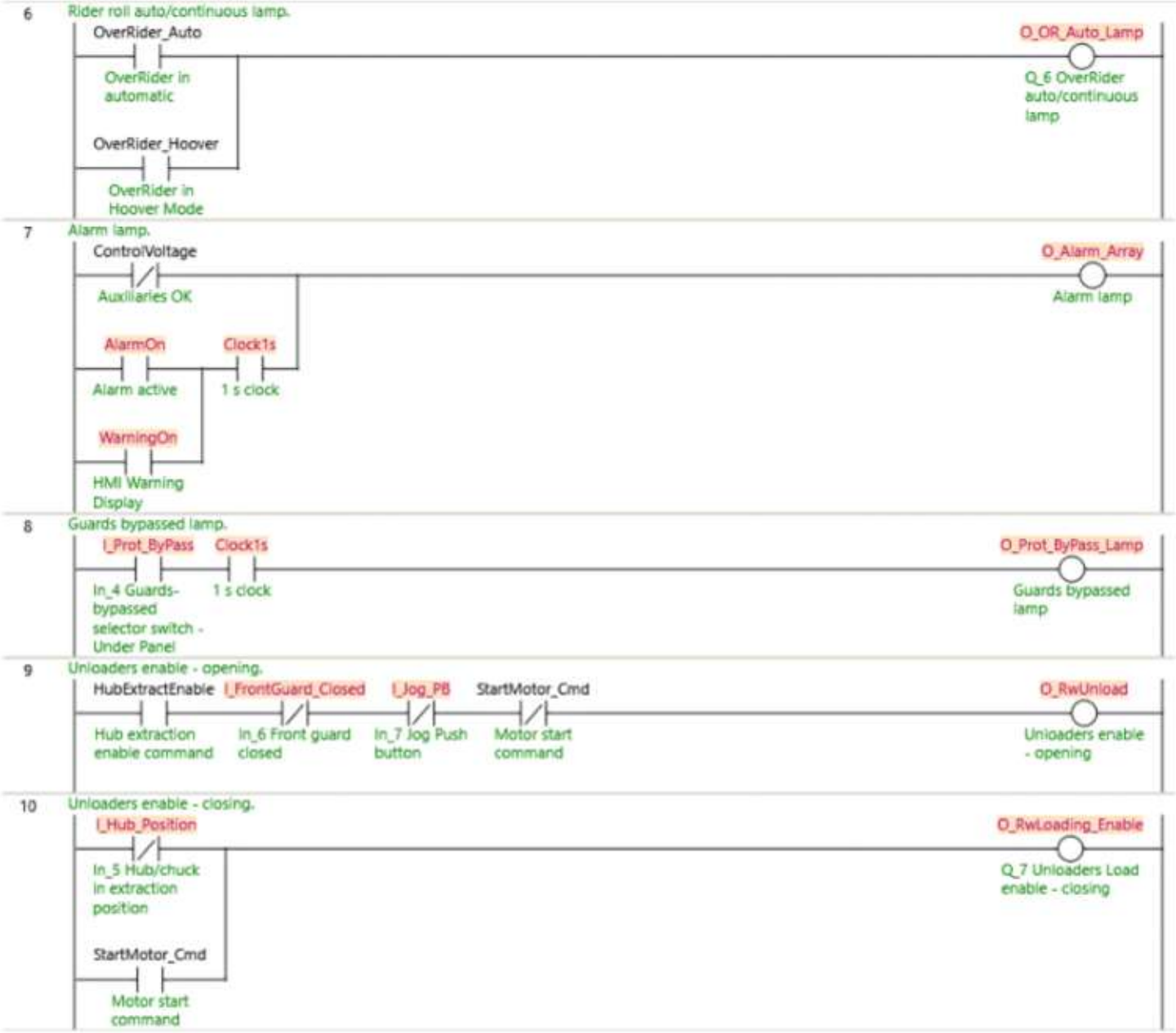



26
Front Gate Not Closed ! - Warning - Reset
I_RsetAlarm
Alarm reset
Alarm_Reset
Alarm reset from panel
Alarm_Array[19]
R
Alarms Front Gate Not Closed !
27
AutoStop Done ! - Warning
ControlVoltage
Auxiliaries OK
AutoStopDone
Alarm_Array[20]
Alarms AutoStop Done !
28
AutoStop Set Enable Monitor - Warning
AutoEnRefused
Alarm_Array[21]
Alarms AutoStop Enable Monitor
29
Alarm HMI display.
ControlVoltage
Auxiliaries OK
AlarmStrobe
Alarm message display strobe
T_Display
TON
In Q
T#2000ms
PT ET
AlarmStrobe
R
Alarm message display strobe
Alarm_Array[Index_B]
Alarms
AlarmStrobe
R
Alarm message display strobe
30
Strobe displays alarm message.
Alarm_Array[Index_B]
Alarms
T_Display.Q
Alarm display timer
AlarmStrobe
S
Alarm message display strobe
MOVE
EN ENO
Index_B
Alarm index
In Out
Msg
Message...
31
Alarm presence check cycle.
AlarmStrobe
Alarm message display strobe
<
EN
Index_B
Alarm index
In1
INT#33
In2
Inc
EN ENO
Index_B
Alarm index
InOut InOut
Index_B
Alarm index
32
Initializing alarm message.
=
EN
Index_B
Alarm index
In1
INT#33
In2
MOVE
EN ENO
INT#0
In Out
Index_B
Alarm index
P_First_RunMode
First RUN Period Flag


33
Alarm Reset
AlarmOn
Alarm active
WarningOn
HMI Warning Display
MOVE
EN ENO
INT#33
In
Out
Msg
Message...

1-8-1-1-7.Output_Manager




6
Rider roll auto/continuous lamp.
OverRider_Auto
OverRider in automatic
OverRider_Hoover
OverRider in Hoover Mode
O_OR_Auto_Lamp
Q_6 OverRider auto/continuous lamp
7
Alarm lamp.
ControlVoltage
Auxiliaries OK
AlarmOn
Alarm active
Clock1s
1 s clock
WarningOn
HMI Warning Display
O_Alarm_Array
Alarm lamp
8
Guards bypassed lamp.
I_Prot_ByPass
In_4 Guards-bypassed selector switch - Under Panel
Clock1s
1 s clock
O_Prot_ByPass_Lamp
Guards bypassed lamp
9
Unloaders enable - opening.
HubExtractEnable
Hub extraction enable command
I_FrontGuard_Closed
In_6 Front guard closed
I_Jog_PB
In_7 Jog Push button
StartMotor_Cmd
Motor start command
O_RwUnload
Unloaders enable - opening
10
Unloaders enable - closing.
I_Hub_Position
In_5 Hub/chuck in extraction position
StartMotor_Cmd
Motor start command
O_RwLoading_Enable
Q_7 Unloaders Load enable - closing

11 Horn / acoustic signal.

```
// ==== V9: REPLACE Output_Manager rung 11 ("Horn / acoustic signal.", ladder) by ONE ST rung
// ==== with this code (no contact in front). Comment: "V9: horn, beeps set on the HMI".
// Old: horn blinking with Clock1s (0.5 s ON / 0.5 s OFF) during the whole slow-down
//      and the crawl at final speed -> very annoying.
// New, slow-down / final speed (StartMotor_Cmd AND SlowDown AND NOT Stop):
//      HornStartBeep_s beep when it starts (1 s).
//      then a HornBeepLen_s beep (0.2 s) every HornBeepPeriod (5 s).
//      then ONE HornStopBeep_s beep (3 s) when the machine has stopped (< 2 m/min).
//      Each setting 0 = that beep OFF. Front guard moving: unchanged (Clock1s blinking).
// PrimaryTask 4 ms: 250 scans = 1 s.

// HMI limits

IF HornStartBeep_s < REAL#0 THEN
    HornStartBeep_s := REAL#0;
ELSIF HornStartBeep_s > REAL#5 THEN
    HornStartBeep_s := REAL#5;
END_IF;

IF HornBeepLen_s < REAL#0 THEN
    HornBeepLen_s := REAL#0;
ELSIF HornBeepLen_s > REAL#2 THEN
    HornBeepLen_s := REAL#2;
END_IF;

IF HornBeepPeriod < REAL#0 THEN
    HornBeepPeriod := REAL#0;
ELSIF (HornBeepPeriod > REAL#0) AND (HornBeepPeriod < HornBeepLen_s + REAL#0.5) THEN
    HornBeepPeriod := HornBeepLen_s + REAL#0.5;    // always a pause between beeps
ELSIF HornBeepPeriod > REAL#30 THEN
    HornBeepPeriod := REAL#30;
END_IF;

IF HornStopBeep_s < REAL#0 THEN
    HornStopBeep_s := REAL#0;
ELSIF HornStopBeep_s > REAL#10 THEN
    HornStopBeep_s := REAL#10;
END_IF;

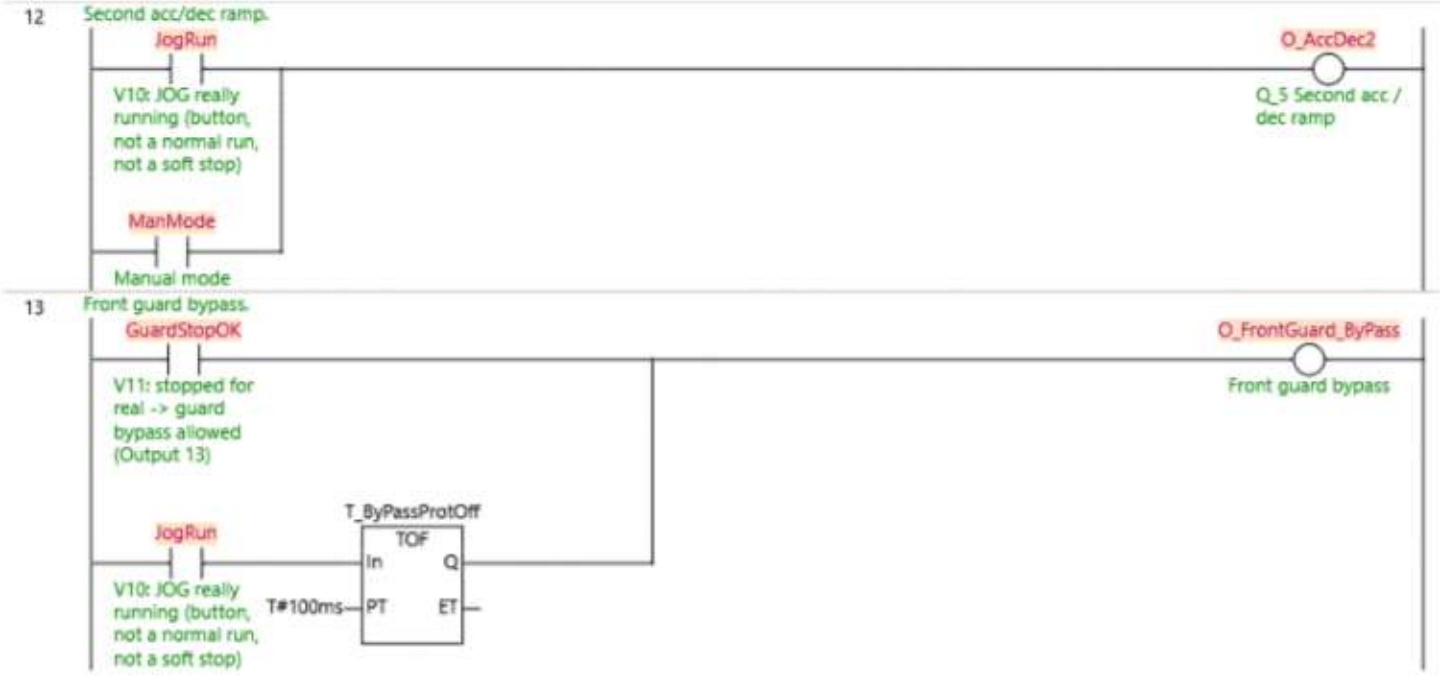
HornStartScans := REAL_TO_INT(HornStartBeep_s * REAL#250);
HornLenScans   := REAL_TO_INT(HornBeepLen_s * REAL#250);
HornPer        := REAL_TO_INT(HornBeepPeriod * REAL#250);

// Slow-down / final speed beeps

HornRun := StartMotor_Cmd AND SlowDown AND NOT Stop AND NOT JogRun;  // no horn while jogging

IF HornRun THEN
    IF NOT HornRunMem THEN
        HornTicks := 0;                    // slow-down starts
```

```
51      END_IF;
52
53      IF HornTicks < 30000 THEN
54          HornTicks := HornTicks + 1;
55      END_IF;
56
57      IF HornTicks <= HornStartScans THEN
58          HornSlowBeep := TRUE;                    // start beep
59          HornCnt := 0;
60      ELSIF (HornPer > 0) AND (HornLenScans > 0) THEN
61          HornCnt := HornCnt + 1;
62          IF HornCnt >= HornPer THEN
63              HornCnt := 0;
64          END_IF;
65
66          HornSlowBeep := (HornCnt >= HornPer - HornLenScans);   // beep at the end
    of each period
67      ELSE
68          HornSlowBeep := FALSE;                   // periodic beeps OFF
69      END_IF;
70  ELSE
71      HornTicks := 0;
72      HornCnt := 0;
73      HornSlowBeep := FALSE;
74  END_IF;
75
76  // End of a slow-down by a stop (auto stop, STOP button): one beep at standstill
77  // (Stop is set in Motor_Manager before StartMotor_Cmd drops on the next scan)
78
79  IF HornRunMem AND NOT HornRun AND (Stop OR NOT StartMotor_Cmd) THEN
80      HornStopPend := TRUE;
81  END_IF;
82
83  IF StartMotor_Cmd AND NOT HornStartMem THEN
84      HornStopPend := FALSE;                       // restarted: no stop beep
85      HornStopTicks := 0;
86  END_IF;
87
88  HornStartMem := StartMotor_Cmd;
89
90  IF HornStopPend AND (Velocity_M_min < REAL#2) THEN
91      HornStopPend := FALSE;
92      HornStopTicks := REAL_TO_INT(HornStopBeep_s * REAL#250);
93  END_IF;
94
95  IF HornStopTicks > 0 THEN
96      HornStopTicks := HornStopTicks - 1;
97  END_IF;
98
99  HornRunMem := HornRun;
100
101 // HMI test button: horn while pressed (machine stopped only)
102
103 O_AcousticSignal := (NOT I_FrontGuard_Closed AND NOT I_FrontGuard_Open AND
    Clock1s)
104                 OR HornSlowBeep OR (HornStopTicks > 0)
105                 OR (HornTest AND NOT StartMotor_Cmd);
```


12
Second acc/dec ramp.
JogRun
V10: JOG really running (button, not a normal run, not a soft stop)
ManMode
Manual mode
O_AccDec2
Q_5 Second acc / dec ramp
13
Front guard bypass.
GuardStopOK
V11: stopped for real -> guard bypass allowed (Output 13)
JogRun
V10: JOG really running (button, not a normal run, not a soft stop)
T_ByPassProtOff
TOF
In
Q
T#100ms
PT
ET
O_FrontGuard_ByPass
Front guard bypass

1-8-1-1-8.AutoStop_Control

0 Auto Stop Process 1-5

```
(*
=============================================================
====================
  AutoStop_Control  V10  -  PART 1 of 2 (rung 0): sections 1 - 5c
  Project : 04820KA001 Mono Canada II  (NX1P2, PrimaryTask 4 ms)
  Rewinder (Rw) : SICK DT35 0-10 V -> AI_Diam_Rewinder (AD2603 Ch2)
  Unwinder (Uw) : meter counter mode + sonar diameter (UwDiam_Act)

  NAMING  Uw_ = unwinder   Rw_ = rewinder
        _mm = millimeters   _m = meters
        _Pls = encoder pulses (counts)   _Pps = pulses per second
        _Pps2 = pulses per second squared

  V10: JogRun (jog really running) instead of the raw jog button (5, 5b):
       long baseline also restarted if the encoder goes 2 m back (4b).
  V9 : RwDiamRev = DT35 averaged over exactly one revolution (2b), used by
       the thickness learning (removes the out-of-round wave), RwWave_mm.
     GSM popup: once per real new roll (empty 3 s, then > stop + 100 mm
       for 3 s): NEW PAPER closes it (section 10).
     paper thickness: long-baseline learning (D^2 growth / length) from the
       start point, LOCKED at RwLockDiam (300 mm): no more change, no
       paper-change detection, until NEW PAPER / GSM popup YES (4b).
  V8 : = V7 + V7c unwinder DIAMETER slow-down follows the speed (6c).
       V7e ManMode slow-down model as before V7 (section 6),
       V7f S-curve extra length in the slow-down (section 6)
  V7 : FEED-FORWARD speed loop (Motor_Manager rung 8): reference = set speed
     x RwRatioD / diameter along a ramp in real web mm/s2 (DecelWeb /
     AccelWeb), RwRatioD adapted slowly -> no hunting on big rolls.
     Section 6: slow-down length = one ramp at DecelWeb (FFEnable), else
     web coarse ramp + trim tail; learning only as fallback.
  v6y: rewinder AUTO stop (RwStopEnable) limits the work speed to
     RwAutoMaxSpd (500 m/min), taken over at standstill only; section 6
     uses the limited speed.
  v6x: new parent roll on the unwinder -> GsmAsk (HMI popup "did you change
     the paper?"); GsmAskYes forgets the learned thickness.
  v6w: paper thickness window is LENGTH-based instead of 10 s, modulated by
     the diameter (RwWinGrow_mm 2.5 mm of growth, 4 m .. RwWinLen_m 30 m):
     same accuracy at every speed and roll size; stopped window = dropped.
     ALARM RESET also clears LearnFailCnt / StopFailCnt (alarm 16).
  v6v: BrakeHold released on the SIGNED web speed (Velocity_M_min < 5),
     so a web pulled back after the stop releases the T-one at once.
  v6u: stop learning: a stop shorter than the reaction distance or faster
     than RwDecelMax_mm is ignored (not a fault). Only a too slow stop
     (< RwDecelMin_mm) counts toward alarm 16.
  v6t: rewinder slow-down fallback while the paper thickness is not
     learned: RwSlowReq at RwTargetDiam - RwSlowFallbackMm (30 mm).
  v6s: paper change: a learning window more than RwMatChangePct (25 %) away
     from RwMat_K is ignored once, taken directly the 2nd time in a row
     (automatic). RwMatReset (HMI "NEW PAPER") forgets RwMat_K.
  v6p: section 5c: RwLearnReset clears RwStopDecelK, RwStopDecel and
     RwSlowDecelK (the v6f seed that refilled RwStopDecelK is removed).
  v6l: previous value saved before every TEACH CORE (RwAxisDist_Prev,
       UwCoreDist_Prev, Retain) + UNDO buttons (RwTeachUndo,
       UwTeachUndo): swap current <-> previous. State 4 = RESTORED.
  v6k: unwinder TEACH CORE (Core_Measure) works like the rewinder one:
       same checks, 2 s stable with sonar AVERAGE, status UwTeachState.
```

```
55          Replaces General rung 15 (one immediate sonar reading).
56      v6j: TEACH CORE (section 14): HMI button RwTeachCore, empty core on the
57          rewinder, machine stopped -> after 2 s stable:
58          RwAxisDist := RwDist + RwCoreDiameter / 2. Status in RwTeachState.
59      v6i: auto: pots scaled to WorkSpd_Max / FinalSpd_Max (Motor_Manager rung 6,
60          Pot1_Mem / Pot2_Mem). ManMode keeps the raw pots (percent),
61          so section 6 uses the raw pots in ManMode.
62      v6h: BrakeHold keeps O_Brake (T-one "reel brake automatic") ON during a
63          normal stop until the web is almost stopped (< 5 m/min, max 15 s).
64          Before: T-one switched to "stop" with the run command, while the
65          rewinder was decelerating hard -> unwinder overran, web on the
66          floor (field 10/20/2026). Used in Output_Manager rung 2.
67      v6g: slow-down AND stop decel proportional to the rewinder diameter.
68          The ATV930 ramp is in Hz/s (motor = rewind shaft), so the web
69          decel [mm/s2] = shaft decel x radius. Field data 10/20/2026:
70          24 mm/s2 at 120 mm -> 90 mm/s2 at 600 mm.
71          Learned: RwSlowDecelK, RwStopDecelK [mm/s2 per mm of diameter].
72          Live:    RwSlowDecelLrn [mm/s2], RwStopDecel [pulses/s2] = K x DecelDiam.
73      v6f: learn the REAL slow-down deceleration (section 5b) -> RwSlowDecelLrn
74          (mm/s2). The ATV930 stretches its ramp with a heavy roll (no
75          braking resistor), so the PLC ramp (DecelCoarse) was too
76          optimistic: the stop came before final speed was reached.
77          Priority in section 6: RwSlowDecel (manual) > RwSlowDecelLrn
78          (learned) > DecelCoarse/DecelTrim model.
79      v6e: GSM_Learned_OK (replaces global RwLearnFault): TRUE only when
80          RwMat_K is learned, RwStopDecel is learned (if RwLearnStop) and
81          there is no learning fault. Alarm 16 uses GSM_LearnFault.
82      v6d: exact 100 ms tick (25 x 4 ms scan counter, the self-resetting
83          TON ran ~104-108 ms -> speed ~4-8 % too high)
84          standstill = <= 2 counts / 100 ms (encoder jitter)
85          stop learning skipped after jog / manual / slow-button runs and
86            when the auxiliaries drop (E-stop): RunWasJog, StopLearnValid
87          work / final speed from the potentiometers in BOTH modes
88            (Speed_Display / LowSpd_Display are not m/min in ManMode)
89          slow-down length includes the DecelTrim tail inside the
90            SpdDelta_Percent band (Motor_Manager rungs 14/15)
91      v6c: learning supervision -> RwLearnFault (alarm 16), learned paper
92          thickness RwThick_mm, plausibility limits for K and stop decel
93      v6b: RwMaxSlowAtDiam display + RwMaxSlowFallback (slow-down warning
94          while RwMat_K is not learned yet)
95      v6 : consistent internal names, units in names.
96          Globals renamed: RwRemain_m, RwStopLen_m, RwSlowLen_m,
97                      RwSlowMargin_m, UwSlowMargin_m
98      v5 : max diameter protection (always active), sensor plausibility,
99          12SA5 master (section 9), alarms in section Alarms (ladder)
100     v4b: AUTO interlock + single HMI display (SlowAtDisp / SlowAtMode)
101
        ================================================================
        ==================== *)
102
103 (* ---- 1. 100 ms tick, mm per encoder pulse ------------------------------ *)
104 (* v6d: 25 scans x 4 ms PrimaryTask = exactly 100 ms.                    *)
105 (*      Change 25 if the PrimaryTask period is changed.                  *)
106
107 TickCnt := TickCnt + 1;
108
109 IF TickCnt >= 25 THEN
110     TickCnt := 0;
111     Tick := TRUE;
```

```
ELSE
    Tick := FALSE;
END_IF;

IF (Encoder_Pulse > 0) AND (EncoderWheel_Circumf > 0.0) THEN
    MmPerPulse := EncoderWheel_Circumf / (INT_TO_REAL(Encoder_Pulse) * 4.0);
ELSE
    MmPerPulse := 0.0;
END_IF;

(* ---- 2. Rewinder DT35 -> filtered distance -> diameter ---------- *)

RwDistRaw_mm := RwDist0V + (INT_TO_REAL(AI_Diam_Rewinder) / 4000.0)
              * (RwDist10V - RwDist0V);

IF Tick THEN
    IF NOT RwFiltInit THEN
        RwDist := RwDistRaw_mm;
        RwFiltInit := TRUE;
    ELSE
        RwDist := RwDist + RwFilter_K * (RwDistRaw_mm - RwDist);
    END_IF;
END_IF;

RwSensFault := (AI_Diam_Rewinder < RwRawMin) OR (AI_Diam_Rewinder >
RwRawMax)
            OR (RwDist > RwAxisDist);

RwDiam := 2.0 * (RwAxisDist - RwDist);

IF RwDiam < 0.0 THEN
    RwDiam := 0.0;
END_IF;

(* ---- 2b. V9: diameter averaged over exactly ONE revolution ---------- *)
(* The out-of-round roll gives a wave at the rotation frequency (seen in   *)
(* the 10/29 traces: 2.4 Hz at 444 m/min / 973 mm). Averaging the raw      *)
(* DT35 distance over exactly one revolution cancels it (and all its       *)
(* harmonics), whatever the phase -> no shaft position sensor needed:      *)
(* one revolution = PI x D of web on the encoder. Each scan is weighted    *)
(* by its encoder pulses = by roll ANGLE, so ramps and standstill do not   *)
(* bias it (stopped: nothing added). Used only by the thickness learning. *)
(* RwWave_mm = peak-peak diameter wave in the last revolution (HMI).       *)

RevDP := ABS(Encoder_Pos - RevPrevPos);
RevPrevPos := Encoder_Pos;

IF RwSensFault OR (MmPerPulse <= 0.0) OR (RwDiam < 50.0) OR (RevDP > 100000)
THEN
    RevW := 0.0;
    RevSum := 0.0;
    RwDiamRevOk := RwDiamRevOk AND NOT RwSensFault;
ELSIF RevDP > 0 THEN

    IF RevW <= 0.0 THEN
        RevMin := RwDistRaw_mm;
        RevMax := RwDistRaw_mm;
    END_IF;

```

```
169     RevSum := RevSum + RwDistRaw_mm * DINT_TO_REAL(RevDP);
170     RevW := RevW + DINT_TO_REAL(RevDP);
171
172     IF RwDistRaw_mm < RevMin THEN
173         RevMin := RwDistRaw_mm;
174     END_IF;
175
176     IF RwDistRaw_mm > RevMax THEN
177         RevMax := RwDistRaw_mm;
178     END_IF;
179
180     IF RevW >= 3.14159 * RwDiam / MmPerPulse THEN
181         RwDiamRev := 2.0 * (RwAxisDist - RevSum / RevW);
182         RwWave_mm := 2.0 * (RevMax - RevMin);
183         RwDiamRevOk := TRUE;
184         RevW := 0.0;
185         RevSum := 0.0;
186     END_IF;
187 END_IF;
188
189 (* v6g: diameter used by the decel model (last valid on sensor fault)      *)
190
191 IF NOT RwSensFault THEN
192     DecelDiam := RwDiam;
193 ELSE
194     DecelDiam := RwLastValidDiam;
195 END_IF;
196
197 IF DecelDiam < 100.0 THEN
198     DecelDiam := 100.0;                    (* core *)
199 END_IF;
200
201 (* ---- 3. Actual speed from encoder, 100 ms window ---------------------- *)
202
203 IF Tick THEN
204
205     IF NOT SpdInit THEN
206         EncPrev := Encoder_Pos;
207         SpdInit := TRUE;
208     END_IF;
209
210     SpdAct_Pps := DINT_TO_REAL(ABS(Encoder_Pos - EncPrev)) * 10.0;
211     EncPrev := Encoder_Pos;
212 END_IF;
213
214 RwSpeed := SpdAct_Pps * MmPerPulse * 0.06;        (* m/min *)
215
216 (* ---- 4a. v6s: NEW PAPER button: forget the learned thickness ---------- *)
217 (* RwMat_K = 0 -> the first valid 10 s window sets it directly.           *)
218 (* HMI momentary button: must be HELD 3 s, then the reset is done once and *)
219 (* the PLC clears the bit (HMI lamp goes off = done).                     *)
220
221 T_MatReset(In := RwMatReset, PT := T#3s);
222
223 IF T_MatReset.Q THEN
224     RwMat_K := 0.0;
225     LearnFailCnt := 0;
226     MatChangeCnt := 0;
227     RwMatLocked := FALSE;                  (* V9 *)
```

```
228     LongActive := FALSE;
229     GsmAsk := FALSE;                    (* V9: NEW PAPER answers the popup *)
230     GsmArmed := FALSE;
231     RwMatReset := FALSE;
232   END_IF;
233
234   (* ---- 4b. V9: long-baseline thickness learning + LOCK ---------- *)
235   (* From the start point (first run after NEW PAPER / GSM YES / power-on   *)
236   (* while not locked): K = (D^2 - D0^2) / (2 x pulses). The waving of the  *)
237   (* roll (+/-0.4 mm) only matters at the two ends, so the error falls with *)
238   (* the growth (50 mm growth: ~1 %). After 20 mm growth it replaces the    *)
239   (* 30 m windows. At RwLockDiam (and >= RwLockMinGrow_mm growth) the value
      *)
240   (* is LOCKED (Retain): windows and paper-change detection stop.          *)
241   (* RwLockDiam = 0 -> never locked (windows + long baseline as learning).  *)
242
243   IF Tick AND NOT RwMatLocked THEN
244
245     IF LongActive AND (RwDiamRev < LongStartD - 30.0) AND RwDiamRevOk THEN
246       LongActive := FALSE;              (* new core: restart *)
247     END_IF;
248
249     IF Encoder_Reset OR (Encoder_Pos < LongStartPos - 20000) THEN
250       LongActive := FALSE;              (* metre counter reset / 2 m back *)
251     END_IF;
252
253     IF NOT LongActive AND O_MotorStart AND RwDiamRevOk
254       AND (RwDiamRev > 0.0) THEN
255       LongStartPos := Encoder_Pos;
256       LongStartD := RwDiamRev;
257       LongActive := TRUE;
258     END_IF;
259
260     LongValid := FALSE;
261
262     IF LongActive AND RwDiamRevOk AND (MmPerPulse > 0.0) THEN
263       LongGrow_mm := RwDiamRev - LongStartD;
264       LongPls := DINT_TO_REAL(ABS(Encoder_Pos - LongStartPos));
265
266       IF (LongGrow_mm >= 20.0) AND (LongPls > 0.0) THEN
267         LongK := (RwDiamRev * RwDiamRev - LongStartD * LongStartD) / (2.0 *
    LongPls);
268         LongThick_mm := LongK * 3.14159 / (2.0 * MmPerPulse);
269
270         IF (LongThick_mm >= RwThickMin_mm) AND (LongThick_mm <=
    RwThickMax_mm) THEN
271           LongValid := TRUE;
272           RwMat_K := LongK;
273           MatChangeCnt := 0;
274           LearnFailCnt := 0;
275
276           IF (RwLockDiam > 0.0) AND (RwDiamRev >= RwLockDiam)
277             AND (LongGrow_mm >= RwLockMinGrow_mm) THEN
278             RwMatLocked := TRUE;        (* solid from now on *)
279             LongActive := FALSE;
280           END_IF;
281         END_IF;
282       END_IF;
283     END_IF;
```

```
284 END_IF;
285
286 IF RwMatLocked THEN
287     LongValid := FALSE;
288     LearnFailCnt := 0;                    (* no thickness alarm *)
289 END_IF;
290
291 (* ---- 4. Learn material factor  K = dDiam/dPulse * Diam ---------- *)
292
293 IF Tick THEN
294     LearnWinTicks := LearnWinTicks + 1;
295
296     IF NOT LearnInit THEN
297         LearnWinStartPos := Encoder_Pos;  LearnWinStartDiam := RwDiamRev;  (* V9 *)
298         LearnWinTicks := 0;
299         LearnWinRunning := O_MotorStart;
300         LearnInit := TRUE;
301     ELSE
302         LearnWinRunning := LearnWinRunning AND O_MotorStart;
303         (* v6w: LENGTH-based window, modulated by the diameter: closes    *)
304         (* when the roll should have grown RwWinGrow_mm (2.5 mm), so the  *)
305         (* accuracy vs DT35 noise is the same at every speed and size.    *)
306         (* Pulses = growth x diameter / K (K = learned RwMat_K).          *)
307         (* Limits: min 4 m, max RwWinLen_m (30 m). K not learned: max.    *)
308         (* Motor stopped during the window -> window dropped (no fault).  *)
309
310         IF MmPerPulse > 0.0 THEN
311
312             IF (RwMat_K > 0.0) AND (RwWinGrow_mm > 0.0) THEN
313                 LearnWinLen_Pls := RwWinGrow_mm * MAX(RwDiam, REAL#100.0) /
RwMat_K;
314             ELSE
315                 LearnWinLen_Pls := RwWinLen_m * 1000.0 / MmPerPulse;
316             END_IF;
317
318             IF LearnWinLen_Pls < 4000.0 / MmPerPulse THEN
319                 LearnWinLen_Pls := 4000.0 / MmPerPulse;          (* 4 m *)
320             END_IF;
321
322             IF (RwWinLen_m > 0.0)
323               AND (LearnWinLen_Pls > RwWinLen_m * 1000.0 / MmPerPulse) THEN
324                 LearnWinLen_Pls := RwWinLen_m * 1000.0 / MmPerPulse;
325             END_IF;
326         ELSE
327             LearnWinLen_Pls := 300000.0;
328         END_IF;
329
330         IF (DINT_TO_REAL(ABS(Encoder_Pos - LearnWinStartPos)) >= LearnWinLen_Pls)
331           OR NOT LearnWinRunning THEN
332             LearnWin_Pls := ABS(Encoder_Pos - LearnWinStartPos);
333             LearnWinDiamDelta := RwDiamRev - LearnWinStartDiam;  (* V9 *)
334             (* A window SHOULD learn: ran all the time, enough length,  *)
335             (* DT35 OK. Then it must give a plausible paper thickness. *)
336
337             IF LearnWinRunning AND (LearnWin_Pls >= RwMinPulses)
338               AND NOT RwSensFault
339               AND NOT RwMatLocked AND NOT LongValid     (* V9 *)
340               AND RwDiamRevOk AND (LearnWinStartDiam > 0.0) THEN
341                 LearnWinOk := FALSE;
```

```
342
343             IF (LearnWinDiamDelta > 0.0) AND (MmPerPulse > 0.0) THEN
344                 MatK_Window := (LearnWinDiamDelta / DINT_TO_REAL(LearnWin_Pls))
345                             * ((RwDiamRev + LearnWinStartDiam) / 2.0);
346                 (* K = 2 * thickness * mm-per-pulse / PI         *)
347                 ThickWin_mm := MatK_Window * 3.14159 / (2.0 * MmPerPulse);
348
349                 IF (ThickWin_mm >= RwThickMin_mm)
350                   AND (ThickWin_mm <= RwThickMax_mm) THEN
351                     LearnWinOk := TRUE;
352
353                     IF RwMat_K <= 0.0 THEN
354                         RwMat_K := MatK_Window;
355                         MatChangeCnt := 0;
356                     ELSIF (RwMatChangePct > 0.0)
357                         AND (ABS(MatK_Window - RwMat_K)
358                             > RwMat_K * RwMatChangePct / 100.0) THEN
359                         (* v6s: far from the learned value: 1 window = ignored *)
360                         (* (outlier), 2 in a row = paper changed -> take it.  *)
361                         MatChangeCnt := MatChangeCnt + 1;
362
363                         IF MatChangeCnt >= 2 THEN
364                             RwMat_K := MatK_Window;
365                             MatChangeCnt := 0;
366                         END_IF;
367                     ELSE
368                         MatChangeCnt := 0;
369                         RwMat_K := RwMat_K + RwLearn_K * (MatK_Window - RwMat_K);
370                     END_IF;
371                 END_IF;
372             END_IF;
373
374             IF LearnWinOk THEN
375                 LearnFailCnt := 0;
376             ELSIF LearnFailCnt < 1000 THEN
377                 LearnFailCnt := LearnFailCnt + 1;  (* no growth / implausible *)
378             END_IF;
379         END_IF;
380
381         LearnWinStartPos := Encoder_Pos;  LearnWinStartDiam := RwDiamRev;  (* V9
    *)
382         LearnWinTicks := 0;
383         LearnWinRunning := O_MotorStart;
384       END_IF;
385     END_IF;
386 END_IF;
387
388 (* ---- 5. Learn real stop deceleration on every stop ---------------------- *)
389 (* v6d: a run with jog / manual / slow button uses the 2nd ramp (O_AccDec2) *)
390 (*      -> its stop must not be learned. Latched for the whole run because  *)
391 (*      the jog button is already released on the stop edge.                *)
392
393 IF O_MotorStart AND NOT MotorRunMem THEN          (* start edge *)
394     RunWasJog := FALSE;
395 END_IF;
396
397 IF O_MotorStart AND (ManMode OR JogRun) THEN
398     RunWasJog := TRUE;
399 END_IF;
```

```
400
401 IF MotorRunMem AND NOT O_MotorStart THEN          (* stop command edge *)
402     StopMeasActive := TRUE;
403     StopStartPos := Encoder_Pos;
404     StopStartSpd_Pps := SpdAct_Pps;
405     StopStartDiam := DecelDiam;
406     StandstillTicks := 0;
407     (* STOP button and automatic stop take the same path (drive DI1 and    *)
408     (* O_Brake -> T-one drop together, electrical plan sheets 1/9/15),     *)
409     (* so both are learned.                                                *)
410     StopLearnValid := ControlVoltage AND NOT RunWasJog AND NOT ManMode;
411 END_IF;
412
413 MotorRunMem := O_MotorStart;
414
415 IF O_MotorStart THEN
416     StopMeasActive := FALSE;                  (* restarted: abort *)
417 ELSIF StopMeasActive AND Tick THEN
418
419     IF NOT ControlVoltage THEN
420         StopLearnValid := FALSE;              (* E-stop during the stop *)
421     END_IF;
422
423     IF SpdAct_Pps <= 20.0 THEN                (* <= 2 counts / 100 ms *)
424         StandstillTicks := StandstillTicks + 1;
425     ELSE
426         StandstillTicks := 0;
427     END_IF;
428
429     IF StandstillTicks >= 3 THEN              (* standstill 300 ms *)
430         StopMeasActive := FALSE;
431         StopMeas_Pls := DINT_TO_REAL(ABS(Encoder_Pos - StopStartPos));
432
433         IF RwLearnStop AND StopLearnValid
434           AND (StopStartSpd_Pps * MmPerPulse * 0.06 >= RwMinLearnSpd) THEN
435           (* This stop SHOULD learn: result must be plausible       *)
436           (* v6u: a stop shorter than the reaction distance, or     *)
437           (* faster than RwDecelMax_mm, is NOT a fault: the web     *)
438           (* stopped within a few mm, nothing to learn (trace       *)
439           (* 10/28/2026: 23 m/min -> stopped in 102 pulses = 10 mm). *)
440           (* Only a too SLOW stop (< RwDecelMin_mm) counts as fail.  *)
441           StopLearnOk := FALSE;
442
443           IF StopMeas_Pls > StopStartSpd_Pps * RwTreact THEN
444             StopDecelMeas := (StopStartSpd_Pps * StopStartSpd_Pps)
445                       / (2.0 * (StopMeas_Pls - StopStartSpd_Pps * RwTreact));
446             StopDecelMeas_mm := StopDecelMeas * MmPerPulse;  (* mm/s2 *)
447
448             IF (StopDecelMeas_mm >= RwDecelMin_mm)
449               AND (StopDecelMeas_mm <= RwDecelMax_mm) THEN
450               StopLearnOk := TRUE;
451               (* v6g: learn per mm of diameter *)
452               DecelK_Meas := StopDecelMeas_mm / ((StopStartDiam + DecelDiam) /
    2.0);
453
454               IF RwStopDecelK <= 0.0 THEN
455                 RwStopDecelK := DecelK_Meas;
456               ELSE
```

```
                RwStopDecelK := RwStopDecelK + 0.5 * (DecelK_Meas -
RwStopDecelK);
            END_IF;

            StopFailCnt := 0;
          ELSIF StopDecelMeas_mm < RwDecelMin_mm THEN

            IF StopFailCnt < 1000 THEN
              StopFailCnt := StopFailCnt + 1;   (* stop too long *)
            END_IF;
          END_IF;                          (* too fast: ignored *)
        END_IF;                            (* too short: ignored *)
      END_IF;
    END_IF;
END_IF;

(* ---- 5b. v6f: learn real slow-down deceleration ------------------------ *)
(* From the SlowDown edge (at speed) until final speed is reached, or     *)
(* until the stop comes first. a = (v0^2 - v1^2) / (2 * distance).        *)
(* Not learned in jog / manual / slow button (2nd ramp).                  *)

IF SlowDown AND NOT SlowMem AND O_MotorStart
   AND NOT ManMode AND NOT JogRun
   AND (SpdAct_Pps > SpdFinal_Pps + SpdBand_Pps) THEN
    SlowMeasActive := TRUE;
    SlowStartPos := Encoder_Pos;
    SlowStartSpd_Pps := SpdAct_Pps;
    SlowStartDiam := DecelDiam;
END_IF;
SlowMem := SlowDown;

IF SlowMeasActive THEN

    IF NOT SlowDown OR ManMode OR JogRun THEN
      SlowMeasActive := FALSE;                (* aborted *)
    ELSIF Tick AND ((SpdAct_Pps <= SpdFinal_Pps + 0.25 * SpdBand_Pps)
              OR NOT O_MotorStart) THEN      (* final reached or stopped *)
      SlowMeasActive := FALSE;
      SlowMeas_Pls := DINT_TO_REAL(ABS(Encoder_Pos - SlowStartPos));

      IF (SlowMeas_Pls > 0.0) AND (MmPerPulse > 0.0)
        AND ((SlowStartSpd_Pps - SpdAct_Pps) * MmPerPulse * 0.06 >= 10.0) THEN
        SlowDecelMeas_mm := (SlowStartSpd_Pps * SlowStartSpd_Pps
                           - SpdAct_Pps * SpdAct_Pps)
                          / (2.0 * SlowMeas_Pls) * MmPerPulse;  (* mm/s2 *)

        IF (SlowDecelMeas_mm >= RwDecelMin_mm)
          AND (SlowDecelMeas_mm <= RwDecelMax_mm) THEN
          (* v6g: learn per mm of diameter *)
          DecelK_Meas := SlowDecelMeas_mm / ((SlowStartDiam + DecelDiam) /
2.0);

          IF RwSlowDecelK <= 0.0 THEN
            RwSlowDecelK := DecelK_Meas;
          ELSE
            RwSlowDecelK := RwSlowDecelK + 0.5 * (DecelK_Meas -
RwSlowDecelK);
          END_IF;
        END_IF;
```

```
513         END_IF;
514     END_IF;
515 END_IF;
516
517 (* ---- 5c. v6g: live decels = learned K x current diameter ---------- *)
518 (* v6p: the one-time v6f seed is removed (it re-filled RwStopDecelK from    *)
519 (* RwStopDecel, so the learning could never be reset).                     *)
520 (* RwLearnReset (HMI / Watch tab, PLC resets it): clear the learned decels. *)
521 (* Both are re-learned at the next auto slow-down / stop >= RwMinLearnSpd.  *)
522
523 IF RwLearnReset THEN
524     RwStopDecelK := 0.0;
525     RwStopDecel := 0.0;
526     RwSlowDecelK := 0.0;
527     StopFailCnt := 0;
528     RwLearnReset := FALSE;
529 END_IF;
530
531 IF (RwStopDecelK > 0.0) AND (MmPerPulse > 0.0) THEN
532     RwStopDecel := RwStopDecelK * DecelDiam / MmPerPulse;     (* pulses/s2 *)
533 END_IF;
534
535 IF RwSlowDecelK > 0.0 THEN
536     RwSlowDecelLrn := RwSlowDecelK * DecelDiam;               (* mm/s2 *)
537 ELSE
538     RwSlowDecelLrn := 0.0;
539 END_IF;
540
```

1 Auto Stop Process 6-14

```
(*
 ================================================================
 ====================
   AutoStop_Control  V12  -  PART 2 of 2 (rung 1): sections 6 - 14c
   V12: END-OF-ROLL STOP RAMP: auto / diameter stops ramp down at
FinalStopDecel
        (soft stop mechanism) instead of the drive DEC ramp (3 s = 20 Hz/s,
        up to ~8 m/s2 of web on a big roll): stop lengths use it (6, 11b).
   V11: 3-state unwinder display + stop command at UwStop_at_D_Val (section 8).
   (Sysmac: an inline ST rung holds max 1000 rows -> split in 2 rungs.
    Rung 0 = part 1 runs first, then this rung, every scan.)

 ================================================================
 ==================== *)

(* ---- 6. Lengths in pulses: stop, slow-down, remaining ---------- *)
(* Stop length uses the ACTUAL speed (precision)                    *)

IF RwStopDecel > 0.0 THEN
   StopLen_Pls := SpdAct_Pps * RwTreact
                + (SpdAct_Pps * SpdAct_Pps) / (2.0 * RwStopDecel);
ELSE
   StopLen_Pls := SpdAct_Pps * (RwTreact + RwTstopDef / 2.0);
END_IF;

(* V12: end-of-roll stop ramp at FinalStopDecel (web mm/s2) + S-curve    *)

IF (FinalStopDecel > 0.0) AND FFEnable AND NOT ManMode AND (MmPerPulse >
0.0) THEN
   StopLen_Pls := SpdAct_Pps * RwTreact
                + (SpdAct_Pps * SpdAct_Pps) / (2.0 * FinalStopDecel / MmPerPulse)
                + SpdAct_Pps * 0.5 * SCurveTime;
END_IF;

StopLen_Pls := StopLen_Pls * RwTrim;

(* Slow-down uses SpdRef = MAX(actual, work-speed setpoint):            *)
(* stable value, shown to the operator even before start                *)

SlowLenBase_Pls := 0.0;  RwSlowLen_Pls := 0.0;  UwSlowLen_Pls := 0.0;

IF MmPerPulse > 0.0 THEN
   (* v6f: manual > learned > PLC ramp model                           *)
   (* V7: manual > feed-forward (one ramp at DecelWeb, real web mm/s2)  *)
   (*     > old loop with web ramps (coarse DecelWeb + trim tail)      *)
   (*     > learned > old PLC ramp model                               *)

   IF RwSlowDecel > 0.0 THEN
      SlowDecel_Pps2 := RwSlowDecel / MmPerPulse;
      SlowOneRamp := TRUE;
   ELSIF FFEnable AND (DecelWeb > 0.0) AND (WebRatio > 0.0) AND NOT
ManMode THEN
      SlowDecel_Pps2 := DecelWeb / MmPerPulse;
      SlowOneRamp := TRUE;
   ELSIF (DecelWeb > 0.0) AND (WebRatio > 0.0) AND NOT ManMode THEN  (* V7e
*)
      SlowDecel_Pps2 := DecelWeb / MmPerPulse;
      SlowOneRamp := FALSE;
```

```
51    ELSIF RwSlowDecelLrn > 0.0 THEN
52       SlowDecel_Pps2 := RwSlowDecelLrn / MmPerPulse;  (* real, incl. trim tail *)
53       SlowOneRamp := TRUE;
54    ELSE
55       SlowDecel_Pps2 := DecelCoarse * 250.0 / MmPerPulse; (* 1 = 250 mm/s2 *)
56       SlowOneRamp := FALSE;
57    END_IF;
58
59    (* v6d: DecelTrim ramp used inside the +/- SpdDelta_Percent band *)
60
61    IF (DecelTrim > 0.0) AND (WebRatio > 0.0) AND (Vel_Max_mms > 0.0) THEN
62       (* V7: trim ramp (reference units) converted to the real web decel *)
63       TrimDecel_Pps2 := DecelTrim * 1000000.0 / (0.06 * WebRatio * Vel_Max_mms)
64                         / MmPerPulse;
65    ELSIF DecelTrim > 0.0 THEN
66       TrimDecel_Pps2 := DecelTrim * 250.0 / MmPerPulse;
67    ELSE
68       TrimDecel_Pps2 := SlowDecel_Pps2;
69    END_IF;
70
71    SpdBand_Pps := SpdDelta_Percent * 1000.0 / 60.0 / MmPerPulse;
72
73    (* v6d: speeds from the potentiometers, valid in auto AND ManMode     *)
74    (*      (m/min = Pot / Input_Ratio, same as Motor_Manager rung 8)     *)
75
76    IF (Input_Ratio > 0.0) AND ManMode THEN
77       (* v6i: ManMode drives the inverter with the raw pots (rung 17) *)
78       SpdFinal_Pps := INT_TO_REAL(Pot2_Slow_Velocity) / Input_Ratio * 1000.0 /
   60.0 / MmPerPulse;
79       SpdWork_Pps  := INT_TO_REAL(Pot1_Work_Velocity) / Input_Ratio * 1000.0 /
   60.0 / MmPerPulse;
80    ELSIF Input_Ratio > 0.0 THEN
81       SpdFinal_Pps := INT_TO_REAL(Pot2_Mem) / Input_Ratio * 1000.0 / 60.0 /
   MmPerPulse;
82       SpdWork_Pps  := INT_TO_REAL(Pot1_Mem) / Input_Ratio * 1000.0 / 60.0 /
   MmPerPulse;
83    ELSE
84       SpdFinal_Pps := LowSpd_Display * 1000.0 / 60.0 / MmPerPulse;
85       SpdWork_Pps  := Speed_Display  * 1000.0 / 60.0 / MmPerPulse;
86    END_IF;
87
88    (* v6y: rewinder AUTO stop -> work speed limited to RwAutoMaxSpdAct *)
89    (* (limit + value taken over only at standstill, Motor_Manager rung 5) *)
90
91    IF RwAutoLimAct AND NOT ManMode
92       AND (SpdWork_Pps > RwAutoMaxSpdAct * 1000.0 / 60.0 / MmPerPulse) THEN
93       SpdWork_Pps := RwAutoMaxSpdAct * 1000.0 / 60.0 / MmPerPulse;
94    END_IF;
95
96    SpdRef_Pps := MAX(SpdAct_Pps, SpdWork_Pps);
97
98    IF RwStopDecel > 0.0 THEN
99       StopLenFinal_Pls := SpdFinal_Pps * RwTreact
100                         + (SpdFinal_Pps * SpdFinal_Pps) / (2.0 * RwStopDecel);
101   ELSE
102      StopLenFinal_Pls := SpdFinal_Pps * (RwTreact + RwTstopDef / 2.0);
103   END_IF;
104
105   (* V12: final stop from the final speed on the end-of-roll ramp *)
```

```
106
107   IF (FinalStopDecel > 0.0) AND FFEnable AND NOT ManMode THEN
108      StopLenFinal_Pls := SpdFinal_Pps * RwTreact
109                        + (SpdFinal_Pps * SpdFinal_Pps) / (2.0 * FinalStopDecel /
      MmPerPulse)
110                        + SpdFinal_Pps * 0.5 * SCurveTime;
111   END_IF;
112
113   SlowLenBase_Pls := StopLenFinal_Pls * RwTrim;
114
115   IF (SpdRef_Pps > SpdFinal_Pps) AND (SlowDecel_Pps2 > 0.0) THEN
116
117      IF SlowOneRamp OR (TrimDecel_Pps2 <= 0.0) THEN
118         (* manual or learned decel: one constant ramp *)
119         SlowLenBase_Pls := SlowLenBase_Pls
120            + (SpdRef_Pps * SpdRef_Pps - SpdFinal_Pps * SpdFinal_Pps)
121             / (2.0 * SlowDecel_Pps2);
122      ELSE
123         (* auto: coarse ramp down to the band, trim ramp inside it *)
124         SpdKnee_Pps := SpdFinal_Pps + SpdBand_Pps;
125
126         IF SpdRef_Pps > SpdKnee_Pps THEN
127            SlowLenBase_Pls := SlowLenBase_Pls
128               + (SpdRef_Pps * SpdRef_Pps - SpdKnee_Pps * SpdKnee_Pps)
129                / (2.0 * SlowDecel_Pps2)
130               + (SpdKnee_Pps * SpdKnee_Pps - SpdFinal_Pps * SpdFinal_Pps)
131                / (2.0 * TrimDecel_Pps2);
132         ELSE
133            SlowLenBase_Pls := SlowLenBase_Pls
134               + (SpdRef_Pps * SpdRef_Pps - SpdFinal_Pps * SpdFinal_Pps)
135                / (2.0 * TrimDecel_Pps2);
136         END_IF;
137      END_IF;
138   END_IF;
139
140   (* V7f: S-curve rounding: the slow-down ends SCurveTime later ->      *)
141   (*      extra length = (SpdRef - SpdFinal) / 2 x SCurveTime          *)
142
143   IF FFEnable AND NOT ManMode AND (RwSlowDecel <= 0.0) AND (DecelWeb >
      0.0)
144      AND (WebRatio > 0.0) AND (SCurveTime > 0.0)
145      AND (SpdRef_Pps > SpdFinal_Pps) THEN
146      SlowLenBase_Pls := SlowLenBase_Pls
147                        + (SpdRef_Pps - SpdFinal_Pps) * 0.5 * SCurveTime;
148   END_IF;
149
150   RwSlowLen_Pls := SlowLenBase_Pls + RwSlowMargin_m * 1000.0 / MmPerPulse;
151
152   IF UwSlowMargin_m > 0.0 THEN
153      UwMarginUsed_m := UwSlowMargin_m;
154   ELSE
155      UwMarginUsed_m := RwSlowMargin_m;
156   END_IF;
157
158   UwSlowLen_Pls := SlowLenBase_Pls + UwMarginUsed_m * 1000.0 / MmPerPulse;
159 END_IF;
160
161 IF RwDiam >= RwTargetDiam THEN
```

```
ELSIF RwMat_K > 0.0 THEN
    RwRemain_Pls := (RwTargetDiam * RwTargetDiam - RwDiam * RwDiam)
                  / (2.0 * RwMat_K);
ELSE
    RwRemain_Pls := 1.0E9;                    (* K not learned: plain target *)
END_IF;

RwRemain_m  := RwRemain_Pls  * MmPerPulse / 1000.0;
RwStopLen_m := StopLen_Pls   * MmPerPulse / 1000.0;
RwSlowLen_m := RwSlowLen_Pls * MmPerPulse / 1000.0;

(* ---- 6b. Live operator displays ------------------------------------------ *)
(* Rewinder: diameter where slow-down / stop command will happen          *)

IF RwMat_K > 0.0 THEN
    SqrtArg := RwTargetDiam * RwTargetDiam - 2.0 * RwMat_K * RwSlowLen_Pls;
    IF SqrtArg > 0.0 THEN
        RwSlow_at_D_Val := SQRT(SqrtArg);
    ELSE
        RwSlow_at_D_Val := 0.0;
    END_IF;

    SqrtArg := RwTargetDiam * RwTargetDiam - 2.0 * RwMat_K * StopLen_Pls;

    IF SqrtArg > 0.0 THEN
        RwStop_at_D_Val := SQRT(SqrtArg);
    ELSE
        RwStop_at_D_Val := 0.0;
    END_IF;
ELSE
    (* v6t: thickness not learned: fixed slow-down point before the target *)

    IF RwUseSlow AND (RwSlowFallbackMm > 0.0) THEN
        RwSlow_at_D_Val := RwTargetDiam - RwSlowFallbackMm;
    ELSE
        RwSlow_at_D_Val := RwTargetDiam;
    END_IF;
    RwStop_at_D_Val := RwTargetDiam;
END_IF;

(* Unwinder: meter count where the slow-down will start                   *)

IF MmPerPulse > 0.0 THEN
    UwSlow_at_M_Val := UwStop_at_M_Val - UwSlowLen_Pls * MmPerPulse / 1000.0;
ELSE
    UwSlow_at_M_Val := UwStop_at_M_Val;
END_IF;

IF UwSlow_at_M_Val < 0.0 THEN
    UwSlow_at_M_Val := 0.0;
END_IF;

(* Unwinder slow-down request, latched                                    *)

IF (Current_Metre < UwSlow_at_M_Val - 50.0) OR (UwStop_at_M_Val <= 0.0) THEN
    UwSlow_at_M_Req := FALSE;                 (* counter reset / longer roll *)
END_IF;
```

```
221 IF (UwStop_at_M_Val > 0.0) AND (Current_Metre >= UwSlow_at_M_Val) THEN
222     UwSlow_at_M_Req := TRUE;
223 END_IF;
224
225 (* ---- 6c. V7c: unwinder DIAMETER slow-down follows the speed ---------- *)
226 (* Works without AutoStop (only Measure_ON, like General rung 16).       *)
227 (* Paper left above the stop diameter, in pulses:                         *)
228 (*     (UwDiam_Act^2 - UwStopDiam_Limit^2) / (2 x RwMat_K)                *)
229 (* (same paper as the rewinder -> same learned K). Slow down when it is   *)
230 (* shorter than UwSlowLen_Pls = decel from the speed (DecelWeb) + final    *)
231 (* stop + UwSlowMargin_m. Solved for the diameter:                        *)
232 (*     UwSlowDiamCalc = SQRT(UwStopDiam_Limit^2 + 2 x RwMat_K x
    UwSlowLen_Pls)    *)
233 (* K not learned / UwSlow_at_D_Req OFF -> fixed UwSlow_at_D_Val as before. *)
234 (* Latched: once slowed down, the point is frozen (+10 mm hysteresis), so  *)
235 (* the lower speed does not move it back and speed up again. Released by   *)
236 (* a new roll (diameter 30 mm above the frozen point).                     *)
237
238 IF UwSlow_at_D_Req AND (RwMat_K > 0.0) AND (UwStopDiam_Limit > 0.0)
239    AND (UwSlowLen_Pls > 0.0) THEN
240     SqrtArg := UwStopDiam_Limit * UwStopDiam_Limit + 2.0 * RwMat_K *
    UwSlowLen_Pls;
241     UwSlowDiamCalc := SQRT(SqrtArg);            (* same pattern as 6b *)
242 ELSE
243     UwSlowDiamCalc := UwSlow_at_D_Val;
244 END_IF;
245
246 IF UwSlowDiamCalc < UwStopDiam_Limit + 10.0 THEN
247     UwSlowDiamCalc := UwStopDiam_Limit + 10.0;
248 ELSIF UwSlowDiamCalc > 1500.0 THEN
249     UwSlowDiamCalc := 1500.0;
250 END_IF;
251
252 IF UwSlowDiamLatch AND (UwDiam_Act > UwSlowDiamAct + 30.0) THEN
253     UwSlowDiamLatch := FALSE;                   (* new roll *)
254 END_IF;
255
256 IF NOT UwSlowDiamLatch THEN
257     UwSlowDiamAct := UwSlowDiamCalc;            (* live point, HMI *)
258     IF Measure_ON AND (UwDiam_Act > 0.0) AND (UwDiam_Act <=
    UwSlowDiamCalc) THEN
259         UwSlowDiamLatch := TRUE;
260         UwSlowDiamAct := UwSlowDiamCalc + 10.0;    (* frozen + hysteresis *)
261     END_IF;
262 END_IF;
263
264 (* ---- 7. Rewinder requests: latched, reset when a new core is loaded ---- *)
265
266 IF NOT RwStopEnable
267    OR ((RwDiam < RwTargetDiam - RwResetHyst) AND NOT RwSensFault) THEN
268     RwStop_at_D_Req := FALSE;
269     RwSlow_at_D_Req := FALSE;
270 END_IF;
271
272 IF RwStopEnable THEN
273
274     IF (RwRemain_Pls <= StopLen_Pls) OR (RwSensFault AND RwStopOnFault) THEN
275         RwStop_at_D_Req := TRUE;
276     END_IF;
```

```
277
278     IF RwUseSlow AND (RwRemain_Pls <= RwSlowLen_Pls) THEN
279         RwSlow_at_D_Req := TRUE;
280     END_IF;
281
282     (* v6t: thickness not learned (RwMat_K = 0): slow down at a fixed      *)
283     (* distance before the target instead of stopping from full speed.    *)
284
285     IF RwUseSlow AND (RwMat_K <= 0.0) AND (RwSlowFallbackMm > 0.0)
286       AND NOT RwSensFault
287       AND (RwDiam >= RwTargetDiam - RwSlowFallbackMm) THEN
288         RwSlow_at_D_Req := TRUE;
289     END_IF;
290 END_IF;
291
292 (* ---- 8. One AUTO at a time + single "SLOW DOWN AT" display ------------ *)
293
294 // ---- Refuse AUTO STOP when its target is already reached (HMI warning 20) ----
295 // AutoEnWhy (HMI bits): 1 = unwinder metres already reached, 2 = rewinder roll
    >= target
296
297 IF I_AutoStop_OS AND RwStopEnable AND NOT RwArmMem AND NOT RwSensFault
298     AND (RwTargetDiam > REAL#0.0) AND (RwDiam >= RwTargetDiam -
    RwResetHyst) THEN
299     RwStopEnable := FALSE;                    // refused
300     AutoEnRefused := TRUE;
301     AutoEnWhy := 2;
302 END_IF;
303
304 IF I_AutoStop_OS AND UwStopEnable AND NOT UwArmMem
305     AND (UwStop_at_M_Val > REAL#0.0) AND (Current_Metre >= UwStop_at_M_Val)
    THEN
306     UwStopEnable := FALSE;                    // refused
307     AutoEnRefused := TRUE;
308     AutoEnWhy := 1;
309 END_IF;
310
311 IF UwStopEnable AND NOT UwEnMem THEN          (* unwinder AUTO switched
    ON *)
312     RwStopEnable := FALSE;
313 END_IF;
314
315 IF RwStopEnable AND NOT RwEnMem THEN          (* rewinder AUTO switched
    ON *)
316     UwStopEnable := FALSE;
317 END_IF;
318
319 T_AutoEnRef(In := AutoEnRefused, PT := T#5s);     // warning shown 5 s
320
321 IF T_AutoEnRef.Q OR I_RsetAlarm OR Alarm_Reset THEN
322     AutoEnRefused := FALSE;
323 END_IF;
324
325 RwArmMem := I_AutoStop_OS AND RwStopEnable;       // armed = selector 12SA5
    AND enable
326 UwArmMem := I_AutoStop_OS AND UwStopEnable;
327 UwEnMem := UwStopEnable;
328 RwEnMem := RwStopEnable;
329
```

```
IF RwStopEnable THEN
    SlowAtDisp := RwSlow_at_D_Val;                (* mm *)
    SlowAtMode := 2;
ELSIF UwStopEnable THEN
    SlowAtDisp := UwSlow_at_M_Val;                (* m  *)
    SlowAtMode := 1;
ELSE
    SlowAtDisp := 0.0;
    SlowAtMode := 0;                              (* hide display *)
END_IF;

SlowAtShow := (SlowAtMode <> 0);

(* V11: unwinder column, same place, 3 states:                          *)
(*   AUTO STOP UNWINDER (12SA5 + unwinder AUTO) -> STOP LENGTH [m]      *)
(*   normal run, before the slow-down          -> SLOW AT Ø [mm]        *)
(*   slow-down reached (decel / final speed)   -> STOP AT Ø [mm]        *)
(* V11: stop by diameter EARLIER, so the web is stopped AT UwStopDiam_Limit:  *)
(* the stop command comes at UwStop_at_D_Val = UwStopDiam_Limit + the diameter  *)
(* of the paper unwound during the stop from the ACTUAL speed           *)
(* (StopLen_Pls, section 6, learned stop decel):                        *)
(*   UwStop_at_D_Val = SQRT(UwStopDiam_Limit^2 + 2 x RwMat_K x StopLen_Pls)  *)
(* General rung 18 (UwStopByDiam) compares with UwStop_at_D_Val.        *)
(* K not learned -> UwStopDiam_Limit (as before). Max UwStopDiam_Limit + 100 mm.  *)

UwStopLenShow := I_AutoStop_OS AND (UwStopEnable OR (AutoEnRefused AND (AutoEnWhy = 1)));
UwSlowDiamShow := NOT UwStopLenShow AND Measure_ON AND NOT UwSlowDiamLatch;
UwStopDiamShow := NOT UwStopLenShow AND Measure_ON AND UwSlowDiamLatch;

IF (RwMat_K > 0.0) AND (StopLen_Pls > 0.0) THEN
    SqrtArg := UwStopDiam_Limit * UwStopDiam_Limit + 2.0 * RwMat_K * StopLen_Pls;
    UwStop_at_D_Val := SQRT(SqrtArg);
ELSE
    UwStop_at_D_Val := UwStopDiam_Limit;
END_IF;

IF UwStop_at_D_Val > UwStopDiam_Limit + 100.0 THEN
    UwStop_at_D_Val := UwStopDiam_Limit + 100.0;
ELSIF UwStop_at_D_Val < UwStopDiam_Limit THEN
    UwStop_at_D_Val := UwStopDiam_Limit;
END_IF;

(* ---- 9. Automatic stop master = selector 12SA5 (I_AutoStop_OS) -------- *)

AutoSlowCmd := I_AutoStop_OS AND
           ((UwStopEnable AND UwSlow_at_M_Req) OR (RwStopEnable AND RwSlow_at_D_Req));

AutoStopCmd := I_AutoStop_OS AND
           ((UwStopEnable AND (UwStop_at_M_Val > 0.0)
             AND (Current_Metre >= UwStop_at_M_Val))
            OR (RwStopEnable AND RwStop_at_D_Req));
```

```
381
382 (* ---- 10. Unwinder roll empty (for alarm 15) ---------------------------- *)
383
384 IF UwDiam_Act > (UwStopDiam_Limit + UwEmptyHyst) THEN
385     UnWinder_Empty := FALSE;                    (* new roll loaded *)
386 END_IF;
387
388 IF (UwDiam_Act > 0.0) AND (UwDiam_Act <= UwStopDiam_Limit) THEN
389     UnWinder_Empty := TRUE;                     (* roll empty *)
390 END_IF;
391
392 (* v6x: new parent roll loaded (empty -> not empty) -> HMI popup        *)
393
394 (* "DID YOU CHANGE THE PAPER (GSM)?". YES = forget the learned thickness *)
395 (* (same as NEW PAPER). NO = the HMI resets GsmAsk.                      *)
396 (* V9: ask only ONCE per REAL new roll: the roll must have been empty for *)
397 (* 3 s (arms the question), then the new roll must read more than        *)
398 (* UwStopDiam_Limit + 100 mm for 3 s. The sonar bouncing around
    UwStopDiam_Limit     *)
399 (* (empty core, hands, arms) no longer re-opens the popup.              *)
400
401 IF Tick THEN
402
403     IF UnWinder_Empty THEN
404
405         IF UwEmptyTicks < 1000 THEN
406             UwEmptyTicks := UwEmptyTicks + 1;
407         END_IF;
408
409         IF UwEmptyTicks >= 30 THEN
410             GsmArmed := TRUE;
411         END_IF;
412
413         UwFullTicks := 0;
414     ELSE
415         UwEmptyTicks := 0;
416
417         IF GsmArmed AND Measure_ON AND (UwDiam_Act > UwStopDiam_Limit +
    100.0) THEN
418             UwFullTicks := UwFullTicks + 1;
419
420             IF UwFullTicks >= 30 THEN
421                 GsmAsk := TRUE;
422                 GsmArmed := FALSE;
423                 UwFullTicks := 0;
424             END_IF;
425         ELSE
426             UwFullTicks := 0;
427         END_IF;
428     END_IF;
429 END_IF;
430
431 UwEmptyMem := UnWinder_Empty;
432
433 IF GsmAskYes THEN
434     RwMat_K := 0.0;
435     RwMatLocked := FALSE;                       (* V9 *)
436     LongActive := FALSE;
437     LearnFailCnt := 0;
```

```
438     MatChangeCnt := 0;
439     GsmAskYes := FALSE;
440     GsmAsk := FALSE;
441   END_IF;
442
443   (* V10: HMI popup "UNWINDER DIAMETER SENSOR SONAR IS OFF ! DO YOU WANT
      TO  *)
444   (* TURN IT ON ?" - only at standstill: at power-up and after every stop    *)
445   (* while Measure_ON = FALSE. YES (SonarAskYes, PLC resets it) = sonar ON:  *)
446   (* NO = the HMI resets SonarAsk (asked again after the next stop).        *)
447
448   IF Measure_ON THEN
449     SonarAsk := FALSE;                    (* sonar ON: no popup *)
450   ELSIF NOT StartMotor_Cmd AND (SonarRunMem OR NOT SonarInit) THEN
451     SonarAsk := TRUE;                     (* stopped / power-up *)
452   END_IF;
453
454   SonarInit := TRUE;
455   SonarRunMem := StartMotor_Cmd;
456
457   IF StartMotor_Cmd THEN
458     SonarAsk := FALSE;                    (* never while running *)
459   END_IF;
460
461   IF SonarAskYes THEN
462     Measure_ON := TRUE;
463     SonarAsk := FALSE;
464     SonarAskYes := FALSE;
465   END_IF;
466
467   (* ---- 11. Rewinder max diameter protection - ALWAYS active ---------- *)
468   (* Independent of selector 12SA5 and of the AUTO buttons.                *)
469   (* Mechanical limit 680 mm radius = 1360 mm                             *)
470
471   IF NOT RwSensFault THEN
472     RwLastValidDiam := RwDiam;
473   END_IF;
474
475   IF RwDiam >= RwMaxDiam THEN
476     RwRemainMax_Pls := 0.0;
477   ELSIF RwMat_K > 0.0 THEN
478     RwRemainMax_Pls := (RwMaxDiam * RwMaxDiam - RwDiam * RwDiam)
479                        / (2.0 * RwMat_K);
480   ELSE
481     RwRemainMax_Pls := 1.0E9;             (* K not learned: plain limit *)
482   END_IF;
483
484   (* Display: diameter where the max-protection slow-down will start       *)
485
486   IF RwMat_K > 0.0 THEN
487     SqrtArg := RwMaxDiam * RwMaxDiam - 2.0 * RwMat_K * RwSlowLen_Pls;
488     IF SqrtArg > 0.0 THEN
489       RwMaxSlowAtDiam := SQRT(SqrtArg);
490     ELSE
491       RwMaxSlowAtDiam := 0.0;
492     END_IF;
493
494   ELSIF RwMaxSlowFallback > 0.0 THEN
495     RwMaxSlowAtDiam := RwMaxSlowFallback;  (* fallback in use *)
```

```
496  ELSE
497      RwMaxSlowAtDiam := RwMaxDiam;
498  END_IF;
499
500 (* Reset only after the roll is removed (diameter well below the limit)    *)
501
502 IF (RwDiam < RwMaxDiam - RwResetHyst) AND NOT RwSensFault THEN
503      RwMaxStop := FALSE;
504      RwMaxSlow := FALSE;
505  END_IF;
506
507 IF NOT RwSensFault THEN
508
509      IF RwRemainMax_Pls <= StopLen_Pls THEN
510          RwMaxStop := TRUE;                    (* speed-compensated stop *)
511      END_IF;
512
513      IF RwRemainMax_Pls <= RwSlowLen_Pls THEN
514          RwMaxSlow := TRUE;                    (* warning slow-down + horn *)
515      END_IF;
516
517      IF (RwMat_K <= 0.0) AND (RwMaxSlowFallback > 0.0)
518        AND (RwDiam >= RwMaxSlowFallback) THEN
519          RwMaxSlow := TRUE;                    (* fallback while K not learned *)
520      END_IF;
521
522  ELSIF (AI_Diam_Rewinder < RwRawMin) AND (RwLastValidDiam > RwMaxDiam -
     100.0) THEN
523      RwMaxStop := TRUE;                        (* roll grew into the DT35 near zone *)
524  END_IF;
525
526 (* Max diameter protection added OUTSIDE the 12SA5 master (section 9)      *)
527
528  AutoSlowCmd := AutoSlowCmd OR RwMaxSlow;
529  AutoStopCmd := AutoStopCmd OR RwMaxStop;
530
531 (* ---- 11b. V12: END-OF-ROLL STOP RAMP ---------------------------------- *)
532 (* Normal end-of-roll stops (AutoStopCmd: unwinder metres, rewinder target,*)
533 (* rewinder max diameter; UwStopByDiam: unwinder diameter) no longer drop  *)
534 (* the run command at once (drive DEC ramp 3 s = 20 Hz/s -> up to 8 m/s2  *)
535 (* of web on a big roll, hard on the rewinder gearbox). With the feed-    *)
536 (* forward in AUTO they start the SOFT STOP at FinalStopDecel (General    *)
537 (* rung 10): the run command drops below SoftStopEndSpd. START cannot     *)
538 (* cancel it. The stop lengths (section 6) use the same ramp, so the web  *)
539 (* still stops at the target / UwStopDiam_Limit.                          *)
540 (* Immediate stop (EndStopNow -> Motor_Manager rung 4 Stop):              *)
541 (*   ramp not usable (FinalStopDecel 0, FFEnable OFF, ManMode, JOG, the   *)
542 (*   motor stopped = start blocked as before), or safety net: unwinder    *)
543 (*   10 mm below UwStopDiam_Limit / rewinder at RwMaxDiam during the ramp,*)
544
545 IF FinalStopDecel < 0.0 THEN
546      FinalStopDecel := 0.0;
547  ELSIF (FinalStopDecel > 0.0) AND (FinalStopDecel < 50.0) THEN
548      FinalStopDecel := 50.0;
549  ELSIF FinalStopDecel > 2000.0 THEN
550      FinalStopDecel := 2000.0;
551  END_IF;
552
553  EndStopReq := AutoStopCmd OR (Measure ON AND UwStopByDiam);
```

```
554  EndRampOK := (FinalStopDecel > 0.0) AND FFActive AND StartMotor_Cmd;
555
556  IF EndStopReq AND EndRampOK THEN
557      FinalStopRamp := TRUE;                    (* -> soft stop, General 10 *)
558  END_IF;
559
560  IF NOT StartMotor_Cmd THEN
561      FinalStopRamp := FALSE;
562  END_IF;
563
564  EndRampHard := FinalStopRamp
565              AND ((Measure_ON AND (UwDiam_Act > 0.0)
566                  AND (UwDiam_Act < UwStopDiam_Limit - 10.0))
567                  OR ((RwDiam >= RwMaxDiam) AND NOT RwSensFault));
568
569  EndStopNow := (EndStopReq AND NOT FinalStopRamp) OR EndRampHard;
570
571  // ---- 11c. Auto stop DONE (HMI warning 19), latched until START / RESET ----
572  // AutoStopWhy: 1 unwinder metres, 2 rewinder diameter, 4 rewinder MAX, 8
     unwinder diameter
573
574  IF StartMotor_Cmd AND NOT AS_RunMem THEN          // new START: clear
575      AutoStopDone := FALSE;
576      AutoStopSeen := FALSE;
577      AutoStopWhy := 0;
578  END_IF;
579
580  IF StartMotor_Cmd AND EndStopReq AND NOT AutoStopSeen AND NOT JogRun
     THEN
581      AutoStopSeen := TRUE;                     // auto stop started while running
582
583      IF RwMaxStop THEN
584          AutoStopWhy := 4;                     // rewinder MAX (protection, highest)
585      ELSIF I_AutoStop_OS AND RwStopEnable AND RwStop_at_D_Req THEN
586          AutoStopWhy := 2;                     // rewinder diameter reached
587      ELSIF Measure_ON AND UwStopByDiam THEN
588          AutoStopWhy := 8;                     // unwinder diameter (sonar)
589      ELSE
590          AutoStopWhy := 1;                     // unwinder metres
591      END_IF;
592  END_IF;
593
594  IF AS_RunMem AND NOT StartMotor_Cmd AND AutoStopSeen THEN
595      AutoStopDone := TRUE;                     // stopped by the auto stop
596  END_IF;
597
598  IF I_RsetAlarm OR Alarm_Reset THEN
599      AutoStopDone := FALSE;
600  END_IF;
601
602  T_AutoStopDisp(In := AutoStopDone, PT := T#8s);   // show the warning 8s
603
604  IF T_AutoStopDisp.Q THEN
605      AutoStopDone := FALSE;
606  END_IF;
607
608  AS_RunMem := StartMotor_Cmd;
609
610  (* ---- 12. Learning supervision -> alarm 16 (section Alarms) ---------- *)
```

```
(* Learned paper thickness, for display / plausibility check          *)

IF MmPerPulse > 0.0 THEN
    RwThick_mm := RwMat_K * 3.14159 / (2.0 * MmPerPulse);
ELSE
    RwThick_mm := 0.0;
END_IF;

(* Fault: too many consecutive windows / stops that should have       *)
(* learned but did not. Clears by itself on the next good result.     *)
(* v6w: ALARM RESET (panel button or HMI) also clears the fail counters, *)
(* so alarm 16 resets in one press. It comes back only if the learning  *)
(* fails again RwLearnFailMax / RwStopFailMax times in a row.         *)

IF I_RsetAlarm OR Alarm_Reset THEN
    LearnFailCnt := 0;
    StopFailCnt := 0;
END_IF;

GSM_LearnFault := (LearnFailCnt >= RwLearnFailMax)
          OR (StopFailCnt >= RwStopFailMax);      (* -> alarm 16 *)

(* v6e: learning status for the HMI. FALSE until learned.             *)

GSM_Learned_OK := (RwMat_K > 0.0)
              AND ((RwStopDecel > 0.0) OR NOT RwLearnStop)
              AND NOT GSM_LearnFault;

(* ---- 13. v6h: keep T-one in automatic tension control during the stop - *)
(* While running: BrakeHold = Brake_Cmd (no change in running behaviour).  *)
(* After the stop: stays ON until the web is below 5 m/min, after 15 s.    *)
(* or on E-stop. Already TRUE on the stop scan (Output_Manager runs before *)
(* this section), so O_Brake never drops for one scan.                *)
(* Output_Manager rung 2:  O_Brake := Brake_Cmd OR BrakeHold          *)

IF O_MotorStart THEN
    BrakeHold := Brake_Cmd AND ControlVoltage;
    BrakeHoldTicks := 0;
ELSIF BrakeHold THEN

    IF Tick THEN
        BrakeHoldTicks := BrakeHoldTicks + 1;
    END_IF;

    (* v6v: signed web speed: a web pulled BACK (negative speed) also     *)
    (* releases the T-one (trace 10/28 15:00: -6.6 m/min, about 20 cm).  *)

    IF NOT ControlVoltage OR (Velocity_M_min < 5.0) OR (BrakeHoldTicks >= 150)
THEN
        BrakeHold := FALSE;
    END_IF;
END_IF;

(* ---- 14. v6j: TEACH CORE - automatic offset with an empty core -------- *)
(* HMI: set RwTeachCore (button, Set ON). Conditions: motor stopped, web  *)
(* not moving, DT35 OK. After 2 s stable (filter settled):            *)
(*    RwAxisDist := RwDist + RwCoreDiameter / 2   -> RwDiam = RwCoreDiameter
*)
(* RwTeachState: 0 idle, 1 teaching (wait 2 s), 2 done OK             *)
```

```
(* RwTeachState: 0 idle, 1 teaching (wait 2 s), 2 done OK,                *)
(*               3 refused (motor running / web moving / DT35 fault).     *)
(* Only the offset is taught: the scale is RwDist10V - RwDist0V           *)
(* (= real distance between the two DT35 taught points).                  *)

IF RwTeachCore THEN

   IF O_MotorStart OR (RwSpeed > 1.0) OR RwSensFault
     OR NOT RwFiltInit OR (RwCoreDiameter <= 0.0) THEN
      RwTeachState := 3;                    (* refused *)
      RwTeachCore := FALSE;
      TeachTicks := 0;
   ELSE
      RwTeachState := 1;                    (* teaching *)

      IF Tick THEN
         TeachTicks := TeachTicks + 1;
      END_IF;

      IF TeachTicks >= 20 THEN              (* 2 s stable *)
         RwAxisDist_Prev := RwAxisDist;     (* v6l: keep previous *)
         RwAxisDist := RwDist + RwCoreDiameter / 2.0;
         RwTeachState := 2;                 (* done *)
         RwTeachCore := FALSE;
         TeachTicks := 0;
      END_IF;
   END_IF;
ELSE
   TeachTicks := 0;

   IF O_MotorStart THEN
      RwTeachState := 0;                    (* clear status on next run *)
   END_IF;
END_IF;

(* ---- 14b. v6k: unwinder TEACH CORE (HMI button Core_Measure) ---------- *)
(* Same behaviour as section 14. Sonar is not filtered, so the 2 s are    *)
(* averaged:  UwCoreDistance := average(UwDiam_Measure)  (sonar -> core   *)
(* surface). UwDiam_Act = (UwCoreDistance - UwDiam_Measure) * 2           *)
(*                      + UwCoreDiameter   (General rung 14, unchanged). *)
(* UwTeachState: 0 idle, 1 teaching, 2 done OK, 3 refused.                *)

IF Core_Measure THEN

   IF O_MotorStart OR (RwSpeed > 1.0) OR (UwCoreDiameter <= 0.0)
     OR (UwDiam_Measure <= 0.0) THEN
      UwTeachState := 3;                    (* refused *)
      Core_Measure := FALSE;
      UwTeachTicks := 0;
      UwTeachSum := 0.0;
   ELSE
      UwTeachState := 1;                    (* teaching *)

      IF Tick THEN
         UwTeachTicks := UwTeachTicks + 1;
         UwTeachSum := UwTeachSum + UwDiam_Measure;
      END_IF;

      IF UwTeachTicks >= 20 THEN            (* 2 s, 20 samples *)
```

```
726         UwCoreDist_Prev := UwCoreDistance;    (* vbc keep previous *)
727         UwCoreDistance := UwTeachSum / INT_TO_REAL(UwTeachTicks);
728         UwTeachState := 2;                    (* done *)
729         Core_Measure := FALSE;
730         UwTeachTicks := 0;
731         UwTeachSum := 0.0;
732       END_IF;
733     END_IF;
734   ELSE
735     UwTeachTicks := 0;
736     UwTeachSum := 0.0;
737
738     IF O_MotorStart THEN
739       UwTeachState := 0;                      (* clear status on next run *)
740     END_IF;
741   END_IF;
742
743 (* ---- 14c. v6l: UNDO teach = swap current <-> previous (machine stopped) *)
744 (* Pressing UNDO twice goes back to the taught value. State 4 = RESTORED. *)
745
746 IF RwTeachUndo THEN
747
748     IF NOT O_MotorStart AND (RwAxisDist_Prev > 0.0) THEN
749       TeachSwap := RwAxisDist;
750       RwAxisDist := RwAxisDist_Prev;
751       RwAxisDist_Prev := TeachSwap;
752       RwTeachState := 4;                      (* restored *)
753     ELSE
754       RwTeachState := 3;                      (* refused *)
755     END_IF;
756     RwTeachUndo := FALSE;
757   END_IF;
758
759 IF UwTeachUndo THEN
760
761     IF NOT O_MotorStart AND (UwCoreDist_Prev > 0.0) THEN
762       TeachSwap := UwCoreDistance;
763       UwCoreDistance := UwCoreDist_Prev;
764       UwCoreDist_Prev := TeachSwap;
765       UwTeachState := 4;                      (* restored *)
766     ELSE
767       UwTeachState := 3;                      (* refused *)
768     END_IF;
769     UwTeachUndo := FALSE;
770   END_IF;
```

1-8-1-1-9.Tension_Manager

0

```
// ====================== TENSION MANAGER (kg only) ======================
// Section "Tension_Manager" (was Tension_Helper), rung 0, ST, no contact.
// 1. Tension Helper: recommended tension (kg) from paper GSM, roll width, paper type
// 2. T-one remote set point (AO Ch1 -> isolator -> T-one pins 20/19, F.37 = 1):
//    helper x core boost x taper on the rewinder diameter + operator trim (+/-10 kg),
//    x JOG boost while threading (JOG ... START), rate-limited,
//    1..10 V = TensMin_kg..TensMax_kg (T-one F.38 / F.39)
// TensAOEnable = FALSE -> AO Ch1 = fixed 10 V (old Fix10Vdc, pots supply)

// ---- 1. Paper type: KRAFT / BLEACHED selects the factor (N/mm width per 100 g/m2)

IF (TensFactorBleached < REAL#0.1) OR (TensFactorBleached > REAL#2.0) THEN
    TensFactorBleached := REAL#0.54;
END_IF;

IF (TensFactorKraft < REAL#0.1) OR (TensFactorKraft > REAL#2.0) THEN
    TensFactorKraft := REAL#0.80;
END_IF;

IF PaperKraft THEN
    TensFactor := TensFactorKraft;
ELSE
    TensFactor := TensFactorBleached;
END_IF;

// ---- 2. Helper inputs (limits)

IF PaperGSM < REAL#10 THEN
    PaperGSM := REAL#10;
ELSIF PaperGSM > REAL#500 THEN
    PaperGSM := REAL#500;
END_IF;

IF WebWidth_mm < REAL#100 THEN
    WebWidth_mm := REAL#100;
ELSIF WebWidth_mm > REAL#3000 THEN
    WebWidth_mm := REAL#3000;
END_IF;

// ---- 3. Recommended tension (kg)

TensIdeal_N := PaperGSM / REAL#100 * WebWidth_mm * TensFactor;
TensIdeal_Disp := TensIdeal_N / REAL#9.81;

// ---- 4. Setup limits (maintenance popup)

IF TensTaper < REAL#0.0 THEN
    TensTaper := REAL#0.0;
ELSIF TensTaper > REAL#60.0 THEN
    TensTaper := REAL#60.0;
END_IF;

IF TensCoreBoost < REAL#0.0 THEN
```

```
54      TensCoreBoost := REAL#0.0;
55  ELSIF TensCoreBoost > REAL#50.0 THEN
56      TensCoreBoost := REAL#50.0;
57  END_IF;
58
59  IF TensJogBoost < REAL#0.0 THEN
60      TensJogBoost := REAL#0.0;
61  ELSIF TensJogBoost > REAL#100.0 THEN
62      TensJogBoost := REAL#100.0;
63  END_IF;
64
65  IF TensRate_kg_s < REAL#1.0 THEN
66      TensRate_kg_s := REAL#1.0;
67  ELSIF TensRate_kg_s > REAL#20.0 THEN
68      TensRate_kg_s := REAL#20.0;
69  END_IF;
70
71  IF TensMin_kg < REAL#0.0 THEN
72      TensMin_kg := REAL#0.0;
73  END_IF;
74
75  IF TensMax_kg < TensMin_kg + REAL#5.0 THEN
76      TensMax_kg := TensMin_kg + REAL#5.0;
77  END_IF;
78
79  // ---- 5. Operator trim (HMI up/down, +/-10 kg), back to 0 when the paper changes
80
81  IF TensTrim_kg < REAL#-10.0 THEN
82      TensTrim_kg := REAL#-10.0;
83  ELSIF TensTrim_kg > REAL#10.0 THEN
84      TensTrim_kg := REAL#10.0;
85  END_IF;
86
87  IF (PaperGSM <> TensGSMMem) OR (WebWidth_mm <> TensWidthMem) OR
    (PaperKraft <> TensKraftMem) THEN
88      TensTrim_kg := REAL#0.0;
89  END_IF;
90
91  TensGSMMem := PaperGSM;
92  TensWidthMem := WebWidth_mm;
93  TensKraftMem := PaperKraft;
94
95  // ---- 6. Rewinder diameter fraction 0 (core) .. 1 (max diameter)
96
97  IF (RwMaxDiam > RwCoreDiameter + REAL#50.0) AND NOT RwSensFault THEN
98      TensDiaFrac := (RwDiam - RwCoreDiameter) / (RwMaxDiam - RwCoreDiameter);
99  ELSE
100     TensDiaFrac := REAL#0.0;                    // DT35 fault: no taper
101 END_IF;
102
103 IF TensDiaFrac < REAL#0.0 THEN
104     TensDiaFrac := REAL#0.0;
105 ELSIF TensDiaFrac > REAL#1.0 THEN
106     TensDiaFrac := REAL#1.0;
107 END_IF;
108
109 // ---- 7. Profile preview (HMI): core / mid / max
110
111 TensAtCore := TensIdeal_Disp * (REAL#1.0 + TensCoreBoost / REAL#100.0);
```

```
112  TensAtMid  := TensAtCore * (REAL#1.0 - TensTaper / REAL#100.0 * REAL#0.5);
113  TensAtMax  := TensAtCore * (REAL#1.0 - TensTaper / REAL#100.0);
114
115  // ---- 7b. Threading / knife adjustment preset: JOG raises the tension, START
     resumes normal
116
117  IF JogRun THEN
118      TensJogLatch := TRUE;
119  END_IF;
120
121  IF StartPB_OK AND NOT StartBlocked THEN          // START accepted -> normal
     tension
122      TensJogLatch := FALSE;
123  END_IF;
124
125  // ---- 8. Target: profile (taper) + trim, or manual; x JOG boost while threading
126
127  IF TensAuto THEN
128      TensTarget := TensAtCore * (REAL#1.0 - TensTaper / REAL#100.0 * TensDiaFrac) +
     TensTrim_kg;
129  ELSE
130      TensTarget := TensManual_kg;
131  END_IF;
132
133  IF TensJogLatch THEN
134      TensTarget := TensTarget * (REAL#1.0 + TensJogBoost / REAL#100.0);
135  END_IF;
136
137  IF TensTarget < TensMin_kg THEN
138      TensTarget := TensMin_kg;
139  ELSIF TensTarget > TensMax_kg THEN
140      TensTarget := TensMax_kg;
141  END_IF;
142
143  // ---- 9. Rate limit (4 ms scan): smooth change of GSM / width / type / mode /
     trim / JOG boost
144
145  TensStep := TensRate_kg_s * REAL#0.004;
146
147  IF TensSet_kg < TensTarget - TensStep THEN
148      TensSet_kg := TensSet_kg + TensStep;
149  ELSIF TensSet_kg > TensTarget + TensStep THEN
150      TensSet_kg := TensSet_kg - TensStep;
151  ELSE
152      TensSet_kg := TensTarget;
153  END_IF;
154
155  // ---- 10. Analog output Ch1
156
157  IF TensAOEnable THEN
158
159      // 1..10 V (400..4000) = TensMin_kg..TensMax_kg ; 0 V = wire broken (below
     range)
160      TensAOReal := REAL#400.0 + REAL#3600.0 * (TensSet_kg - TensMin_kg) /
     (TensMax_kg - TensMin_kg);
161
162      IF TensAOReal < REAL#400.0 THEN
163          TensAOReal := REAL#400.0;
164      ELSIF TensAOReal > REAL#4000.0 THEN
```

```
165         TensAOReal := REAL#4000.0;
166     END_IF;
167
168     AO_TOne_SetPoint := REAL_TO_INT(TensAOReal);
169 ELSE
170     AO_TOne_SetPoint := INT#4000;              // fixed 10 V (pots supply, as Fix10Vdc)
```

1-9.Data

1-9-1.Data Types

Name	Type	Comment

1-9-2.Global Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Network Publish	Comment
VAR_GLOBAL							
Pot1_Work_Velocity	INT		BuiltInIO://opt/#0/Ch1_Analog_Input_Value	False	False	Publish Only	Potentiometer 1 value - Work speed -
InvRef	INT		BuiltInIO://opt/#0/Ch2_Analog_Output_Value	False	False	Publish Only	Inverter speed reference
AO_TOne_SetPoint	INT		BuiltInIO://opt/#0/Ch1_Analog_Output_Value	False	False	Publish Only	Analog O to T-one Tension SetPoint [Kg]
VelMax	REAL	800		True	False	Publish Only	Maximum speed
VelSet	REAL			False	False	Publish Only	Set speed
RwSlowDecelK	REAL	0.15		True	False	Publish Only	Learned slow-down decel per mm of rewinder diameter (mm/s2 per mm)
RwMaxSlowFallback	REAL	1285.0		True	False	Publish Only	Rewinder max: slow-down diameter while K not learned (mm)
Language	DINT			True	False	Publish Only	Language
RwThickMin_mm	REAL	0.02		True	False	Publish Only	Learning: min plausible paper thickness (mm)
RwThickMax_mm	REAL	2.0		True	False	Publish Only	Learning: max plausible paper thickness (mm)
RwDecelMin_mm	REAL	20.0		True	False	Publish Only	Learning: min plausible stop decel (mm/s2)
RwDecelMax_mm	REAL	5000.0		True	False	Publish Only	Learning: max plausible stop decel (mm/s2)
RwLearnFailMax	INT	12		True	False	Publish Only	Learning: failed windows before alarm (12 = 2 min)
RwStopFailMax	INT	3		True	False	Publish Only	Learning: failed stops before alarm
RwThick_mm	REAL			True	False	Publish Only	Learned paper thickness (mm)
OverSpeed	BOOL			False	False	Publish Only	Speed too high
WorkSpd_Max	REAL	500.0		True	False	Publish Only	Work speed pot full scale (m/min), 50..VelMax
WorkSpd_Max_Set	REAL	400.0		True	False	Publish Only	HMI request: work pot full scale (m/min), applied only with motor stopped
PaperKraft	BOOL	False		True	False	Publish Only	Paper type: TRUE = kraft, FALSE = bleached (tension factor)
TensFactorBleached	REAL	0.54		True	False	Publish Only	Tension factor bleached paper (N/mm width per 100 g/m2)
TensFactorKraft	REAL	0.80		True	False	Publish Only	Tension factor kraft paper (N/mm width per 100 g/m2), tune on machine
TensAuto	BOOL	False		True	False	Publish Only	T-one set point from the PLC profile (FALSE = TensManual_kg)
TensAOEnable	BOOL	False		True	False	Publish Only	AO Ch1 = T-one set point (FALSE = fixed 10 V as old Fix10Vdc)
TensManual_kg	REAL	40.0		True	False	Publish Only	T-one manual set point (kg) when TensAuto is OFF

TensTaper	REAL	30.0		True	False	Publish Only	Tension taper core -> max diameter (%), 0..60
TensCoreBoost	REAL	0.0		True	False	Publish Only	Extra tension at the core (%), 0..50
TensRate_kg_s	REAL	5.0		True	False	Publish Only	T-one set point max change (kg/s), 1..20
TensMin_kg	REAL	10.0		True	False	Publish Only	T-one set point at 1 V (= T-one F.38)
TensMax_kg	REAL	100.0		True	False	Publish Only	T-one set point at 10 V (= T-one F.39)
TensSet_kg	REAL	0.0		False	False	Publish Only	Live T-one set point sent on AO Ch1 (kg)
TensAtCore	REAL	0.0		False	False	Publish Only	Profile preview: tension at the core (kg)
TensAtMid	REAL	0.0		False	False	Publish Only	Profile preview: tension at mid diameter (kg)
TensAtMax	REAL	0.0		False	False	Publish Only	Profile preview: tension at max diameter (kg)
UwRawMin	INT	10		True	False	Publish Only	Unwinder sonar raw min (below = sensor fault, alarm 17)
UwRawMax	INT	3990		True	False	Publish Only	Unwinder sonar raw max (above = sensor fault / no echo, alarm 17)
TensCheckEnable	BOOL	False		True	False	Publish Only	Tension check ON (HMI switch); OFF = start as before
TensWaitSpd	REAL	10.0		True	False	Publish Only	Speed held after START until tension established (m/min)
TensionOK	BOOL	False		False	False	Publish Only	Tension established (debounced T-one Alarm 0)
TensWait	BOOL	False		False	False	Publish Only	Waiting for tension (speed held at TensWaitSpd, HMI lamp)
SonarAsk	BOOL	False		False	False	Publish Only	V10 HMI popup: sonar is OFF, turn it on? (HMI NO resets it)
UwSlow_at_D_Req	BOOL	True		True	False	Publish Only	V7c: unwinder diameter slow-down follows the speed (FALSE = fixed UwSlowDown_Diam)
UwSlowDiamAct	REAL	0.0		False	False	Publish Only	V7c: unwinder slows down at this diameter (mm), HMI display
FinalStopDecel	REAL	500.0		True	False	Publish Only	V12: end-of-roll stop ramp, web decel (mm/s2), 0 = drive DEC ramp as before, 50..2000
FinalStopRamp	BOOL	False		False	False	Publish Only	V12: end-of-roll stop ramp running (HMI lamp)
RwJogMaxDiam	REAL	400.0		True	False	Publish Only	V7d: jog blocked above this rewinder diameter (mm), 0 = no limit
JogInhibit	BOOL	False		False	False	Publish Only	V7d: jog blocked (rewinder roll too big), HMI lamp
SCurveTime	REAL	0.5		True	False	Publish Only	V7f: S-curve rounding of the AUTO ramps (s), 0 = off, max 2
TOneDecel	BOOL	False		False	False	Publish Only	V7f: line decelerating (for a T-one stop/boost input)
HornStartBee	REAL	1.0		True	False	Publish Only	V9 horn: beep when the slo

p_s							w-down starts (s), 0 = off, max 5
HornBeepLen_s	REAL	0.2		True	False	Publish Only	V9 horn: length of the short beeps during slow-down (s), 0 = off, max 2
HornBeepPeriod	REAL	5.0		True	False	Publish Only	V9 horn: one short beep every ... s during slow-down, 0 = off, max 30
HornStopBeep_s	REAL	3.0		True	False	Publish Only	V9 horn: beep when stopped after a slow-down (s), 0 = off, max 10
HornTest	BOOL	False		False	False	Publish Only	V9 horn: HMI test button (momentary), machine stopped only
RwLockDiam	REAL	300.0		True	False	Publish Only	V9: paper thickness LOCKED at this rewinder diameter (mm), 0 = never lock
RwLockMinGrow_mm	REAL	50.0		True	False	Publish Only	V9: min diameter growth since the start of learning before locking (mm)
RwMatLocked	BOOL	False		True	False	Publish Only	V9: paper thickness locked (HMI lamp), NEW PAPER / GSM YES unlocks
LongThick_mm	REAL	0.0		False	False	Publish Only	V9: thickness from the long baseline (mm), HMI display
RwDiamRev	REAL	0.0		False	False	Publish Only	V9: rewinder diameter averaged over one revolution (mm)
RwWave_mm	REAL	0.0		False	False	Publish Only	V9: diameter wave peak-peak in the last revolution (mm), HMI
UwStopLenShow	BOOL	False		False	False	Publish Only	V9 HMI: show unwinder STOP LENGTH (AUTO STOP UNWINDER active)
UwSlowDiamShow	BOOL	False		False	False	Publish Only	V9 HMI: show unwinder SLOW AT diameter (normal operation, sonar ON)
SoftCancelAccel	REAL	100.0		True	False	Publish Only	V9: then ramp up at this web accel (mm/s2), 0 = AccelWeb
RwMatReset	BOOL	False		False	False	Publish Only	HMI NEW PAPER: forget the learned paper thickness RwMat_K (PLC resets it)
FinalSpd_Max_Set	REAL	80.0		True	False	Publish Only	HMI request: final pot full scale (m/min), applied only with motor stopped
FinalSpd_Max	REAL	80.0		True	False	Publish Only	Final speed pot full scale (m/min), 5..WorkSpd_Max
UnderSpeed	BOOL			False	False	Publish Only	Speed too low
RwLearnReset	BOOL	False		False	False	Publish Only	HMI / Watch: clear learned decels RwStopDecelK, RwStopDecel, RwSlowDecelK (PLC resets it)
RwCoreDiameter	REAL	100.0		True	False	Publish Only	Rewinder core outside diameter for TEACH CORE (mm)
RwTeachCore	BOOL	False		False	False	Publish Only	HMI: TEACH CORE request (PLC resets it)
RwTeachStat	INT	0		False	False	Publish Only	TEACH CORE: 0 idle 1 tea

e							ching 2 done 3 refused
SpdDelta_Percent	REAL	12.0		True	False	Publish Only	Speed difference percentage
Encoder_Pos	DINT		IOBus://unit#3/Ch1 Encoder Present Position	False	False	Publish Only	Encoder counter
UwTeachState	INT	0		False	False	Publish Only	Unwinder TEACH CORE: 0 idle 1 teaching 2 done 3 refused
Encoder_On	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Counter Enable	False	False	Publish Only	Encoder enable
Encoder_Reset	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Internal Reset Execution	False	False	Publish Only	Encoder reset
HMI_Screen	ARRAY[0..1] OF DINT			False	False	Publish Only	HMI panel Screen
Velocity_M_min	REAL			True	False	Publish Only	Speed in m/min
Current_Metre	REAL			True	False	Publish Only	Current metres
Speed_Display	REAL			True	False	Publish Only	Set speed (displayed)
I_ControlVoltage	BOOL		BuiltInIO://cpu/#0/Input_Bit_00	False	False	Publish Only	In_0 ControlVoltageON
I_Start_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_01	False	False	Publish Only	In_1 Start button
I_Stop_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_02	False	False	Publish Only	In_2 Stop button
I_Motor_Thermal	BOOL		BuiltInIO://cpu/#0/Input_Bit_03	False	False	Publish Only	In_3 Motor thermal switch
I_Prot_ByPass	BOOL		BuiltInIO://cpu/#0/Input_Bit_04	False	False	Publish Only	In_4 Guards-bypassed selector switch - Under Panel
I_Hub_Position	BOOL		BuiltInIO://cpu/#0/Input_Bit_05	False	False	Publish Only	In_5 Hub/chuck in extraction position
I_FrontGuard_Closed	BOOL		BuiltInIO://cpu/#0/Input_Bit_06	False	False	Publish Only	In_6 Front guard closed
I_Jog_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_07	False	False	Publish Only	In_7 Jog Push button
I_VFD_Alarm	BOOL		BuiltInIO://cpu/#0/Input_	False	False	Publish Only	In_8 Inverter in alarm

			Bit_08				
I_Side_Door_Closed	BOOL		BuiltInIO://cpu/#0/Input_Bit_13	False	False	Publish Only	In_Side Door is Closed
I_OR_Position_Sens	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN01 (Pin 1.1) _6000_01	False	False	Publish Only	OverRider position sens
I_OR_Hoover_Mode	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN02 (Pin 2.1) _6000_02	False	False	Publish Only	OverRider Hoover Mode
I_OR_Auto_OS	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN03 (Pin 1.2) _6000_03	False	False	Publish Only	OverRider in automatic
I_OR_Cont_OS	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN04 (Pin 2.2) _6000_04	False	False	Publish Only	OverRider in continuous
O_MotorStart	BOOL		BuiltInIO://cpu/#0/Output_Bit_00	False	False	Publish Only	Q_0 Motor start
I_FanTemp_Ok	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN06 (Pin 2.3) _6000_06	False	False	Publish Only	Motor (fan) thermal OK
O_Alarm_Array	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 01 (Pin 1.1) _7000_01	False	False	Publish Only	Alarm lamp
O_Prot_ByPass_Lamp	BOOL			False	False	Publish Only	Guards bypassed lamp
O_Motor_Lamp	BOOL		BuiltInIO://cpu/#0/Output_Bit_01	False	False	Publish Only	Q_1 Motor ON lamp
O_RwUnload	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 03 (Pin 1.2) _7000_03	False	False	Publish Only	Unloaders enable - opening
O_FrontGuard_ByPass	BOOL		ECAT://node#[1,1]/IB IL	False	False	Publish Only	Front guard bypass

			24 DO 8/HD-ECO Output_OUT 04 (Pin 2.2) _7000_04				
O_Brake	BOOL		BuiltInIO://cpu/#0/Output_Bit_02	False	False	Publish Only	Q_2 Brake command
O_OR_Auto_Lamp	BOOL		BuiltInIO://cpu/#0/Output_Bit_06	False	False	Publish Only	Q_6 OverRider auto/continuous lamp
O_OverRider_Up	BOOL		BuiltInIO://cpu/#0/Output_Bit_09	False	False	Publish Only	Q_9 Raise OverRider ~10mm Over Round Rewinder roll
O_OverRider_Hold	BOOL		BuiltInIO://cpu/#0/Output_Bit_04	False	False	Publish Only	Q_4 OverRider Lock
O_CoolingFan	BOOL		BuiltInIO://cpu/#0/Output_Bit_03	False	False	Publish Only	Q_3 Brake and motor fans
Fan_Timeout	REAL	0.5		True	False	Publish Only	Fans stop timeout [min]
OverRider_TimeOFF	REAL	0.01		True	False	Publish Only	OverRider photocell OFF timeout (panel)
OverRider_TimeON	REAL	0.05		True	False	Publish Only	OverRider photocell ON timeout (panel)
AlarmOn	BOOL			False	False	Publish Only	Alarm active
BuzzerOn	BOOL			False	False	Publish Only	Panel buzzer command
Msg	INT			False	False	Publish Only	Message: Warnings / Alarms
Index_B	INT			False	False	Publish Only	Alarm index
MotStop	BOOL			False	False	Publish Only	Motor stopped
Clock1s	BOOL			False	False	Publish Only	1 s clock
ManMode	BOOL			True	False	Publish Only	Manual mode
Counter_Reset	BOOL			False	False	Publish Only	Counter reset from panel
I_FrontGuard_Open	BOOL		BuiltInIO://cpu/#0/Input_Bit_09	False	False	Publish Only	In_9 Front guard open
O_RwLoading_Enable	BOOL		BuiltInIO://cpu/#0/Output_Bit_07	False	False	Publish Only	Q_7 Unloaders Load enable - closing
JogSpd_SP	REAL	10.0		True	False	Publish Only	Speed in manual
O_AccDec2	BOOL		BuiltInIO://cpu/#0/Output_Bit_05	False	False	Publish Only	Q_5 Second acc / dec ramp
AccelTrim	REAL	0.08		True	False	Publish Only	Acceleration coefficient in run
DecelTrim	REAL	0.08		True	False	Publish Only	Deceleration coefficient in run
DecelCoarse	REAL	0.3		True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests decel)
AccelCoarse	REAL	0.3		True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests accel)
UwStopDiamShow	BOOL	False		False	False	Publish Only	V11 HMI: show unwinder STOP AT diameter (slow-down reached, decel / final speed)
O_AcousticSignal	BOOL		BuiltInIO://cpu/#0/Output_Bit_08	False	False	Publish Only	Q_8 Horn / Acoustic signal

Pot2_Slow_Velocity	INT		BuiltInIO://opt/#0/Ch2_Analog_Input_Value	False	False	Publish Only	Potentiometer 2 value - Final speed -
I_AutoStop_OS	BOOL		BuiltInIO://cpu/#0/Input_Bit_10	False	False	Publish Only	In_10 Stop on metre counter
I_Input_11_Spare_1255	BOOL		BuiltInIO://cpu/#0/Input_Bit_11	False	False	Do not publish	In_11 SPARE (wire 12.55) Under Panel
I_WebBreak_Monitor	BOOL		BuiltInIO://cpu/#0/Input_Bit_12	False	False	Publish Only	In_12 Paper break monitoring ON
I_Vacuum_OK	BOOL		IOBus://unit#3/Ch1 Reset/External Input Status/Ch1 External Input 0 Status	False	False	Publish Only	Extractor fan thermal OK
I_T_one_TensionOK	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN05 (Pin 1.3) _6000_05	False	False	Publish Only	T-one Alarm 0 (15-16): tension established (replaces I_SlowSens_PH)
I_WebBreak_PH	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN07 (Pin 1.4) _6000_07	False	False	Publish Only	Paper break photocell
UwStop_at_M_Val	REAL	5100		True	False	Publish Only	Unwinder Stop at metres
UwSlowMargin_m	REAL	10		True	False	Publish Only	Unwinder Slow-down metres
LowSpd_Display	REAL			True	False	Publish Only	Slow-down speed (displayed)
RwMaxDiam	REAL	1315.0		True	False	Publish Only	Rewinder max diameter protection (mm)
RwMaxStop	BOOL			False	False	Publish Only	Rewinder max diameter STOP (latched)
Encoder_Pulse	INT	1000		True	False	Publish Only	Encoder pulses/Rotation
RwSlowDeceILrn	REAL	0.0		True	False	Publish Only	Learned real slow-down decel (mm/s2) - 0 = not learned
CountReset_Req	BOOL			False	False	Publish Only	Counter reset requested
Psw	INT			False	False	Publish Only	Password Entry
I_RsetAlarm	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN08 (Pin 2.4) _6000_08	False	False	Publish Only	Alarm reset
EncoderWheel_Circumf	REAL	402		True	False	Publish Only	Encoder wheel (Cylinder / Roller) Circumference [mm]

AlarmCodeSV	ARRAY[0..32] OF BOOL			False	False	Publish Only	Panel alarms
SafeGuard_ByPass	BOOL			False	False	Publish Only	Guards bypassed
FrontGuard_Closed	BOOL			False	False	Publish Only	Front guard closed
SoftStop	BOOL	False		False	False	Publish Only	Soft stop in progress (HMI)
SoftStopEnable	BOOL	True		True	False	Publish Only	STOP button = soft stop (False = immediate as before)
SoftStopEndSpd	REAL	8.0		True	False	Publish Only	Soft stop: run command drops below this speed (m/min)
UwDiam_Measure	REAL			True	False	Publish Only	Measured distance
UwCoreDiameter	REAL	100		True	False	Publish Only	Core Outside diameter [mm]
Core_Measure	BOOL	False		True	False	Publish Only	Stored core value
UwCoreDistance	REAL	785.56		True	False	Publish Only	Sonar distance to empty core. [mm]
AI_Diam_Unwinder	INT		IOBus://unit#2/Ch1 Analog Input Value	False	False	Publish Only	Analog value - UnWinder Diameter
Measure_ON	BOOL	True		True	False	Publish Only	Unwinder Diameter measurement enabled
UwDiam_Act	REAL			True	False	Publish Only	Measured roll diameter
UwSlow_at_D_Val	REAL	200		True	False	Publish Only	Unwinder Slow-down diameter [mm]
UwStopDiam_Limit	REAL	110		True	False	Publish Only	Stop diameter
Alarm_Reset	BOOL			False	False	Publish Only	Alarm reset from panel
Psw22	BOOL			False	False	Publish Only	Password 22 for Popup 22
Psw44	BOOL			False	False	Publish Only	Password 44 for Popup 44
AI_Diam_Rewinder	INT		IOBus://unit#2/Ch2 Analog Input Value	False	False	Publish Only	Analog value - ReWinder Diameter
RwStopEnable	BOOL	False		True	False	Publish Only	Rewinder diameter stop enabled
RwTargetDiam	REAL	300.0		True	False	Publish Only	Rewinder final diameter (mm)
RwUseSlow	BOOL	True		True	False	Publish Only	Slow down before target
RwSlowMargin_m	REAL	10.0		True	False	Publish Only	Rewinder Metres at slow speed before stop
RwSlowDecel	REAL	0.0		True	False	Publish Only	Rewinder Slow-down decel mm/s2 - 0 = auto from DecStart
RwResetHyst	REAL	50.0		True	False	Publish Only	Reset requests below target - this [mm]
RwDist0V	REAL	0.0		True	False	Publish Only	DT35 distance at 0 V (mm)
RwDist10V	REAL	782.0		True	False	Publish Only	DT35 distance at 10 V (mm)
RwRawMin	INT	10		True	False	Publish Only	Raw below = sensor fault
RwRawMax	INT	3900		True	False	Publish Only	Raw above = sensor fault
RwStopOnFault	BOOL	True		False	False	Publish Only	Stop on sensor fault
RwAxisDist	REAL	727.0		True	False	Publish Only	Sensor to shaft centre (mm) - teach
RwFilter_K	REAL	0.2		True	False	Publish Only	Distance filter per 100 ms (

							0..1)
RwMinPulses	DINT	1000		True	False	Publish Only	Min pulses in window to learn
RwLearn_K	REAL	0.3		True	False	Publish Only	Material factor filter weight
RwMat_K	REAL	0.0		True	False	Publish Only	Learned material factor
RwLearnStop	BOOL	True		True	False	Publish Only	Learn stop decel on every stop
RwMinLearn Spd	REAL	15.0		True	False	Publish Only	Min speed to learn a stop (m/min)
RwStopDecel	REAL	0.0		True	False	Publish Only	Learned stop decel (pulses/s2)
RwTreact	REAL	0.03		True	False	Publish Only	Stop reaction time (s)
RwTstopDef	REAL	3.0		True	False	Publish Only	Stop time until decel learned (s)
RwTrim	REAL	1.0		True	False	Publish Only	Stop length trim (>1 = earlier)
RwDist	REAL			False	False	Publish Only	Filtered distance (mm)
RwDiam	REAL			False	False	Publish Only	Rewinder diameter (mm)
RwSpeed	REAL			False	False	Publish Only	Line speed from encoder (m/min)
RwRemain_ m	REAL			False	False	Publish Only	Metres to target diameter
RwStopLen_ m	REAL			False	False	Publish Only	Stopping length now (m)
RwSlowLen_ m	REAL			False	False	Publish Only	Rewinder Slow-down length [m]
RwStop_at_D _Req	BOOL			False	False	Publish Only	Rewinder diameter STOP request
RwSlow_at_ D_Req	BOOL			False	False	Publish Only	Rewinder diameter SLOW request
RwSensFault	BOOL			False	False	Publish Only	Rewinder DT35 sensor fault
UwStopEnabl e	BOOL	False		True	False	Publish Only	Unwinder Auto Stop - Enable
SlowAtDisp	REAL			False	False	Publish Only	HMI: slow-down starts at (m or mm, see SlowAtMode)
SlowAtMode	INT			False	False	Publish Only	HMI: 0=none 1=unwinder [m] 2=rewinder [mm]
SlowAtShow	BOOL			False	False	Publish Only	HMI: show slow-down display
UwSlow_at_ M_Val	REAL			False	False	Publish Only	Unwinder: slow-down starts at [m]
UwSlow_at_ M_Req	BOOL			False	False	Publish Only	Unwinder metre slow-down request (latched)
RwSlow_at_ D_Val	REAL			False	False	Publish Only	Rewinder slow-down starts at diameter [mm]
RwStop_at_D _Val	REAL			False	False	Publish Only	Rewinder: stop command at diameter (mm)
RwMaxSlow	BOOL			False	False	Publish Only	Rewinder max diameter SLOW (latched)
UwEmptyHy st	REAL	30.0		True	False	Publish Only	Unwinder Empty Hysteresis
GSM_Learne d_OK	BOOL			False	False	Publish Only	Paper [Grams / Square Meter] FALSE until GSM learned
RwMaxSlow AtDiam	REAL	1370.0		False	False	Publish Only	Rewinder max: slow-down starts at diameter (mm)
RwStopDecel K	REAL	0.0		True	False	Publish Only	Learned stop decel per mm of rewinder diameter (mm/s2 per mm)
SoftStopDece	REAL	500.0		True	False	Publish Only	Soft stop decel (mm/s2), 0

l							= DecelCoarse
SpdMax_Pending	BOOL	False		False	False	Publish Only	New pot range waiting for the motor to stop
RwAxisDist_Prev	REAL	727.0		True	False	Publish Only	Rewinder: RwAxisDist before the last TEACH (for UNDO)
UwCoreDist_Prev	REAL	785.56		True	False	Publish Only	Unwinder: UwCoreDistance before the last TEACH (for UNDO)
RwTeachUndo	BOOL	False		False	False	Publish Only	HMI: UNDO rewinder TEACH (PLC resets it)
UwTeachUndo	BOOL	False		False	False	Publish Only	HMI: UNDO unwinder TEACH (PLC resets it)
Psw33	BOOL			False	False	Publish Only	Password 33 for Popup 33
RwMatChangePct	REAL	35.0		True	False	Publish Only	Paper change detection: window deviation (%), 2 windows in a row = new paper, 0 = off
RwSlowFallbackMm	REAL	30.0		True	False	Publish Only	Rewinder: slow-down this many mm before the target while the paper thickness is not learned (0 = off)
RwAutoMaxSpd	REAL	500.0		True	False	Publish Only	Rewinder AUTO stop ON: maximum work speed (m/min), 0 = no limit
RwWinLen_m	REAL	60.0		True	False	Publish Only	Paper thickness learning: MAX window length (m of paper)
RwWinGrow_mm	REAL	5		True	False	Publish Only	Paper thickness learning: window = this diameter growth (mm), 4 m .. RwWinLen_m
GsmAsk	BOOL	False		False	False	Publish Only	HMI popup: new parent roll, did you change the paper? (HMI NO resets it)
GsmAskYes	BOOL	False		False	False	Publish Only	HMI popup YES: forget the learned thickness (PLC resets it)
DecelWeb	REAL	250.0		True	False	Publish Only	V7: AUTO decel ramp, real web mm/s2 (0 = DecelCoarse)
AccelWeb	REAL	200.0		True	False	Publish Only	AUTO accel ramp, real web mm/s2 (0 = AccelCoarse)
RwRatioD	REAL	3366.0		True	False	Publish Only	V7: drive counts per m/min x rewinder diameter (mm), adapted while running
FFEnable	BOOL	True		True	False	Publish Only	V7: feed-forward speed loop ON (FALSE = old loop of rung 16)
SpdHold	BOOL	False		False	False	Publish Only	V9: speed held after START during a soft stop, START again = ramp up (HMI lamp)
SonarAskYes	BOOL	False		False	False	Publish Only	V10 HMI popup YES: sonar ON (PLC resets it)
UwStop_at_D_Val	REAL	0.0		False	False	Publish Only	V11: unwinder stop command diameter (mm) so the web stops at UwDiamStop, HMI "STOP AT"
Clock10ms	BOOL			False	False	Publish Only	10mS clock
Clock100ms	bool			False	False	Publish Only	100mS clock

	Sec_to_Min	REAL			False	False	Do not publish	
	Psw48	BOOL			False	False	Publish Only	Password 48 for Popup 48
	JogRun	BOOL			False	False	Publish Only	V10: JOG really running (button, not a normal run, not a soft stop)
	GuardStopOK	BOOL			False	False	Publish Only	V11: stopped for real -> guard bypass allowed (Output 13)
	SlowDown	BOOL			False	False	Publish Only	Slow down
	WarningOn	bool			False	False	Publish Only	HMI Warning Display
	AlarmSilenced	BOOL			False	False	Publish Only	
	PaperGSM	REAL			True	False	Publish Only	
	WebWidth_mm	REAL			True	False	Publish Only	
	TensFactor	REAL			True	False	Publish Only	Tension Helper
	TensIdeal_Disp	REAL			True	False	Publish Only	Tension Helper
	TensLostWarn	BOOL	False		False	False	Publish Only	Tension lost while running (warning)
	UwSensTest	BOOL	False		False	False	Publish Only	Unwinder sonar fault (raw out of range 1 s)
	UwSensFault	BOOL			False	False	Publish Only	UnWinder Sensor Fault
	Thickness_Fault	BOOL			False	False	Publish Only	Thickness Not Learned !
	AutoStopWhy	INT			False	False	Publish Only	
	AutoStopDone	BOOL			False	False	Publish Only	
	AutoEnRefused	BOOL			False	False	Publish Only	
	AutoEnWhy	INT			False	False	Publish Only	
	TensWaitHz	REAL	1.0		True	False	Publish Only	
	StartBlocked	BOOL	True		False	False	Publish Only	
	GuardOpen	BOOL	False		False	False	Publish Only	
	SideDoorOpen	BOOL	False		False	False	Publish Only	
	TensDiaFrac	REAL	0.0		False	False	Publish Only	Rewinder diameter fraction 0 (core) .. 1 (max)
	Psw40	BOOL			False	False	Publish Only	Password 40 for Popup 40
	TensJogBoost	REAL	30.0		True	False	Publish Only	Threading preset: % extra tension from JOG until START, 0..100
	TensJogLatch	BOOL	False		False	False	Publish Only	Threading tension active (JOG -> START), HMI lamp
	TensTrim_kg	REAL	0.0		True	False	Publish Only	Operator trim on the AUTO tension (kg), +/-10, reset when paper changes